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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 322 — Past Questions & Materials
by Renance

Course: GET 322
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 300

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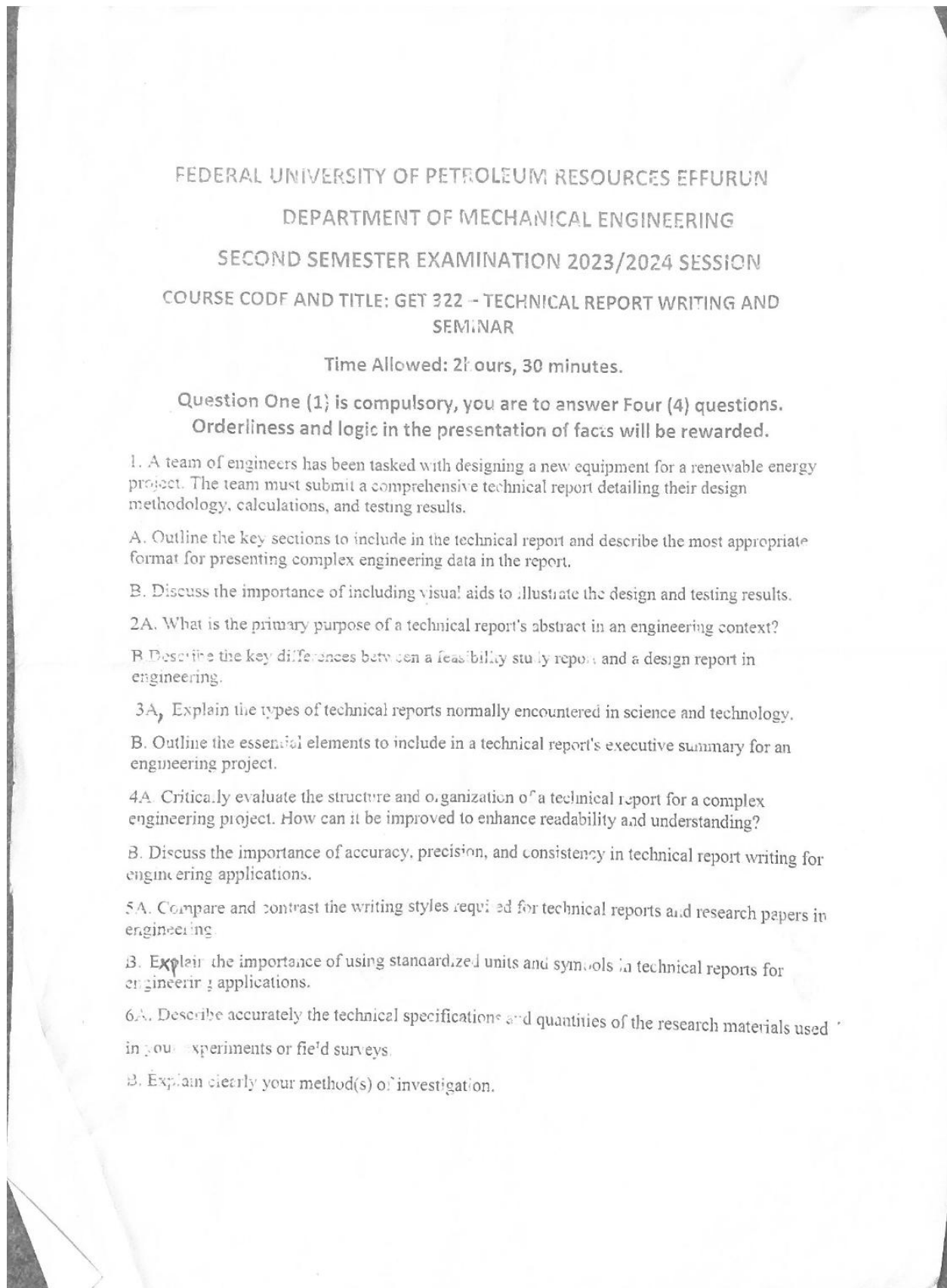
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Paper 1 of 5 — GET 322 Technical Report Writing and Seminar

Level 300 • First Semester • PastQuestion

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Paper 2 of 5 — GET 322 Technical Report Writing and Seminar

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
SECOND SEMESTER EXAMINATION 2023/2024 SESSION
COURSE CODE AND TITLE: GET 322 – TECHNICAL REPORT WRITING AND SEMINAR
Time Allowed: 2 hours, 30 minutes.

Question One (1) is compulsory, you are to answer Four (4) questions.
Orderliness and logic in the presentation of facts will be rewarded.

1. A team of engineers has been tasked with designing a new equipment for a renewable energy project. The team must submit a comprehensive technical report detailing their design methodology, calculations, and testing results.
 - A. Outline the key sections to include in the technical report and describe the most appropriate format for presenting complex engineering data in the report.
 - B. Discuss the importance of including visual aids to illustrate the design and testing results.
- 2A. What is the primary purpose of a technical report's abstract in an engineering context?
 - B. Describe the key differences between a feasibility study report and a design report in engineering.
- 3A. Explain the types of technical reports normally encountered in science and technology.
 - B. Outline the essential elements to include in a technical report's executive summary for an engineering project.
- 4A. Critically evaluate the structure and organization of a technical report for a complex engineering project. How can it be improved to enhance readability and understanding?
 - B. Discuss the importance of accuracy, precision, and consistency in technical report writing for engineering applications.
- 5A. Compare and contrast the writing styles required for technical reports and research papers in engineering.
 - B. Explain the importance of using standardized units and symbols in technical reports for engineering applications.
- 6A. Describe accurately the technical specifications and quantities of the research materials used in your experiments or field surveys.
 - B. Explain clearly your method(s) of investigation.

Paper 3 of 5 — GET 322 Past Question 2020 2021 Second Semester

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY SECOND SEMESTER EXAMINATION -
TECHNICAL WRITING AND PRESENTATION (GET 322) – JULY 2021

Instructions: The questions are divided into two sections A and B. Section A (Question 1) is compulsory. Answer any other three questions in Section B (4 QUESTIONS IN ALL).

Do Not Write on your question paper!

Duration: ONE HOUR

SECTION A [40 marks]

- 1 A) How critical is the consideration for the "Audience" in Technical report writing and presentation?
- 1B) List and write short notes on *at least FIVE MAJOR components* of a technical report.
- 1C) List three attributes for effective technical presentation.
- 1D) Explain **brevity** as it relates to Technical Writing and Presentation. Give an example.

SECTION B [20 marks PER QUESTION]

- 2) State "True" or "False" to each of the following statements:

- i. A formal technical report must have a Table of Content
- ii. All technical reports must have appendices
- iii. An "Abstract" is the same as summary
- iv. List of tables should come before "List of figures"
- v. All tables in a report must not be numbered
- vi. The "Title" of a figure should be at the top of the figure
- vii. The "Title" of a table should be at the bottom of the table
- viii. It does not matter if a Table exceeds one page
- ix. The word "Table" can be abbreviated in a report
- x. The section on bibliography is not necessary in a proposal for project

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- 3) What is the fundamental purpose of technical writing? List five critical technical writing -best practices.

- 4) Give three examples of processes or operations where technical writing and presentations are necessary.

- 5) What is plagiarism? List at least three ways to avoid plagiarism in Technical Writing and Presentation?

- 6 A) What is information literacy? List at least three attributes of information literacy.

- 6 B) Differentiate between Vancouver and APA referencing formats.

- 6 C) State the name of your "GET 322 Project Team" and describe your role in the preparation and presentation of the team project.

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Paper 4 of 5 — GET 322 TECHNICAL REPORT WRITING AND SEMINAR

Level 300 • Second Semester • PastQuestion

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DEPARTMENT OF MECHANICAL ENGINEERING

SECOND SEMESTER EXAMINATION 2023/2024 SESSION

COURSE CODF AND TITLE: GET 322 ~ TECHNICAL REPORT WRITING AND SENANAR

Time Allowed: 2h ours, 30 minutes.

Question One (1} is compulsory, you are to answer Four (4) questions.

Orderiiness and logic in the presentation of facis will be rewarded.

i. A team of engineers has been tasked with designing a new equipment for a renewable energy proeet. The team must submit a comprehensive technical report detailing their design methodology, calculations, and testing results.

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Sa. Desenbe accurately the technic! specifications snd quantities of the research materials used in; Ou "xpperiments or fie'd surveys

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©. Explain clearly your methods of investigation.

Paper 5 of 5 — GET 322

Level 300 • Second Semester • Handout

GET 322 TECHNICAL WRITING AND PRESENTATION

GET 322: TECHNICAL WRITING AND PRESENTATION

Dr. C.O. Mgbemena

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GET 322 TECHNICAL WRITING AND PRESENTATION

Topics to consider:

1. Data gathering and presentation
2. Technical correspondence such as letters of inquiry and replies; Letters of application and memoranda
3. Illustrating technical writing using tables, graphs, diagrams, equations and appendices
4. Writing progress reports, proposals, student projects, thesis and dissertations
5. Oral and visual presentations
6. Computer aided technical writing and presentation using word processing and presentation software packages

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GET 322 TECHNICAL WRITING AND PRESENTATION

CHAPTER THREE

ILLUSTRATING TECHNICAL WRITING USING TABLES, GRAPHS, DIAGRAMS, EQUATIONS AND APPENDICES

3.1 Reasons for Using Graphics

Graphics are invaluable aids in written and oral reports. They are used to clarify relationships, condense text and highlight patterns. Good graphics display the significance of your data (which may be more exactly displayed in a table) and allow the reader to follow your discussion. Drafting graphics may also be a very effective way to help you draft a long written report or oral presentation.

3.2 Common Graphics

Graphics commonly used in technical documents include the following:

- o Working drawings
- o Tables
- o Bar graphs
- o Line graphs
- o Pie graphs

- o Illustrations
- o Diagrams
- o Photographs
- o Exploded views
- o Schematic diagrams
- o Flowcharts
- o Timetables or Gantt charts

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i. Working Drawings

Working drawings may be used to give detailed descriptions of the current state of your experimental apparatus and as common graphics in oral design reviews.

Working drawings may be highly detailed and may present exact specifications for the construction of apparatus. Identify working drawings with an explicit title and the date. Be sure to show tolerances and exact measurements and, if necessary, a scale of measurement on all figures.

Figure 3.1 A typical working drawing

ii. Tables

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Tables present data in a highly condensed form. They present data more exactly than a graph, but they do not readily display the trends within your data. (They should rarely be used in an oral presentation--rely on figures instead.)

Every table is identified in a written report by a number and a title, placed above the table. In contrast, the number and title of a figure in a report are usually given below the figure.

The parts of a table include

- Title and number (above the table).
- Box head, the horizontal region across the top of a table containing column headings with units of measurement clearly identified.
- Stub, the vertical column to the far left of a table in which you list the various line headings that identify the horizontal rows of data in the body of the table.
- Body, all the data, presented in columns below the box head, describing items in the stub. If you want the reader to see a comparison between any items in a table, then place those items close to one another.

If any table or figure is taken from another source, proper credit must be given in a source note below the table or figure.

Table 2. The volume of solution in ml delivered from a random sample of 1.00 ml capacity tuberculin syringes with Luer slip tips and 9.5 mm 27

gauge needles with intradermal bevels.

OBS GRADUATION

0.05 ml 0.10 ml 0.15 ml 0.20 ml 0.25 ml 1.00 ml

1 0.0466 0.0895 0.1405 0.1949 0.2429 1.0012

2 0.0474 0.0998 0.1505 0.2033 0.2514 0.9994

3 0.0487 0.0972 0.1473 0.1978 0.2451 1.0052

4 0.0505 0.0935 0.1472 0.1978 0.2502 1.0016

5 0.0442 0.0935 0.1443 0.1982 0.2450 1.0031

--Seth Frisbie, The Determination of Total Dissolved Inorganic Carbon in Pure Systems

iii. Bar Graphs

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Bar graphs display relationships among data by means of vertical or horizontal bars of different lengths. Sometimes a single bar of a set height is used to break down percentages of the whole, in much the same fashion as a pie graph.

Single Bar Graph

Single bar graphs visualize the effects of varying limiting conditions on one object of study, as in Figure 5, where heart rate is shown under various burden conditions of exercise.

Multiple Bar Graph

Multiple bar graphs visualize the different objects of study under one limiting condition, as in Figure 6, where three items are compared over time.

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Stacked Bar Graph

Stacked bar graphs visualize various items as percentages of the whole for ease of comparison and contrast. Figure 7 is a single bar graph showing percentages of one object of study. (Compare this figure with Figure 11, a pie graph drawn from the same data.) Figure 2 is an example of a multiple stacked bar graph where various items are broken down into percentages of the whole for comparison.

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iv. Line Graphs

Line graphs visualize trends among dense data sets, which are sometimes listed in an accompanying table in a report. Data points are plotted with relation to a vertical axis showing the dependent variable and a horizontal axis showing the independent

variable. A line is then drawn through these points to display significant trends, as shown in Figure 8, an example of a single line graph.

The intersection of the x and y axes is always the zero point. Put a break point on the y axis if the data span a range too large to fit in your graph, as shown in Figure 9.

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If for some reason zero is not the starting point for your axes (because the values are too high, for example), then state that explicitly in prose within the figure or indicate it on the axes themselves.

To use line graphs effectively, follow these guidelines:

- Label each axis clearly and show units of measurement by means of regularly spaced ticks.
- Show significant data points plotted between the x and y axes clearly. If more than one set of data is plotted, you may want to distinguish data points by means of different symbols (circle, square, triangle), which are then identified in an accompanying key or legend.
- Keep your graph clear of excessive grid lines which may obscure your data points or curves.
- Clearly label plotted lines.

Multiple Line Graphs

Multiple line graphs show comparison between two (or more) data sets for the same value, as shown in Figure 10, in which data for the item of study in Figure 8 are not compared with data for another item.

v. Pie Graphs

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Pie graphs pictorially represent percentages of the whole by showing these percentages as "slices" of a complete circle (the complete circle represents 100 percent of whatever item or quantity you are discussing). Be sure that the percentages of the whole represented by the slices total 100 percent. The number of percentage slices drawn on a pie graph may vary. Too many segments may make your graph unreadable; too few segments may make it useless for discussion. A carefully sized and composed pie graph will permit you to include more than the usually recommended five items, affording you greater scope in making comparisons, as shown in Figure 11.

Label all sections clearly with the percentage and the name of the item being depicted.

Pie graphs give a striking and quick representation of simplified data. They are most often used before nonexpert audiences because the relationships of data

are not highly detailed

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vi. Illustrations

Illustrations can provide your reader with a large amount of information about an object or topic in a very small space. An illustration can accurately depict the form of an object, help the reader to visualize how the object functions, or show the relationship of one object to another. It allows you to focus your audience's attention precisely on the details that you are describing, as shown in Figure 12.

To use illustrations effectively, follow these guidelines:

□ Choose appropriate objects or topics to illustrate. Illustrations are excellent forms of graphics for situations in which colour distinctions are limited or irrelevant; if colour distinctions are the focus of your graphic, you may find a photograph to be more effective. On the other hand, if you will be printing your document in black and white or photocopying it onto acetates

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for overhead projection, an illustration may result in a graphic that is clearer and easier to interpret.

□ Make your illustrations clear and easy to understand. Incorporate only those details that will be relevant.

□ If the size of the elements in your illustration is relevant to your discussion, provide the scale for your readers.

□ Label the elements of your illustration clearly.

□ Identify each illustration with a figure number and a caption.

Figure 13 presents several overviews of an object, illustrating the overall form of the object as well as the relationship of parts to whole.

vii. Diagrams

Diagrams are used to show the relationships between several objects or to portray the function of an object. Diagrams are similar to illustrations in that they are often pictorial representations of an object, but they are often more abstract than illustrations. For instance, the elements that make up a diagram may not be drawn to

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scale with respect to each other; rather than give an accurate picture of the appearance of an object, the goal of a diagram is to help the reader understand how the elements in the diagram are connected.

Figure 14 is an abstract, pictorial diagram visualizing a transfer line setup connecting

a holding tank to a tank car.

Figure 15 is a pictorial representation of an experimental design for a coolable nozzle, showing the interrelationships of parts.

To use diagrams effectively, follow these guidelines:

- Keep your diagrams simple. Include only those details that will be relevant to your discussion.
- Label the elements of your diagram clearly.
- Identify each diagram with a figure number and a caption

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viii. Photographs

Photographs can provide an overall view of an object, especially when the object is new or strange, or they may record results that rely on visual inspection, such as X-rays or the effects of destructive testing for failure in materials. To use photographs in a report effectively, you must

- Frame your photograph to include only the elements you need to discuss. Extraneous items will only confuse the reader.
- Use only clearly focused photographs, free of shadows. Poor photography frequently mars otherwise effective report presentations. Remember, photographs are hard to reproduce well in printed reports, so you may want to limit your use of them.
- Provide a way of showing relative size in a photograph (unless your photograph is of something very large-scale, such as a landscape or a whole factory). Show relative size in a photograph by placing some commonplace object of fixed size (such as a ruler, a coin or a pencil) next to the object being photographed.
- Identify each photograph with a figure number and a caption, as you would any other kind of figure in a written report.

If you need photographs for an oral presentation, use a 35-millimeter projector to display them. Photographs transferred to overhead acetates by photocopying machines are usually too dark to be easily viewed by the audience.

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ix. Exploded Views

Exploded views present the interrelated parts of a complex object in near proximity so that your audience can see at a glance the total effect of the placement or interaction of part to whole, as shown in Figure 18.

All parts of an object in an exploded view are identified by name. The figure is given a number and a caption in a written report. Figures of this type are useful for quick,

at-a-glance introduction of complex objects (sometimes, the point of an exploded view is to show just how complex a familiar object can be). Exploded views do not
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allow you to focus on parts of the apparatus, however. Therefore, combine this kind of illustration with more detailed illustrations for a complete analysis.

x. Schematic Diagrams

As with other kinds of diagrams, which stress the general or abstract over the realistic, schematic diagrams employ highly specialized, technical symbols to depict the workings of a system or apparatus, as shown in Figure 19.

Schematic diagrams should only be used to communicate with a technical and expert audience.

xi. Flowcharts

A flowchart is traditionally defined as a detailed diagram of the operations or equipment through which material passes. But a flowchart can be any kind of schematic diagram which shows pictorially or by means of symbolic representations a sequence of operations for any process. A common form of flowchart depicts the sequence of subroutines and flow of information within computer programs, as shown in Figure 20.

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xii. Timetables (Gantt charts)

Timetables, or Gantt charts, give a visual presentation of the relative timing of the separate tasks that make up a unified project. With just a glance at a timetable, a reader can ascertain the variety of tasks involved in a project and how the tasks are related chronologically, as shown in Figure 21.

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There are a variety of software programs you can use to generate and update your timetables automatically.

3.3 General Guidelines for Graphics

To use graphics effectively, the following steps are taken:

□ Label them accurately

All parts of figures and graphs must be labelled accurately. Use callouts (clarifying text connected to relevant part of figure by means of an arrow) to name the parts of your figure or data curves on a graph. Text should be printed on the horizontal when possible. Figures 22 and 23 provide examples.

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□ Number them consecutively

Give all figures and tables in written reports an identifying number and title so you can refer to them within reports by means of that number. Number your figures sequentially, starting from 1. In large reports with many sections, use a double numeration system. In this system, the number consists of the number of the section in which the graphic appears plus the sequential number of the graphic within that section (for example, "Fig. 4.3"). This will prevent you from having to give figures confusingly high reference numbers. In general, number and title go below graphs and figures but above tables. In an oral report, figures and graphs need not be numbered. Give every viewgraph in an oral report a title placed at the top of

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the viewgraph. Also, give numbers and titles to figures and graphs placed in an appendix to a report. Use a number that is a composite of the appendix number or letter and the sequential number of the figure.

□ Integrate graphics and text

Place each graphic within the text of your report as close as possible to its text reference. This integration of text and figure will help your reader understand the content and argument of your report, as, for example, when you are explaining an apparatus or discussing results. Avoid referring to a page number for any graphic: it is likely that published reports will not maintain the integrity of your pagination

3.4 Illustrating Graphics: Some key examples

Example 1

The charts below show the levels of participation in education and science in developing and industrialised countries in 1980 and 1990. Write a report for a university lecturer not less than 150 words describing the information shown below.

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Sample Answer 1:

The given bar charts compare the educational involvements of students in terms of total years of education and their participation in science-related subjects and study in developing and industrialised countries from 1980 to 1990.

According to the first bar graph, the average years of schooling in developing countries was only about 2 while this was over 8 years in industrialised countries in 1980. After ten years, the number of year in school on an average increased both in the developing and industrialised countries reaching just over 3 years in developing countries while over 10 years in developed countries.

According to the second graph, the number of Scientist and Technician was much higher in developed countries than the developing countries. The percentage of Scientist and Technician in developing countries both in 1980 and 1990 was below 20 while this was around 50 on an average in developed countries.

Finally, the third graph represents that the amount spent on Research and Development in industrially developed countries was much higher than the spending in developing countries. In 1980 the spending was around 50 billion US dollars in developing countries while this amount was more than 150 billion US dollar in developed countries. Interestingly, after 10 years the spending in Research &

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Development in developed countries was doubled while that amount decreased almost by 50% in developing countries.

Sample Answer 2:

The given bar graphs compare the educational involvements of students in term of total years of education and their participation in science-related subjects in developing and industrial countries from the year 1980 to 1990.

As is presented in the first bar graph, average years of schooling in developing countries were approximately 2 years while in industrial countries schooling years were over 8 years in 1980. Interestingly after one decade schooling years in developing countries magnified to over 3 years while it surged to over 10 years in industrial countries.

As is observed from the second bar graph, the number of scientists and technicians were much higher in industrial countries than in developing countries. Percentage of scientists and technicians were less than 20 percent in developing countries from 1980 to 1990 while it was just over 40 percent in 1980 which soared to approximately 70 percent in 1990.

According to the third bar graph, it can be seen clearly, spending on research and development in industrial countries was far higher than developing countries. The spending in 1980 in developing countries was around 50 billion US dollars while it was 150 billion dollars in industrial countries. Ultimately, after a decade spending on research and development slumped significantly almost 50 percent while spending in industrial countries peaked to approximately 350 billion US dollars respectively.

Example 2

The graph shows Internet Usage in Taiwan by Age Group, 1998-2000.

Summarise the information by selecting and reporting the main features, and make comparisons where relevant. You should write at least 150 words.

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Sample Answer 1:

The given data present the information about the internet users in four age groups in Taiwan between the years 1998 to 2000. As is presented in the bar graph, people from 16 to 30 years old of Taiwan used the internet more than people of other age group and over the time internet users from younger and aged people increased. According to the given data, in 1998 more than 50% internet users of Taiwan were above 16 years old and below 30 years old. People from 31 to 50 years old were 41% of total internet users. Very fewer people over 50 years old or less than 15 years used the internet in Taiwan in this year. In next year, the percentage of younger and aged users of internet increased to 8% of total internet users and internet users from other two age groups decreased slightly. In the year 2000, the same trend can be observed and aged internet users increased to 10% while the percentage of internet users among 16 to 50 years decreased by 1-2%. As is observed from the bar graph, the internet is popular among the 16 to 50 years old people in Taiwan and over times the younger and senior citizens' affection and needs towards internet usages increased.

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Sample Answer 2:

The bar chart illustrates the percentages of internet users in Taiwan in three different age groups between 1998 and 2000.

It is clear that the highest internet user was the young age group (aged between 16 and 30 years), though the rate had decreased over the abovementioned period, while the lowest was the junior age people (aged below 15 years old). In 1998, the highest population of internet users in Taiwan was the young people, as their figure was accounted to 53%. Over the years, it had dropped drastically to 44% in 2000. As for the middle age people (aged between 31 and 50 years), their figure had decreased slightly from 41% in 1998 to 37% in 2000.

The senior people (aged 50 years and above) and the junior people figures were 4% and 2% 1998. As the senior people's percentage grew by two-fold, the junior group rate also increased by 6% in 1999. And then both figures continued to increase as they reached 10% and 9% in 2000.

3.5 Handling equations

The equations can be handled using the following:

- Equation editor (Microsoft Equation 3.0)
- LaTeX equation editor
- MathType
- Formular editor
- MathMagic
- Apache Openoffice math

3.6 Appendices

An appendix (one item) OR appendices (more than one item) is information that is NOT ESSENTIAL to explain your findings in the essay or report that you have written. However, this information may support your analysis and validate your conclusions.

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Some of the items may have been written by yourself or printed/photocopied from elsewhere, so there are special rules that you must follow.

3.6.1 When to use appendices

The body of the text must be complete without the appendices, and it must contain all information including tables, diagrams and results necessary to answer the question or support the thesis. Therefore, you will need to know that:

- Appendices are used when the incorporation of material in the body of the work would make it poorly structured or too long and detailed.
- Appendices may be used for helpful, supporting or essential material that would otherwise clutter, break up or be distracting to the text.
- Other people's work in the appendix will be referred to (e.g. see Appendix 3), not quoted (e.g. using short or long quotes) from the appendix.
- Appendices must be referred to in the body of the text, for example, 'details of the questionnaire are given in Appendix B (on page 23)'.
- Appendices are not included in the word count.

3.6.2 Examples of items in appendices

Appendices may include some of the following:

- supporting evidence (e.g. raw data that is referred to in the text)
- contributory facts, specialised data (raw data appear in the appendix, but summarised data appear in the body of the text.)
- sample calculations (referred to in the text)
- technical figures, graphs, tables, statistics (referred to in the text)
- detailed description of research instruments (referred to in the text)
- maps, charts, photographs, drawings (referred to in the text)
- letters, copies of emails (referred to in the text)
- questionnaires/surveys (questionnaire/survey results appear in the body of the text)

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- transcripts of interviews (summarised in the text)
- specification or data sheets (summarised in the text)

3.6.3 Examples of students' understandings about using appendices

Student 1 (reflective writing)

This example is about a two-page questionnaire that I designed and placed in the appendices. My understanding is that an appendix can be information that you used

to inform your writing that you don't want to put directly into the essay; however, you **MUST** refer to this appendix in the body of the essay. Therefore, in my Education essay, I included an appendix that was the actual questionnaire that I gave to my students when I was on teaching practice. In my essay, I wrote for the marker to 'see Appendix 1' when I had mentioned the results of the questionnaire I was required to write and use with my class.

Student 2 (reflective writing)

In my nursing assignment, my appendices were used to supplement the information in my essay. Firstly, we had to demonstrate our knowledge of the human skeletal system. The photocopied diagrams covered several pages, so Appendix 1 was a set of photocopied, labelled diagrams of the skeletal system discussed and referred to in the essay. Because the photocopied diagrams were copied from a book, I placed an in-text reference to this source in my writing. Secondly, we had to summarise the results of interviews with five patients at the local hospital about their recovery from bone breakages in car accidents, so Appendix 2 was a transcript of these interviews. Using appendices like this meant that I could show my lecturer that I had completed the set task and collected the data for the issues that were discussed in the essay.

3.6.4 How to format appendices

The format and positioning of appendices in your essay must follow a few procedures:

1. Appendices may follow the reference list.
2. Each appendix must begin on a new page.

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3. The order they are presented in is dictated by the order they are mentioned in the text of the report.
4. The heading should be written as APPENDIX (upper case) or Appendix (sentence case), followed by a letter or a number. Centre the title of each appendix, and change to bold type e.g. APPENDIX A, Appendix 1. Be consistent in your choice of style.
5. Appendices must be listed in the table of contents (if used).
6. The page number(s) of the appendix/appendices will continue with the numbering from the last page of the text.

WARNING

There is no limit to what can be placed in an appendix, but it must be relevant and referred to at the relevant point in a text (e.g. See Appendix 2). Do not attempt to overuse the appendix for vague or irrelevant information as this additional information will not contribute to your word count or affect essay marks unless it contributes directly to the essay answer.

ASSIGNMENT #3

The graphs shown below give information about computer ownership as a percentage of the population between 2002 and 2010, and by level of education for the years 2002 and 2010.

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Figure 1 Computer ownership, 2002-10

Figure 2 Computer ownership by education level, 2002 and 2010

Task

Summarise the information by selecting and reporting the main features, and make comparisons where relevant (Write at least 150 words).

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GET 322: TECHNICAL WRITING AND PRESENTATION

Dr. C.O. Mgbemena

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Topics to consider:

1. Data gathering and presentation
2. Technical correspondence such as letters of inquiry and replies; Letters of application and memoranda
3. Illustrating technical writing using tables, graphs, diagrams, equations and appendices
4. Writing progress reports, proposals, student projects, thesis and dissertations
5. Oral and visual presentations
6. Computer aided technical writing and presentation using word

processing and presentation software packages

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CHAPTER THREE

ILLUSTRATING TECHNICAL WRITING USING TABLES, GRAPHS, DIAGRAMS, EQUATIONS AND APPENDICES

3.1 Reasons for Using Graphics

Graphics are invaluable aids in written and oral reports. They are used to clarify relationships, condense text and highlight patterns. Good graphics display the significance of your data (which may be more exactly displayed in a table) and allow the reader to follow your discussion. Drafting graphics may also be a very effective way to help you draft a long written report or oral presentation.

3.2 Common Graphics

Graphics commonly used in technical documents include the following:

- o Working drawings
- o Tables
- o Bar graphs
- o Line graphs
- o Pie graphs
- o Illustrations
- o Diagrams
- o Photographs
- o Exploded views
- o Schematic diagrams
- o Flowcharts
- o Timetables or Gantt charts

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i. Working Drawings

Working drawings may be used to give detailed descriptions of the current state of your experimental apparatus and as common graphics in oral design reviews.

Working drawings may be highly detailed and may present exact specifications for the construction of apparatus. Identify working drawings with an explicit title and the date. Be sure to show tolerances and exact measurements and, if necessary, a scale of measurement on all figures.

Figure 3.1 A typical working drawing

ii. Tables

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Tables present data in a highly condensed form. They present data more exactly than a graph, but they do not readily display the trends within your data. (They should rarely be used in an oral presentation--rely on figures instead.)

Every table is identified in a written report by a number and a title, placed above the table. In contrast, the number and title of a figure in a report are usually given below the figure.

The parts of a table include

- Title and number (above the table).
- Box head, the horizontal region across the top of a table containing column headings with units of measurement clearly identified.
- Stub, the vertical column to the far left of a table in which you list the various line headings that identify the horizontal rows of data in the body of the table.
- Body, all the data, presented in columns below the box head, describing items in the stub. If you want the reader to see a comparison between any items in a table, then place those items close to one another.

If any table or figure is taken from another source, proper credit must be given in a source note below the table or figure.

Table 2. The volume of solution in ml delivered from a random sample of 1.00 ml capacity tuberculin syringes with Luer slip tips and 9.5 mm 27 gauge needles with intradermal bevels.

OBS GRADUATION

0.05 ml 0.10 ml 0.15 ml 0.20 ml 0.25 ml 1.00 ml

1 0.0466 0.0895 0.1405 0.1949 0.2429 1.0012

2 0.0474 0.0998 0.1505 0.2033 0.2514 0.9994

3 0.0487 0.0972 0.1473 0.1978 0.2451 1.0052

4 0.0505 0.0935 0.1472 0.1978 0.2502 1.0016

5 0.0442 0.0935 0.1443 0.1982 0.2450 1.0031

--Seth Frisbie, The Determination of Total Dissolved Inorganic Carbon in Pure Systems

iii. Bar Graphs

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Bar graphs display relationships among data by means of vertical or horizontal bars of different lengths. Sometimes a single bar of a set height is used to break down percentages of the whole, in much the same fashion as a pie graph.

Single Bar Graph

Single bar graphs visualize the effects of varying limiting conditions on one object of

study, as in Figure 5, where heart rate is shown under various burden conditions of exercise.

Multiple Bar Graph

Multiple bar graphs visualize the different objects of study under one limiting condition, as in Figure 6, where three items are compared over time.

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Stacked Bar Graph

Stacked bar graphs visualize various items as percentages of the whole for ease of comparison and contrast. Figure 7 is a single bar graph showing percentages of one object of study. (Compare this figure with Figure 11, a pie graph drawn from the same data.) Figure 2 is an example of a multiple stacked bar graph where various items are broken down into percentages of the whole for comparison.

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iv. Line Graphs

Line graphs visualize trends among dense data sets, which are sometimes listed in an accompanying table in a report. Data points are plotted with relation to a vertical axis showing the dependent variable and a horizontal axis showing the independent variable. A line is then drawn through these points to display significant trends, as shown in Figure 8, an example of a single line graph.

The intersection of the x and y axes is always the zero point. Put a break point on the y axis if the data span a range too large to fit in your graph, as shown in Figure 9.

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If for some reason zero is not the starting point for your axes (because the values are too high, for example), then state that explicitly in prose within the figure or indicate it on the axes themselves.

To use line graphs effectively, follow these guidelines:

- Label each axis clearly and show units of measurement by means of regularly spaced ticks.
- Show significant data points plotted between the x and y axes clearly. If more than one set of data is plotted, you may want to distinguish data points by means of different symbols (circle, square, triangle), which are then identified in an accompanying key or legend.
- Keep your graph clear of excessive grid lines which may obscure your data points or curves.
- Clearly label plotted lines.

Multiple Line Graphs

Multiple line graphs show comparison between two (or more) data sets for the same value, as shown in Figure 10, in which data for the item of study in Figure 8 are not compared with data for another item.

v. Pie Graphs

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Pie graphs pictorially represent percentages of the whole by showing these percentages as "slices" of a complete circle (the complete circle represents 100 percent of whatever item or quantity you are discussing). Be sure that the percentages of the whole represented by the slices total 100 percent. The number of percentage slices drawn on a pie graph may vary. Too many segments may make your graph unreadable; too few segments may make it useless for discussion. A carefully sized and composed pie graph will permit you to include more than the usually recommended five items, affording you greater scope in making comparisons, as shown in Figure 11.

Label all sections clearly with the percentage and the name of the item being depicted.

Pie graphs give a striking and quick representation of simplified data. They are most often used before nonexpert audiences because the relationships of data are not highly detailed

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vi. Illustrations

Illustrations can provide your reader with a large amount of information about an object or topic in a very small space. An illustration can accurately depict the form of an object, help the reader to visualize how the object functions, or show the relationship of one object to another. It allows you to focus your audience's attention precisely on the details that you are describing, as shown in Figure 12.

To use illustrations effectively, follow these guidelines:

□ Choose appropriate objects or topics to illustrate. Illustrations are excellent forms of graphics for situations in which colour distinctions are limited or irrelevant; if colour distinctions are the focus of your graphic, you may find a photograph to be more effective. On the other hand, if you will be printing your document in black and white or photocopying it onto acetates

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for overhead projection, an illustration may result in a graphic that is clearer and easier to interpret.

- Make your illustrations clear and easy to understand. Incorporate only those details that will be relevant.
- If the size of the elements in your illustration is relevant to your discussion, provide the scale for your readers.
- Label the elements of your illustration clearly.
- Identify each illustration with a figure number and a caption.

Figure 13 presents several overviews of an object, illustrating the overall form of the object as well as the relationship of parts to whole.

vii. Diagrams

Diagrams are used to show the relationships between several objects or to portray the function of an object. Diagrams are similar to illustrations in that they are often pictorial representations of an object, but they are often more abstract than illustrations. For instance, the elements that make up a diagram may not be drawn to
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scale with respect to each other; rather than give an accurate picture of the appearance of an object, the goal of a diagram is to help the reader understand how the elements in the diagram are connected.

Figure 14 is an abstract, pictorial diagram visualizing a transfer line setup connecting a holding tank to a tank car.

Figure 15 is a pictorial representation of an experimental design for a coolable nozzle, showing the interrelationships of parts.

To use diagrams effectively, follow these guidelines:

- Keep your diagrams simple. Include only those details that will be relevant to your discussion.
- Label the elements of your diagram clearly.
- Identify each diagram with a figure number and a caption

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viii. Photographs

Photographs can provide an overall view of an object, especially when the object is new or strange, or they may record results that rely on visual inspection, such as X-rays or the effects of destructive testing for failure in materials. To use photographs in a report effectively, you must

- Frame your photograph to include only the elements you need to discuss. Extraneous items will only confuse the reader.
- Use only clearly focused photographs, free of shadows. Poor photography frequently mars otherwise effective report presentations. Remember, photographs are hard to reproduce well in printed reports, so you may want

to limit your use of them.

□ Provide a way of showing relative size in a photograph (unless your photograph is of something very large-scale, such as a landscape or a whole factory). Show relative size in a photograph by placing some commonplace object of fixed size (such as a ruler, a coin or a pencil) next to the object being photographed.

□ Identify each photograph with a figure number and a caption, as you would any other kind of figure in a written report.

If you need photographs for an oral presentation, use a 35-millimeter projector to display them. Photographs transferred to overhead acetates by photocopying machines are usually too dark to be easily viewed by the audience.

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ix. Exploded Views

Exploded views present the interrelated parts of a complex object in near proximity so that your audience can see at a glance the total effect of the placement or interaction of part to whole, as shown in Figure 18.

All parts of an object in an exploded view are identified by name. The figure is given a number and a caption in a written report. Figures of this type are useful for quick, at-a-glance introduction of complex objects (sometimes, the point of an exploded view is to show just how complex a familiar object can be). Exploded views do not

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allow you to focus on parts of the apparatus, however. Therefore, combine this kind of illustration with more detailed illustrations for a complete analysis.

x. Schematic Diagrams

As with other kinds of diagrams, which stress the general or abstract over the realistic, schematic diagrams employ highly specialized, technical symbols to depict the workings of a system or apparatus, as shown in Figure 19.

Schematic diagrams should only be used to communicate with a technical and expert audience.

xi. Flowcharts

A flowchart is traditionally defined as a detailed diagram of the operations or equipment through which material passes. But a flowchart can be any kind of schematic diagram which shows pictorially or by means of symbolic representations a sequence of operations for any process. A common form of flowchart depicts the sequence of subroutines and flow of information within computer programs, as shown in Figure 20.

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xii. Timetables (Gantt charts)

Timetables, or Gantt charts, give a visual presentation of the relative timing of the separate tasks that make up a unified project. With just a glance at a timetable, a reader can ascertain the variety of tasks involved in a project and how the tasks are related chronologically, as shown in Figure 21.

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There are a variety of software programs you can use to generate and update your timetables automatically.

3.3 General Guidelines for Graphics

To use graphics effectively, the following steps are taken:

□ Label them accurately

All parts of figures and graphs must be labelled accurately. Use callouts (clarifying text connected to relevant part of figure by means of an arrow) to name the parts of your figure or data curves on a graph. Text should be printed on the horizontal when possible. Figures 22 and 23 provide examples.

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□ Number them consecutively

Give all figures and tables in written reports an identifying number and title so you can refer to them within reports by means of that number. Number your figures sequentially, starting from 1. In large reports with many sections, use a double numeration system. In this system, the number consists of the number of the section in which the graphic appears plus the sequential number of the graphic within that section (for example, "Fig. 4.3"). This will prevent you from having to give figures confusingly high reference numbers. In general, number and title go below graphs and figures but above tables. In an oral report, figures and graphs need not be numbered. Give every viewgraph in an oral report a title placed at the top of

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the viewgraph. Also, give numbers and titles to figures and graphs placed in an appendix to a report. Use a number that is a composite of the appendix number or letter and the sequential number of the figure.

□ Integrate graphics and text

Place each graphic within the text of your report as close as possible to its text reference. This integration of text and figure will help your reader understand the content and argument of your report, as, for example, when you are explaining

an apparatus or discussing results. Avoid referring to a page number for any graphic: it is likely that published reports will not maintain the integrity of your pagination

3.4 Illustrating Graphics: Some key examples

Example 1

The charts below show the levels of participation in education and science in developing and industrialised countries in 1980 and 1990. Write a report for a university lecturer not less than 150 words describing the information shown below.

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Sample Answer 1:

The given bar charts compare the educational involvements of students in terms of total years of education and their participation in science-related subjects and study in developing and industrialised countries from 1980 to 1990.

According to the first bar graph, the average years of schooling in developing countries was only about 2 while this was over 8 years in industrialised countries in 1980. After ten years, the number of year in school on an average increased both in the developing and industrialised countries reaching just over 3 years in developing countries while over 10 years in developed countries.

According to the second graph, the number of Scientist and Technician was much higher in developed countries than the developing countries. The percentage of Scientist and Technician in developing countries both in 1980 and 1990 was below 20 while this was around 50 on an average in developed countries.

Finally, the third graph represents that the amount spent on Research and Development in industrially developed countries was much higher than the spending in developing countries. In 1980 the spending was around 50 billion US dollars in developing countries while this amount was more than 150 billion US dollar in developed countries. Interestingly, after 10 years the spending in Research &

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Development in developed countries was doubled while that amount decreased almost by 50% in developing countries.

Sample Answer 2:

The given bar graphs compare the educational involvements of students in term of total years of education and their participation in science-related subjects in developing and industrial countries from the year 1980 to 1990.

As is presented in the first bar graph, average years of schooling in developing countries were approximately 2 years while in industrial countries schooling years were over 8 years in 1980. Interestingly after one decade schooling years in

developing countries magnified to over 3 years while it surged to over 10 years in industrial countries.

As is observed from the second bar graph, the number of scientists and technicians were much higher in industrial countries than in developing countries. Percentage of scientists and technicians were less than 20 percent in developing countries from 1980 to 1990 while it was just over 40 percent in 1980 which soared to approximately 70 percent in 1990.

According to the third bar graph, it can be seen clearly, spending on research and development in industrial countries was far higher than developing countries. The spending in 1980 in developing countries was around 50 billion US dollars while it was 150 billion dollars in industrial countries. Ultimately, after a decade spending on research and development slumped significantly almost 50 percent while spending in industrial countries peaked to approximately 350 billion US dollars respectively.

Example 2

The graph shows Internet Usage in Taiwan by Age Group, 1998-2000.

Summarise the information by selecting and reporting the main features, and make comparisons where relevant. You should write at least 150 words.

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Sample Answer 1:

The given data present the information about the internet users in four age groups in Taiwan between the years 1998 to 2000. As is presented in the bar graph, people from 16 to 30 years old of Taiwan used the internet more than people of other age group and over the time internet users from younger and aged people increased. According to the given data, in 1998 more than 50% internet users of Taiwan were above 16 years old and below 30 years old. People from 31 to 50 years old were 41% of total internet users. Very fewer people over 50 years old or less than 15 years used the internet in Taiwan in this year. In next year, the percentage of younger and aged users of internet increased to 8% of total internet users and internet users from other two age groups decreased slightly. In the year 2000, the same trend can be observed and aged internet users increased to 10% while the percentage of internet users among 16 to 50 years decreased by 1-2%. As is observed from the bar graph, the internet is popular among the 16 to 50 years old people in Taiwan and over times the younger and senior citizens' affection and needs towards internet usages increased.

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Sample Answer 2:

The bar chart illustrates the percentages of internet users in Taiwan in three different age groups between 1998 and 2000.

It is clear that the highest internet user was the young age group (aged between 16 and 30 years), though the rate had decreased over the abovementioned period, while the lowest was the junior age people (aged below 15 years old). In 1998, the highest population of internet users in Taiwan was the young people, as their figure was accounted to 53%. Over the years, it had dropped drastically to 44% in 2000. As for the middle age people (aged between 31 and 50 years), their figure had decreased slightly from 41% in 1998 to 37% in 2000.

The senior people (aged 50 years and above) and the junior people figures were 4% and 2% 1998. As the senior people's percentage grew by two-fold, the junior group rate also increased by 6% in 1999. And then both figures continued to increase as they reached 10% and 9% in 2000.

3.5 Handling equations

The equations can be handled using the following:

- Equation editor (Microsoft Equation 3.0)
- LaTeX equation editor
- MathType
- Formular editor
- MathMagic
- Apache Openoffice math

3.6 Appendices

An appendix (one item) OR appendices (more than one item) is information that is NOT ESSENTIAL to explain your findings in the essay or report that you have written. However, this information may support your analysis and validate your conclusions.

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Some of the items may have been written by yourself or printed/photocopied from elsewhere, so there are special rules that you must follow.

3.6.1 When to use appendices

The body of the text must be complete without the appendices, and it must contain all information including tables, diagrams and results necessary to answer the question or support the thesis. Therefore, you will need to know that:

- Appendices are used when the incorporation of material in the body of the work would make it poorly structured or too long and detailed.
- Appendices may be used for helpful, supporting or essential material that would otherwise clutter, break up or be distracting to the text.
- Other people's work in the appendix will be referred to (e.g. see Appendix 3), not quoted (e.g. using short or long quotes) from the appendix.
- Appendices must be referred to in the body of the text, for example, 'details of the questionnaire are given in Appendix B (on page 23)'.
- Appendices are not included in the word count.

3.6.2 Examples of items in appendices

Appendices may include some of the following:

- supporting evidence (e.g. raw data that is referred to in the text)
- contributory facts, specialised data (raw data appear in the appendix, but summarised data appear in the body of the text.)
- sample calculations (referred to in the text)
- technical figures, graphs, tables, statistics (referred to in the text)
- detailed description of research instruments (referred to in the text)
- maps, charts, photographs, drawings (referred to in the text)
- letters, copies of emails (referred to in the text)
- questionnaires/surveys (questionnaire/survey results appear in the body of the text)

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- transcripts of interviews (summarised in the text)
- specification or data sheets (summarised in the text)

3.6.3 Examples of students' understandings about using appendices

Student 1 (reflective writing)

This example is about a two-page questionnaire that I designed and placed in the appendices. My understanding is that an appendix can be information that you used to inform your writing that you don't want to put directly into the essay; however, you MUST refer to this appendix in the body of the essay. Therefore, in my Education essay, I included an appendix that was the actual questionnaire that I gave to my students when I was on teaching practice. In my essay, I wrote for the marker to 'see Appendix 1' when I had mentioned the results of the questionnaire I was required to write and use with my class.

Student 2 (reflective writing)

In my nursing assignment, my appendices were used to supplement the information in my essay. Firstly, we had to demonstrate our knowledge of the human skeletal system. The photocopied diagrams covered several pages, so Appendix 1 was a set of photocopied, labelled diagrams of the skeletal system discussed and referred to in the essay. Because the photocopied diagrams were copied from a book, I placed an in-text reference to this source in my writing. Secondly, we had to summarise the results of interviews with five patients at the local hospital about their recovery from bone breakages in car accidents, so Appendix 2 was a transcript of these interviews. Using appendices like this meant that I could show my lecturer that I had completed the set task and collected the data for the issues that were discussed in the essay.

3.6.4 How to format appendices

The format and positioning of appendices in your essay must follow a few procedures:

1. Appendices may follow the reference list.

2. Each appendix must begin on a new page.

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3. The order they are presented in is dictated by the order they are mentioned in the text of the report.

4. The heading should be written as APPENDIX (upper case) or Appendix (sentence case), followed by a letter or a number. Centre the title of each appendix, and change to bold type e.g. APPENDIX A, Appendix 1. Be consistent in your choice of style.

5. Appendices must be listed in the table of contents (if used).

6. The page number(s) of the appendix/appendices will continue with the numbering from the last page of the text.

WARNING

There is no limit to what can be placed in an appendix, but it must be relevant and referred to at the relevant point in a text (e.g. See Appendix 2). Do not attempt to overuse the appendix for vague or irrelevant information as this additional information will not contribute to your word count or affect essay marks unless it contributes directly to the essay answer.

ASSIGNMENT #3

The graphs shown below give information about computer ownership as a percentage of the population between 2002 and 2010, and by level of education for the years 2002 and 2010.

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Figure 1 Computer ownership, 2002-10

Figure 2 Computer ownership by education level, 2002 and 2010

Task

Summarise the information by selecting and reporting the main features, and make comparisons where relevant (Write at least 150 words).

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<http://web.mit.edu/course/21/21.guide/grfx.txt>

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ASSIGNMENT #3

The graphs shown below give information about computer ownership as a percentage of the population between 2002 and 2010, and by level of education for the years 2002 and 2010.

Figure 1 Computer ownership, 2002-10

Figure 2 Computer ownership by education level, 2002 and 2010

Task

Summarise the information by selecting and reporting the main features and make comparisons where relevant (Write at least 150 words).

University of South Australia

Division of Information Technology, Engineering and the Environment

School of Natural and Built Environments

Report Writing

Style Guide

for Engineering Students

4th edition

by Anne Winckel and Bonnie Hart

revised and updated by Monica Behrend and Bev Kokkinn

Prepared in association with Engineering staff and students, and the Flexible Learning Centre, University of South Australia

Sponsored by the University of South Australia Foundation Inc

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Major revisions in the fourth edition:

- inclusion of a section of Writing a section
- updating of the Referencing section

Major revisions in the third edition:

- inclusion of examples of electronic referencing • additions to Appendix D, Punctuation in technical writing • inclusion of examples in Appendix G, Purposes of paragraphs

Major revisions in the second edition included:

- addition of an index
- variation to the headings for lists of Figures, Tables and Symbols
- addition of a table on the report writing process and related resources (Table 1)
- addition of a section on Starting writing (section 5.1) • inclusion of all referencing examples in section 4 instead of the

appendices

- addition of new diagrams (Figs 10–13) to illustrate the elements involved in a reference list

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Abstract

Students need to develop the skills associated with report writing to communicate appropriately at University and to fulfil the obligations of their future employment. The provision of this Report Writing Style Guide for Engineering Students is to assist students in developing these writing skills. The initial project was funded by the University of South Australia Foundation Inc, the Faculty of Engineering and the Flexible Learning Centre (formerly the Centre for University Teaching and Learning) at the University of South Australia. Learning Advisers, Engineering Staff and personnel from other units collaborated to produce this document which is written in the form of a model report. The Style Guide specifically deals with: formatting guidelines, components of a report, writing a section, referencing of sources, and the technical language appropriate to a quality report. Style is often a matter of personal preference. Report writing styles will sometimes differ according to the purpose of the report and the requirements of a particular school, employer or publisher. It is recommended that particular attention be given to visual impact and readability; referencing of sources; proofreading and editing; and that the selection of style be consistent and appropriate to both audience and purpose.

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Disclaimer

We declare the following to be our own work, unless otherwise referenced, as defined by the University's policy on plagiarism.

Anne Winckel Bonnie Hart Monica Behrend

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Science

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Symbols

G = amplitude of oscillation

t = time

x = displacement

X_{st} = static deflection of the system under the influence of the imposed force β =
 frequency ratio

ϕ = phase angle

ω = natural circular frequency

ω_d = damped natural circular frequency

ζ = damping ratio

x

1 Introduction

Report writing is an essential skill for engineering students. Engineering reports analyse data, present results and conclusions, and make recommendations in a logical, precise and accessible manner. Report writing is both a requirement for completing an engineering qualification at university and a central part of succeeding in future employment. The ability to prepare professional engineering reports is one of the competency standards specified by The Institution of Engineers, Australia (1993) (see Appendix A). Having recognised the need for students to write quality reports, the University of South Australia Foundation Inc, the Faculty of Engineering, and the Flexible Learning Centre initially provided funding to produce this Report Writing Style Guide for Engineering Students (now referred to as Style Guide). Learning Advisers developed the Style Guide in consultation with University of South

Australia Engineering staff, staff from other units within the University and other personnel from various external organisations.

While this Style Guide has been prepared to assist engineering students with their report writing, it is not intended to be an inflexible set of rules but rather guidelines. Style is often a personal choice, and report writing styles and content will differ according to the type of report, the purpose, and the audience. For example, engineering companies have their own ‘in-house’ style. For each report, students should consult their lecturer or school to determine if any specific requirements apply — just as engineers who publish articles and technical reports must conform to the style required by their particular publisher or employer. (Staff may vary guidelines from this document according to school requirements.)

This Style Guide is one of many resources that a student may use when writing a report (see Table 1). In the writing process often the planning, writing and revising stages need to be repeated to achieve a quality document.

Table 1 Resources for report writing process

Report writing process Possible resources

Plan • Analyse requirements/topic ⇒ lecturers; topic guidelines and objectives •
Brainstorm/preliminary plan ⇒ prior knowledge
Study • Research/investigate ⇒ library; computer data; field work; experiments
Plan • Revise plan ⇒ newly informed knowledge; notes • Outline sections and headings ⇒
Style Guide: structuring technical writing
Write • Draft report ⇒ Style Guide: formatting, components, referencing, and technical language
Revise • Edit and proofread ⇒ Style Guide: recommendations, technical language, report writing checklist
Write • Revise draft report ⇒ Style Guide: formatting, components, referencing, and technical language

1

Publish • Publish final report ⇒ other students; staff

Where possible, this Style Guide models the format of an engineering report. At times, however, this document varies from report styles expected of students. (For instance, as this document is intended to provide information in an easily accessible format, there are numerous bullet points.) This Style Guide suggests guidelines for engineering reports in relation to formatting, components, writing a section, referencing of sources, and technical language. It is essential that aspects of style are consistent throughout a report.

1.1 Project reports

Primarily this Style Guide will be useful to undergraduate engineering students who are writing project reports. Project reports must be presented in a well structured and visually attractive manner; the competent use of technical language, and accurate

referencing of all sources is also a requirement.

To logically structure a report, students should understand the purpose of each component. The preliminary pages of a report include an Abstract which summarises the purpose, methods, results and conclusions of the project. The Introduction provides the background, objective(s), approach, scope and limitations of the report. The main sections of the report include the work or research undertaken and the associated information. This must be arranged logically with headings to guide the reader. Visual aids such as figures and tables can assist with the clear communication of material. The Conclusion summarises and identifies the important findings, placing them in context. Recommendations may be included indicating future directions. A report ends with a list of references, and any required appendices.

1.2 Other uses of the Style Guide

Students will be required to write a variety of other reports while at university, such as laboratory/practical reports, field reports, industrial experience and vacation reports. Postgraduate students will also have to produce a thesis. Not every aspect of style detailed in this document will be relevant to all of these reports. The Technical language section however, may assist all students with their writing, while the Referencing and Formatting sections may also be of use to postgraduate students. An example of the requirements of a thesis are set out in the University of South Australia's publication, Research Degree Student Information Guide (1996).

Writers should strive to present their work with clear expression and within an organised structure. Such writing will lead to a professional result which communicates effectively.

2

2 Formatting guidelines

Appropriate formatting of reports improves the readability and accessibility of information. This section outlines essential formatting guidelines as required by your lecturers. An important principle underlying all formatting choices is consistency. Therefore each report should maintain the same internal style and structure, which can be achieved. This Style Guide is formatted as a model and the guidelines assume that reports will be word processed on a computer. If you are unsure about how to apply the formatting guidelines which follow, refer to the Help menu within the word processor. If reports are to be hand-written, students should consult their lecturers.

2.1 Templates

One way in which consistency of a word processed report can be achieved is through the use of a template. The template forms the basis of a word processed document as it determines the basic structure and formatting for the entire document. You can use templates provided with your word processor or use the ones provided by your lecturers.

When using a template, styles can be created and modified to suit your purposes. For example, you can create styles for the several levels of headings. Creating a style saves you the tedium of applying the same font size, appearance, numbering and white space every time you type a heading or sub-heading. The single most significant advantage is that using styles allows you to automatically create a Contents page.

2.2 Format of pages

2.2.1 Paper and margins

- white A4 paper is appropriate, and printing on one side only is often preferred for assessment purposes
 - each major section of the report begins on a new page; this may be inappropriate for very short sections or short reports
 - wide margins are recommended; for example, allowing 2.5 cm on all sides
- bound reports have a left-hand margin of at least 3.5 cm, and 1.5 on the other three sides

To change these features using MSWord 2000, go to the File menu and access Page Setup.

2.2.2 Page numbering

- the title page should not be numbered
- all other pages may be numbered either in the right-hand upper corner, or in the centre at the bottom of the page
- roman numerals — i, ii, iii, iv etc. — are usually used to number the preliminary pages (Abstract, Contents etc.)

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- arabic numerals — 1, 2, 3, 4 etc. — are used to number the text from the Introduction page to the end of the Appendices
- alternatively, pages in the Appendices can be numbered internally, according to the letter of the individual appendix and the number of pages within each appendix — A1, A2, B1, C1, C2, C3 etc. (where Appendix A has two pages; Appendix B has one page, and so on)

To change these features using MSWord 2000, go to the View menu and access Header and Footer.

2.2.3 Headers and footers

- headers or footers position page numbers automatically
- two different headers or footers must be created so that the preliminary pages are numbered separately from the text of the report
- student name(s) and ID number(s) can appear in a footer placed against the right margin in 9 point size (see the bottom of this page)

2.3 Format of type and headings

2.3.1 Font

- a serif font (such as Times or Palatino) is usually chosen for the text of a report, as serif fonts tend to be easier to read
- a different font (such as the sans serif fonts, Helvetica or Arial) may be used for headings and tables
- the same font should be used throughout the whole report, unless a second font is chosen for headings and tables

To change these features using MSWord 2000, go to the Format menu and access Font.

2.3.2 Size and style of type

- 12 point type is the normal size for the text of a report
- 10 point may be used for long, indented quotations, figures and tables
- Table 2 suggests a style of headings with specific sizes
- italics can be used to emphasise text where necessary (but it should not be overused, particularly as there may be confusion with italicised titles)
- underlining or bold is avoided in emphasis of text as it is distracting
- the use of precise words may be sufficient to create the required emphasis

2.3.3 Headings

- headings are numbered according to their importance, using the decimal numbering system
- some writers do not use numbering, but differentiate between levels of headings through size and style (readability and access to information are the key considerations)
- headings without text should never appear on the bottom line of a page

4

- size and style of headings should vary according to the importance of the heading; the suggested style in Table 2 may prove helpful (to change the style of headings, go to the Format menu and access Style)
- headings can be differentiated using bold type, size and italics (overuse of underlining should be avoided as it makes reading more difficult)
- capital letters are for the first letter of the first word in each heading only; except for any acronyms (e.g. IEEE), trade names, or personal names and places, which may require more than one capital (aim for minimal capitalization in headings as overuse of capital letters slows reading)

Table 2 Size and style of headings

Heading Point Size Style Example

Acknowledgments

First-level (includes

Preliminaries) 6 Environmental

18 bold

impact

Second-level 14 bold

6.1 Chemical waste Third-level 12 bold

6.1.1 Cost analysis

Fourth-level 12 bold 6.1.1.1 Equipment costs

italics

- the best headings are:
 - summary statements rather than questions
 - not too brief (e.g. 'Inspection of contaminated site' is more explicit than 'Results') – not judgmental (e.g. 'Equipment for review' is more neutral than 'Poor equipment') – not too long (e.g. 'The equipment on the Whyalla factory floor' is unhelpfully long)

2.3.4 Decimal numbering system

- decimal numbering of headings is normally used throughout the text up to and including the reference list (excluding the preliminaries)
- Introduction is numbered 1; References have the final number (e.g. 8)
- too many levels of subheadings should be avoided: third-level headings are the preferred limit (e.g. 7.2.1)
- fourth-level headings (e.g. 7.2.1.1) should only be used if necessary, but not if there is only one heading at that fourth-level
- decimal points are placed between numerals, but not after them (e.g. 7.2)
- Appendices are not numbered using the decimal system; they can be internally labelled with letters (Appendix A, Appendix B etc.)
- omission of decimal numbering in shorter reports or where there are multiple subheadings, may increase readability

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2.4 Format of text

2.4.1 Justification

- all text and headings can be aligned against the left margin except for where indentation is appropriate (see 2.3.3 Indentation and centring)
- text may also be justified on the right-hand margin, and full justification (right and left margins) is often preferred for formal documents and theses

2.4.2 Line spacing

- 1.5 or double line spacing is generally used for work submitted for assessment; automatic line spacing (as provided with word processing) may be acceptable
- line spacing should be greater at the end of a section so it assists in separating sections
- line spacing may be decreased between headings and text so it assists in grouping information; the smallest line space (i.e. a four point space) may be appropriate before text such as a list of bullet points
- extra line spacing should be used in the following cases: – to

separate paragraphs

– to separate figures, tables, equations and long quotations from the text – to separate a main heading from the section of text which comes before To change line spacing using MSWord 2000, go to the Format menu and access Paragraph.

2.4.3 Indentation and centring

- indenting text at the beginning of a paragraph is unnecessary when an extra line space is being allowed between paragraphs
- indentation is appropriate for long quotations, complex mathematical expressions, lists with bullet points (dots •) and the list of headings on the contents page

- long quotations are usually indented from both the left and right margins •

figures and tables may be centred in the text

- students may choose to indent each subheading with its associated text, as shorter lines of text are easier to read; overall length is then affected To change indentation using MSWord 2000, go to the Format menu and access Paragraph.

2.4.4 Figures and tables

- figures include diagrams, graphs, sketches, photographs and maps • tables summarise data in rows and columns; a dash (—) indicates no data • all figures and tables may be centred
- all should be labelled in bold with the appropriate consecutive number (figures are generally labelled at the bottom, and tables at the top)

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- another method of labelling figures and tables is to number them according to the major section where they appear (as Table 4 is the second table in section 3 of this document, it would become Table 3.2)
- each should have a simple, descriptive caption which explains the figure or table; arrows, symbols, or abbreviations should be explained in the text • all figures and tables must be referred to in the text (e.g. 'see Figure 14'); avoid using words such as 'Figure above' or 'Table below'
- all should be referenced (either author-date or numerical) if copied or adapted from another source
- figures and tables can be separated from the text with extra line spacing; labels and captions can be separated using four point spacing

To design a table using MSWord 2000, go to the Table menu.

2.4.5 Quotations

- quotes of less than two lines (or 30 words) in length should be inserted 'within your text within single inverted commas' (Hadgraft, Robertson & Eaves 1994, p. 23).
- quotes of more than two lines (or 30 words) should be: set off from the body of the text in a separate block, indented...and single spaced. Do

not use inverted commas around block quotations. When using the block format, place the [author-date] citation on the line below the last line of the quote [or if using the numerical system of referencing, place the number directly after the quotation].

(Hadgraft, Robertson & Eaves 1994, p. 23)

- changes or additions to a quote should be placed in square brackets e.g. 'Ambiguity was caused by [financial] statistics being excluded from the report.'
- words left out of a quote are indicated by the three points of ellipsis (...) e.g. 'Ambiguity was caused by...statistics being excluded...'

2.4.6 Equations and formulae

- equations should be numbered consecutively as they appear, with arabic numerals in parentheses () on the right margin, in order to distinguish them from any reference numbers which may appear in square brackets
- equations can begin on a new line and should be centred in the text, e.g.

$x = G e^{-\omega t} \sin(\omega dt + \phi)$ (1) • in a series of consecutive equations, the equal signs (=) should be vertically aligned, e.g.

$$0 = -C_1\beta^2 + C_2(2\zeta\beta) + C_1 \quad (2) \quad X_{st} = -C_2\beta^2 + C_1(2\zeta\beta) + C_2 \quad (3)$$

- equations should then be referred to by either their number only, e.g. (2); or with the abbreviation Eq. and the number, e.g. A combination of Eq. 2 and Eq. 3 allows the evaluation of constants C_1 and C_2 .

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3 Components of a report

An engineering report can be divided into three main sections:

- preliminary pages
- text of the report
- supplements if appropriate

The components of a report will be determined by the purpose and complexity of the report. The possible components and sequence of a report are illustrated in Figure 1, and the significance of the core of the report is highlighted by the double-lined box.

- Symbols ‡

Preliminaries Text of report

- Introduction
- Main sections organised under appropriate headings and subheadings
- Conclusion
- Recommendations ‡
- References
- Title Pa
- Appendices ‡

- Ab
- Disclaim
- Acknow
- Content
- Figures
- Tables ‡

Supplements (‡ indicates an optional section) Figure 1 Components of a report

3.1 Preliminary pages

The preliminary pages are all numbered with roman numerals (i, ii, iii, iv, v, vi etc.) except for the title page which is generally not numbered. Not all the components described below will be required in every report; for example, students may choose

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to omit a list of figures if only one or two diagrams appear in the report (for other optional preliminary components, see Figure 2).

Also, a glossary of technical terms or acronyms might be added to a technical report, or an executive summary may be required in certain cases. Students can adapt the order of their preliminary pages to meet any specific instructions given by lecturers or particular schools. Postgraduate students should order the preliminary pages of a thesis according to the sequence required by their university.

- Title Page

Preliminaries • Abstract

- Disclaimer
- Acknowledgments ‡
- Contents
- Figures ‡
- Tables ‡
- Symbols ‡

(‡ indicates an optional section)

Figure 2 Preliminary components of a report

3.1.1 Title page

A model title page can be seen in Appendix B. A title should be brief but informative.

A title page should be simple, attractive, and include the following information:

- name of the university
- name of the particular engineering school
- name and code of the subject
- title of the report (which is precise and indicates the topic and scope) •
- name of author(s) and ID number(s) (with the family name underlined) •
- name of tutor/supervisor

- date of submission and/or date of experiment as applicable

3.1.2 Disclaimer

A 'disclaimer' or declaration of authenticity is often required in major pieces of work such as large reports, projects and theses. It is a signed statement declaring that the report is the work of the stated author(s). The disclaimer should appear on a page following the abstract, and the acknowledgments may be on the same page. This is a possible disclaimer statement:

I declare the following to be my own work, unless otherwise referenced, as defined by the University's policy on plagiarism ... followed by the author's signature.

Students are not guilty of plagiarism if they have correctly referenced the sources of all material which is not their own work (see Appendix C for a university policy on plagiarism). Reference must be made both in the text of the report and in the

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reference list. A disclaimer may also be used to indicate that the report's results, conclusions and recommendations are those of the author, and not necessarily a reflection of the opinions or policies of the university or sponsoring organisation.

3.1.3 Abstract

The abstract is also known as an overview, synopsis or summary. The abstract is often written last as its purpose is to provide a summary of the report's essential information. All material in the abstract will also be in the report, particularly the Introduction. The abstract should appear on a separate page after the title page, and it is usually about 100–200 words in length. The abstract should include the following elements:

- why? background problem and purpose of the report
- how? brief details of the approach/procedure/methods
- what? important results/findings
- so what? major conclusion(s) and recommendation(s)

In professions such as engineering, the abstract may be the key part of the report that is read. A manager may use the abstract to decide whether or not the content of the report is relevant. For academic purposes, an abstract allows researchers to quickly review literature and significant findings on a particular topic. An executive summary can be required for a more complex report. It is a detailed overview of the essential content, and its length can be several pages or more.

3.1.4 Acknowledgments

The acknowledgments section is optional; however, it is essential that significant assistance, editing or work carried out by another person or organisation be acknowledged. Also, students may wish to thank colleagues or supervisors. In a group-writing situation where different sections of a report are written by various people, writers of separate sections must be clearly identified.

3.1.5 Contents

The heading for the list of contents is Contents. The list should clearly include:

- all major section/subdivision headings; numbered and worded exactly as in the text of the report (minor headings are optional)
- page numbers for each section/subsection

Some students choose to include the preliminary pages on the contents list; however, the title page should be excluded. The contents page may be set out with each level of subheading indented by a tab space. This allows the reader to understand at a glance the structure of the report, and to differentiate between important and less significant information. Figure 3 gives a possible format for a contents list.

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5 Modelling of linear and non-linear network parts 8 5.1 Linear network 8 5.2

Non-linear network 9

5.2.1 The function of saturation 9 5.2.2 The function of hysteresis 9 5.2.2.1 Only use fourth-level headings if essential 10

Figure 3 Indentation of headings in a contents list

3.1.6 Figures

The heading for the list of figures is Figures. The list is only necessary if more than a few figures appear in the main text of the report. The list includes the figure number, caption, and page number, ordered as in the text. For a model of a list of figures, see preliminary page viii.

3.1.7 Tables

The heading for the list of tables is Tables. The list is only necessary if more than a few tables appear in the main text of the report. The list includes the table number, caption, and page number, ordered as in the text. For a model of a list of tables, see preliminary page viii.

3.1.8 Symbols

Where symbols are used extensively, a list of definitions should appear at the beginning of the report. If there is no list, symbols should be defined in the text when first used. The heading for the list is Symbols. The list of symbols should include appropriate information such as the symbol, definition, quantity to which the symbol refers, and the unit of measurement. Use an appropriate number of significant figures or level of accuracy when presenting measurements.

All units of measurement should be in the metric form given by the International System of Units (SI: *Système International d'Unités*). Appendix D covers Punctuating numerals and abbreviations, and care should be taken to include the space between the numeral and the unit (e.g. 55 cm). Table 3 indicates SI base units. If you are using another system of units, convert these into SI units.

Table 3 Base SI units

Quantity	Unit	Symbol
length	metre	m

mass kilogram kg

time second s

electric current ampere A

thermodynamic temperature kelvin K

(from Style Manual for Authors, Editors and Printers 2002, p. 178)

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3.2 Text of report

The pages of the text of the report (beginning with the Introduction) are numbered with arabic numerals (1, 2, 3 etc.).

An effective report has clear objectives and presents information in a logical sequence (see section 5.2 Structuring writing). A structured approach assists the reader by providing a framework for reading the report. An effective structure includes an introduction, followed by material organised under appropriate headings, and ends with a conclusion and possibly recommendations (see Figure 4).

Care should be taken to refer to the sources of all information both in the text and in the reference list at the end (see section 4 Referencing of sources and originality).

- Introduction
- Main sections organised under

Text of report

appropriate headings and subheadings

- Conclusion
- Recommendations ‡
- References

(‡ indicates an optional section)

Figure 4 Components of the text of a report

3.2.1 Introduction

The Introduction gives the reader the necessary background information. It can include:

- a description of purpose(s) and objective(s)/topic(s)
- a statement of the problem(s)
- a survey of background information
- a review of previous work/research and the relationship to the current project •

the method(s) of approach

- an indication of the scope and limitations of study
- an outline of material presented in the rest of the report

(adapted from Michaelson 1990, p. 38)

3.2.2 Main sections

The main sections of the report expand and develop the material in a logical and coherent manner, reflecting the structure outlined in the Introduction. Information in

the report should be placed in context, and well supported with evidence and documentation by using references. In Engineering you will be asked to write two types of reports, scientific and analytical.

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In scientific reports based on experiments, the following areas, with appropriate headings, may be included:

- methods/procedures and materials used
- data collected/information presented
- findings or results/comparisons with previous work
- discussion or analysis of the material

In comparison, analytical reports are based on a review of literature and other evidence. Therefore these reports involve a presentation of research and analysis of the literature. The structure of these analytical reports is determined by the type and purpose of the report (e.g. a feasibility study identifies a problem and evaluates alternative solutions). The structure is also determined by the key ideas or themes which emerge from your thinking about what you have read. Therefore a key skill is to logically order information into sections by using headings and subheadings. In the sections information which came from your reading is explained, expanded on and supported with evidence by using references, figures and tables. How to construct and write a well organized section in a report is explained in detail in Chapter 3. Another key skill is referencing, where you indicate your source of information in the text and in a reference list at the end of the text (Windshuttle 1999, p.295). The skill of referencing is explained in detail in Chapter 4.

The headings and subheadings should divide the report into well defined sections. A careful division of the report and choice of headings creates an outline or framework for the report which allows the reader to progress logically through the material. Lengthy reports are improved by introductions and summaries in each major section. For information about the format of headings, see section 2.3.3 Headings. While words such as 'Introduction' and 'Conclusion' may be appropriate headings, students should never use 'Main body' or 'Main sections' as headings.

Figures (including diagrams, graphs, sketches, photographs and maps) and tables (involving summarised data in rows and columns) can be incorporated into the text to give clear and concise information. Such visual aids support the content of the report and can communicate information more effectively. They must be located in the text as close as possible after the first reference to them. A sentence referring to the figure or table is necessary in the report. For example, 'Over a five year period the performance of groups A, B and C have varied markedly, with no group indicating constant improvement (refer to Table 4)'. For formatting and labelling requirements, see section 2.3.4 Figures and tables.

Consideration should be given as to whether the data is better communicated to the

reader by a table or a figure. Using tables or dotpoint lists will focus the reader on relevant and needed information. This technique is easier for the reader than reading another paragraph. For example, write an introductory comment and then list advantages and disadvantages of two types of dwellings in a table. Also consider that sometimes a figure will demonstrate a numerical trend more effectively than a

table. In the following example, the significance of the data in Table 4 may be more clearly presented in the form of a graph, as can be seen in Figures 5 and 6.

Table 4 Sample Table (from Hadgraft et al. 1994, p. 6) *

Year A B C

1961 97.70 89.93 7.77

1962 1.20 7.74 -6.54

1963 36.10 39.24 -3.14

1964 31.00 30.11 0.89

1965 55.90 13.64 42.26

* Asterisks or other note identifiers (e.g. double daggers ‡) can be used in a footnote to add detail, e.g. to a table.

100 100

C

A

80

80 A

B 60

C B

60 40

20

40 0

-20

20

1961

0 AB

1961 1962 1963 1964 1965 1963

-20

1965C

Figure 5 Sample of 3D columns

Figure 6 Sample of 3D

lines

(from Hadgraft et al. 1994, p. 7)

3.2.3 Conclusion

The conclusion(s) of a report must be related to, and resulting from, the material which appears in the report. The content of the Conclusion will be linked to the

Introduction. The Conclusion places findings in perspective without introducing any new material, and it may include:

- a clear and concise summary of the main points
- the context and significance of the information
- a reference to the original aim(s)/purpose(s) of the report
- the application(s) of the results
- the limitations and advantages of the findings
- the writer's judgment/evaluation

3.2.4 Recommendations

Not all reports include recommendations, but if they are required recommendations should emerge from the conclusions of the report. This section is important to those who must act on the findings. The writer may include a brief, persuasive statement before presenting the recommendations clearly listed in numbered or bullet points. A series of recommendations may be worded in instructional language; for example, each beginning with a verb. There are styles of reports where the Recommendations follow the Introduction. Recommendations may involve:

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- strategies, procedures or techniques for solving the problem(s) •
- an indication of further work which needs to be completed

3.2.5 References

A reference list (not to be confused with a bibliography) must appear at the end of a report, listing all sources that have been referred to in the text. The heading for this list will be References. (Students should ensure that all sources are referenced in the text as well as in the reference list at the end of the report.)

The format of the reference list will depend on the system of referencing chosen for the report. There are two different types of reference lists used in engineering:

- alphabetical reference list according to author – used with the author-date (Harvard) system (see section 4.2.2 Alphabetical reference list)
- numbered reference list in order of their appearance in the text – used with the numerical (endnote) system (see section 4.3.2 Numbered reference list)

The reference list only includes the sources referred to in the report. A bibliography is a wider list of all texts that have been read in preparation for writing. A bibliography is not usually included in an engineering report (check with your lecturer).

3.3 Appendices

Not all engineering reports will need appendices, which are supplements to the text of the report (see Figure 7). The Appendices may include material which is incidental to the report, supportive of the report, or too long or technical to include in the text; for example, maps, folded diagrams, tables of results, letters, questionnaires, statistics, and computer program information. Large maps or figures should be reduced to A3 size where possible, and multiple folding is discouraged.

- Appendices ‡

Supplements (‡ indicates an optional section)

Figure 7 Supplementary components of a report

Each separate appendix should be lettered (Appendix A, Appendix B, Appendix C etc.). The text page numbering can continue into the Appendices, or alternatively, the Appendices can be numbered internally using the letter of the individual appendix and the number of pages within each appendix (A1, B1, B2, B3, C1, C2 etc. where Appendix A has one page, Appendix B has three pages, and so on).

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4 Writing a section of a report

An approach to writing a section of a report is explained in this chapter. This process involves planning, gathering information, reading and noting, further planning, writing the first draft, and finally revising and redrafting. This process is facilitated by the use of many resources, as summarised in Table 1 (page 1).

4.1 Planning

The first step of the writing process always involves clarifying the following expectations: message; purpose; audience, and scope of the report. This also applies when writing a section.

Message: Begin by clarifying the message of the section, i.e. write a precise heading or a summary sentence that summarises the section. This will enable you to focus on the information to communicate.

Purpose: The general purpose of the writing a section of a report is for you to gather information on a specific topic and the using this information to present a concise, organised, logical section. The specific purpose of writing a section relates to the message. For example, you might explain the concept of 'Solar cells' by providing a definition, detailing how they operate, providing an application and discussing research findings.

Audience: Each report is written for a specific person or group of people in mind. For example, often the audience is colleagues and professionals, such as engineers.

Scope: The scope of a section limits the information within the section. You can do this by stating exactly what you will or will not cover in the section. For example, in the section on 'Solar cells' you will not discuss solar energy in general.

Once you have clarified the expectations, you can now plan your section. Useful initial techniques are brainstorming, mindmapping and developing a Contents page.

4.1.1 Brainstorming

Literally allow your brain to storm or generate many ideas. One way to start doing this is by listing all the ideas or keywords that come into your mind, as shown in the example. Do not organise ideas at this stage as that is the next step of mindmapping.

Brainstorm example – Solar energy

Solar cells Why? Suitability of location Weather Financial Amount
of sunshine Advantages Businesses Infrastructure
Disadvantages

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4.1.2 Mindmapping

Now try to develop a logical organisation to the ideas. One way to achieve this is by developing a mindmap, a diagram that shows the linking of all the ideas listed.

Figure 8 An example of a section of a contents page

4.2 Developing a contents page

From the organisation from the mindmap you are now able to prioritise the ideas and generate a series of headings and subheadings. If you can achieve three levels of headings and subheadings, your section is starting to develop a logical argument.

Contents

1 Solar energy

1.1 Sunshine

1.1.1 Amount

1.1.2 Location

1.1.3 Conversion

1.2 Meteorological variables

1.2.1 Cloud cover

1.2.2 Wind

1.2.3 Rain

1.2.4 Seasons

Figure 8 An example of a section of a contents page

Organising information into logical sections is challenging This process can be assisted by having read widely and having obtained more information about the topic.

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5 Referencing of sources and originality

References must be provided in your report if you use someone else's opinions, theories, data or organisation of material, and this includes all occasions when you:

- paraphrase (use their same idea in your own words)
- summarise (use their main points)
- quote (use their exact words)
- copy (use their figures, tables or structure)

It is expected at university that students will present original work for assessment. Where any ideas or written work from other sources/authors appear in your report, you must acknowledge the sources by making reference to them. These references must be in the text as well as in the reference list. Failure to indicate the sources of your ideas may suggest that you are wrongfully claiming their ideas or words to be your

own, which is known as ‘plagiarism’. Universities impose heavy penalties on students who plagiarise (see Appendix C for an example of a plagiarism policy). Referencing of sources is also encouraged because it strengthens a writer’s argument. Referencing shows wide reading and identifies the critical aspects of other research.

5.1 Two systems of referencing

There are two main ways of referencing sources in engineering reports:

- author-date referencing (Harvard system) – see section 4.2 •
- numerical referencing (endnote system) – see section 4.3

In a report, students must use either the author-date or the numerical system, not both. The main differences between the two systems are described in Figure 8.

Numerical referencing

Author-date referencing In the text of report:

In the text of report:

- consecutive numbers in
 - author and date in parentheses square brackets
- e.g. ...validation of results (Smith 1992). e.g. ...validation of results [4].

- page number included if needed

Reference list at the end of report:

Reference list at the end of report:

- numbered list of references
- alphabetical list of references – author’s initials or name can be
- author’s surname first before the surname
- date follows author – title follows author, with date
- no page numbers of quotations – appearing later
- where more than one line in length, – page numbers are last if needed
- the second line is indented

Figure 8 Differences between referencing systems

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5.1.1 Choosing between referencing systems

When deciding on which referencing system to use, students should consider:

- the audience of the report
- the purpose of the report

An author’s referencing style may be determined by the audience of the report — as specific referencing styles are often required by different lecturers, employers and publishers of journals and conference papers. For example, many international journals in Electrical Engineering use the numerical system, while academic journals in Mechanical Engineering often use the author-date system.

The selection of an appropriate system of referencing will also depend on the purpose of the report. Numerical referencing may be used where the purpose of a report is to communicate technical information and the conclusions are of more interest to the

reader than the names of cited sources. The author-date system is useful where the purpose is to present scholarly research, such as in a research or project report, or in an academic article or thesis. The names of those who have completed previous work in the area will be of interest to the reader, and direct textual references to sources will therefore be useful, rather than having the reader turn to a list at the end.

Possible advantages and disadvantages of the two referencing systems are summarised in Figure 9.

Numerical referencing

Author-date referencing Advantages

Advantages

- prevents the text of the report from
- allows author and date to be seen being interrupted by wordy references in context within the text of report
- prevents constant repetition of the
- saves turning to a list at the end to same references as only a number find the name of a cited source needs repeating

- provides an alphabetical reference

list at the end

- means that inserting extra references into the text is easy

Disadvantages

Disadvantages

- creates a non-alphabetical reference
- list at the end
- creates very long author-date entries
- if there are multiple authors and • means turning to reference list to match a numerical reference to its
- sources • creates repetition and
- source • may create complications if an
- disruption to the text when the same
- extra reference needs inserting later
- source is used repeatedly

Figure 9 Advantages and disadvantages of referencing systems

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5.1.2 Consistency in referencing

The numerical system and author-date system outlined below are based on the style that is common for engineering report writing, and the Australian Government Publishing Service (AGPS) Style Manual for Authors, Editors and Printers (1994). Other methods of each referencing system are used by individuals, faculties and publishers. It is therefore important for internal consistency that only one version of a referencing system is used within any report. When referencing sources, writers should also take particular care to maintain consistency with punctuation. For

further guidelines, students can refer to a librarian or the AGPS Style Manual (1994).

5.2 Author-date referencing (Harvard system)

There are two major elements involved in the author-date system of referencing:

- in the text – the author's surname and date appear in parentheses () •

at the end – an alphabetical reference list is included

5.2.1 In-text author and date

To acknowledge a source within the text of your report using the author-date system of referencing, you should state the following items:

- author's surname
- date of the publication
- page numbers, if necessary (e.g. when material is quoted or copied)

This information should appear in parentheses or round brackets () directly after the material quoted (see the first example below). If the author's name has already been used in the text of the report then it does not need to be placed in the parentheses, as seen in the alternative example below.

Author-date references in the text:

a quotation is used It has been argued that 'the relative seriousness of the two kinds of errors differs

from situation to situation' (Smith 1982, p. 47).

or

Smith (1982, p. 47) has argued that 'the relative seriousness of the two kinds of errors differs from situation to situation'.

the author's words are paraphrased A recent study (Jones & Chan 1991) has shown that testing procedures are inadequate.

the same author is cited again Another argument raised by Smith (1982) relates to the error evident

in the conclusion.

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several different sources need to be cited at the same or

time Recent studies (Jones & Chan 1991; Kuwlesky

Jones and Chan (1991), Kuwlesky (1988) and Tan 1988; Tan 1992) all agree...

(1992), all agree that testing procedures are [names of first authors are alphabetical; and separated

inadequate. by semicolons]

Author-date references in the text cont.

there are two or three authors A recent study (Lim, Thompson & King 1993) found... or

Lim, Thompson and King (1993) found the analysis...

[use an ampersand (&) when in the brackets; or 'and' outside]

more than three authors appear A recent study (Boyd et al. 1994) has indicated... or

Boyd et al. (1994) have indicated...

['et al.' means 'and the others']

there are different authors with the same surname idea.

[differentiate between authors by using initials]

Marini (Tan 1992) stated that...

you are referring to an author (primary reference) [Marini is the primary reference]

which you read about in another book or article or

(secondary reference) Marini's study in 1975 (cited in Tan 1992) stated

that... or

Tan (1992), in reporting Marini's 1975 study, states

that...

reference is made to more than one work by the same

author University research (Smith 1982, 1988) has

indicated that... or

University research conducted by Smith (1982,

1988) has indicated that...

the same author has published two or more things in

the same year In recent reports (James 1992, 1993a, 1993b),

recommendations were made...

or

Recommendations by James (1993b) were

implemented... [add a, b, c etc. to distinguish

no author's name appears (but there is a sponsoring between texts in the same year]

body, newspaper or title)

A recent study (CSIRO 1990) has suggested...

or

A recent survey (Advertiser 24 Jan. 1994, p. 5)

the editor's name is the identifying feature indicated that... or

A recent survey (Population Projections 1996)

shows that...

The theory was first put forward in 1970 (Jones,

C. L. 1971), but since then many researchers, Current articles (ed. Harris 1994) on the industry

including A. G. Jones (1983), have rejected the suggest that... or

Current publications edited by Harris (1994) suggest that...

no date can be established Lewis (n.d.) continued the study... or

The study was continued (Lewis n.d.)...

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the source is unpublished Nguyen (unpub.) argues for...

or

An argument (Nguyen, unpub.) has been made for...

or

reference is made to a personal communication The site's safety is proved (King, B. T. 1994, pers. In a telephone conversation on 17 July 1994, King comm., 17 July). [personal communications are not

revealed the site to be safe. included in the reference list]

5.2.2 Alphabetical reference list

When using the author-date system of referencing, you must include an alphabetical reference list at the end of the report (before the Appendices) listing all sources cited. Personal communications such as conversations or personal email messages need not be included in the reference list, as they cannot be checked. When writing the reference list you must take care to include the correct elements, and to use the correct punctuation.

The elements required for a book are: the author's surname(s) and initials, year of publication, title, publisher, and place of publication (other elements may also be relevant — see Appendix E). The correct order of these elements is indicated by the ellipses in Figure 10. The shaded boxes indicate punctuation.

Author's surname(s)Initials Title Publisher Place of publication Year

Muller, B. & Reinhardt, J. 1990, Neural Networks: An Introduction, Springer-Verlag, Berlin.

Ampersand Full stops Commas Capital letters Italics Commas Full stop Figure

10 Book in author-date reference list

The elements required for a journal article are: the author's surname(s) and initials, year of publication, title of the article, title of the journal, volume number, issue number, and page numbers. Some elements will not always apply, and sometimes others will be relevant — see Appendix E. The correct order of these elements is indicated by the ellipses in Figure 11. The shaded boxes indicate punctuation.

Author's surname(s)Initials Title of article

Year

Quotation marks Lower case

Meyers, A. G. & Priest, S. D. 1992, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, pp. 207–214.

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Indent Italics Abbreviations Full stops

Commas Issue number

Volume number Page numbers

Journal title

Figure 11 Journal article in author-date reference list

The following aspects of punctuation should be included (as indicated by the shaded boxes in Figures 10 and 11):

- commas used to separate all elements
- full stops used for the abbreviation of words, and the end of the reference •
- ampersand (&) used to separate last two authors — instead of 'and' • italics or

underlining used to indicate the title of a book or journal • single quotation marks used to indicate the title of an article • titles of unpublished works — such as theses — are not differentiated (by italics, underlining or quotation marks)

- capital letters used for first letters of main words in titles of books and journals, and all other letters are lower case
- capital letters kept to a minimum in titles of articles and unpublished works •

short indentation of the second and subsequent lines of a reference is helpful to show the alphabetical order

A reference list is arranged in alphabetical order according to the authors' surnames:

- if there is no author, then they are listed alphabetically using the first word of the title, excluding 'a' or 'the'
- if there are two or more references by the same author(s), they are listed in order of publication date, with the oldest work first
- if works by the same author(s) are published in the same year, they are arranged alphabetically by title of the work; lower-case letters are added to the publication dates to distinguish each work (e.g. 1992a)
- works where the author is the sole author are listed before works where the author is the senior author with co-authors, regardless of date of publication

For example, this is the correct alphabetical order for an author-date reference list:

Aukland, J. S. 1989, A solar airconditioning system, M Eng Thesis, South Australian Institute of Technology.

Browne, T. E. Jr (ed.) 1984, Circuit Interruption: Theory and Techniques, M. Dekker, New York.

Commission of Inquiry into Poverty 1975, Poverty in Australia, First Main Report (Prof. R. F. Henderson, Chairman), AGPS, Canberra.

Dictionary of Scientific and Technical Terms 1994, 5th edn, McGraw-Hill, New York.

Henderson Report. See Commission of Inquiry into Poverty 1975.

Meyers, A. G. & Priest, S. D. 1992, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, pp. 207-214.

Muller, B. & Reinhardt, J. 1990, Neural Networks: An Introduction, Springer-Verlag, Berlin.

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Subramanian, C. 1992, 'Some considerations towards the design of a wear resistant aluminium alloy',

Wear, vol. 155, pp. 193-205.

Subramanian, C. 1993, 'On mechanical mixing during dry sliding of aluminium-12.3 wt. % silicon alloy against copper', Wear, vol. 161, pp. 53-60.

Subramanian, C. & Strafford, K. N. 1993a, 'Review of multicomponent and multilayer coatings for tribological applications', Wear, vol. 165, pp. 85-95.

Subramanian, C. & Strafford, K. N. 1993b, 'Towards optimization in the selection of surface coatings and treatments to control wear in metal-forming dies and tools', Materials & Design, vol. 14, pp. 291-298.

Subramanian, C., Strafford, K. N., Wilks, T. P., Ward, L. P. & McMillan, W. 1993, 'Influence of

substrate roughness on the scratch adhesion of titanium nitride coatings', *Surface & Coatings Technology*, vol. 62, pp. 529-535.

The sources referred to in a reference list may be books, articles, reports, journals, conference papers, newspapers, audiovisuals, electronic and other non-print material. Referencing of electronic sources is a relatively new activity, and the following examples are adapted from the style suggested by Li and Crane (1993).

Author-date reference lists:

Books:

One or two authors Muller, B. & Reinhardt, J. 1990, *Neural Networks: An Introduction*, Springer-Verlag,

Berlin.

3 or more authors Arrillaga, J., Arnold, C. P. & Harker, B. J. 1983, *Computer Modelling of Electrical Power Systems*, Wiley, Chichester.

No author, 2nd or later edition Computer Integrated Manufacturing 1990, 2nd edn, Architecture Technology Corporation, Minneapolis.

Books sponsored by an institution, corporation or other Australian Manufacturing Council 1993, *Emerging*

organisation Exporters: Australia's High Value-added

Manufacturing Exporters: Final Report of the Study,

Australian Manufacturing Council, Melbourne.

The Electrical Engineering Handbook 1993, ed.

Edited work (role of editor not paramount) Richard C. Dorf, CRC Press, Boca Raton.

Edited work (role of editor paramount) Browne, T. E. Jr (ed.) 1984, *Circuit Interruption:*

McKinsey & Company & the Australian Theory and Techniques, M. Dekker, New York.

Manufacturing Council Secretariat to the

One volume of multi-volume work Gopel, W., Hesse, J. & Zemel, J. N. (eds) 1989, *Sensors: A Comprehensive Survey*, vol. 1, Fundamentals and General Aspects,

eds T. Grandke & W. H. Ko, VCH, Weinheim, Germany.

Chapter in edited book Stead, J. P. & Strutt, J. E. 1987, 'Marine corrosion on offshore pipelines: predicting the problem using computer-aided risk

evaluation', in *Plant Corrosion: Prediction of Materials*

Performance, eds J. E. Strutt & J. R. Nicholls, Ellis Horwood,

Chichester.

Journals:

One or two authors Albrecht, P. & Shabshab, C. 1994, 'Fatigue strength of weathered rolled beam made of A588 steel', *Journal of Materials in Civil*

Engineering, vol. 6, pp. 407-428.

Issue number included (if each issue is paginated *Mechanics and Mining Sciences & Geomechanics* separately) Abstracts, Technical Note, vol. 29, no. 2, pp.

Meyers, A. G. & Priest, S. D. 1992, 'A 187-191.

micro-processor controlled pump for triaxial cell

pressure control', International Journal of Rock

24

No author 'Efficient low cost solar cell developed' 1994, Search, vol. 25, p. 309.

Specialised sources:

Conference papers Hunt, F. L. 1990, 'Ballarat East — from Eureka to the 80's', Proceedings of the Pacific

Rim Congress 90, Gold Coast, Australia,

Vol. III, Australasian Institute of Mining and Metallurgy,

Parkville, Victoria, pp. 473-480.

Maps Mullineaux, D. R. 1976, Preliminary Overview Map of Volcanic Hazards in the 48 Conterminous

United States, U. S. Geological

Survey Miscellaneous Field Studies Map MF-786, scale

1:7 500 000.

Standards American Society for the Testing of Materials 1988, Standard Specification for Cast Steel Wheels for Railway Service, (A 583 -

88), ASTM, Philadelphia.

Standards Association of Australia 1991, Australian Standard: Electrical Installations — Buildings, Structures and Premises,

known as the SAA Wiring Rules, (AS 3000 - 1991), Standards

Australia, North Sydney.

Patents Tan, I. S. & Arnold, F. F. (US Air Force) 1993, In situ Molecular Composites Based on Rigid-rod

Polyamides, US patent 5 247 057.

Author-date reference lists cont.

Government publications Bureau of Transport Economics 1986, Demand for Australian Domestic Aviation Services, Occasional Paper no. 79, AGPS, Canberra.

Australian Bureau of Statistics 1994, Building Approvals Australia, Cat. no. 8731.0, ABS, Canberra.

Ergas, H. 1986, Telecommunications and the Australian Economy, Report to the Department of Communications, AGPS, Canberra.

Government reports Commission of Inquiry into Poverty 1975, Poverty in Australia, First Main Report

(Prof. R. F. Henderson, Chairman), AGPS, Canberra.

Henderson Report. See Commission of Inquiry into Poverty 1975.

Newspaper articles Fist, S. 1994, 'The call to disconnect analogue phones', Australian, 2 Aug., p. 25.

Pamphlet Quit: Give Smoking Away in 5 Days 1987, Victorian Smoking and Health Program booklet, Health Department Victoria, Anti

Cancer Council and National Heart Foundation, May.

Microfiche Herbert, W. G. 1987, The Australian Beef Industry: An Overview, Australian Livestock Council, Canberra, microfiche.

Video What Are We Going to Do With the Mo9ney? (video recording) 8 August 1982, ABC Television.

Unpublished material:

Thesis Wenjiong, Y. 1994, Efficient coding scheme for video-conference applications, M Eng Thesis, University of South Australia.

Unpublished report Janz, J. K. 1993, Orroroo bedrock drilling 1993, South Australian Department of Mines & Energy, Unpublished Report RB

93/042.

Electronic media:

25

Document on World Wide Web Brown, H. 1995, Citing Computer References [Online]. URL: <http://neal.ctstateu.edu/history/cite.html>

Document via FTP or Telnet King, M. L. 1963 August, I Have a Dream [Online]. Available FTP: [mrcnext.cso.uiuc.edu](ftp://mrcnext.cso.uiuc.edu) Directory: [gutenberg/freenet](#) File: [i have-a-dream](#)

Document via email Harnad, S. 1992, Post-Gutenberg galaxy: The fourth revolution in the means of production of knowledge, in Directory of

Electronic Journals, Newsletters and Academic Discussion Lists , 2nd edn [Online]. Available email: LISTSERV@UOTTAWA

Message: Get EJOURNL1 DIRECTORY

Journal article on CD Rom Girotti, T. B., Tweed, N. B. & Houser, N. R. 1990 February, Realtime var control by SCADA, IEEE Transactions on Power

Systems [CD Rom] vol. 5, no. 1, pp. 61-64. Available: UMI

File: IPO (IEEE/IEE Publications Ondisc) Item: 3631953

Journa article via FTP Drew, J. R. 1989 October, So that's why they call it the Big Apple, Qyabta [Online] vol. 1, no. 1. Available FTP: [export.acs](ftp://export.acs.cmu.edu)

.cmu.edu Directory: [pub/quanta](#) File: [quanta-oct.ps.Z](#)

Online databases The Educational Directory [Online] 1992. Available: Knowledge Index File: The Educational Directory (EDUC6)

Lists or bulletin boards Alcohol and Drug Studies [Online]. Available email: ALCOHOL@LMUACAU.BITNET

Computer programs Wu, A. H. & Jenkins, B. 1990 May1, Diagnostic Ordering in Clinical Medicine (DOS version 1.0) [Computer program].

Available Distributor: Health Sciences Consortium, Chapel Hill, NC (Address: 201 Silver Cedar Court, Zip: 27514)

5.3 Numerical referencing (endnote system)

There are two major elements involved in the numerical system of referencing:

- in the text – consecutive numbers appear in square brackets [] • at the end – a numbered reference list is included

5.3.1 In-text numbers

To acknowledge a source within the text of your report using the numerical referencing system, you should place an arabic number in square brackets after the quotation, relevant material, or author's name. This number will then be repeated in the reference list with the corresponding details of the source (e.g. the reference list will include elements such as the author, title, and page numbers).

Consecutive numbers should be used for each new reference. However, where you cite the same source or reference more than once in the report, you can use the same number again each time. However, if a second citing of a source involves a different page number, then a new numerical reference is used — and a corresponding new entry will be included in the reference list.

Numerical references in the text:

a quotation is used It has been argued that 'the relative seriousness of the two kinds of errors differs

from situation to situation' [1].

or

Smith [1] has argued that 'the relative seriousness of the two kinds of errors differs from situation to situation'.

26

the author's words are paraphrased A recent study [2] has shown that testing procedures are inadequate.

the same author is cited again Another argument raised by Smith [1] relates to the error evident in the conclusion.

several different sources need to be cited at the same Jones and Chan [2], Kuwlesky [9] and Tan [7], all

time agree...

Recent studies [2][7][9] all agree that testing procedures are... or

there are two or three authors A recent study [3] found the analysis... or
Lim, Thompson and King [3] found the analysis...

more than three authors appear A recent study [4] has indicated... or
Boyd et al. [4] have indicated...

there are different authors with the same surname

Marini [7] stated that...

you are referring to an author (primary reference) or

which you read about in another book or article Marini's study in 1975 [7] stated that...

(secondary reference) or

The theory was first put forward in 1970 [5], but Tan [7], in reporting Marini's 1975 study, states since then many researchers, including A. G. that...

Jones [6], have rejected the idea.

Numerical references in the text cont.

reference is made to more than one work by the same In recent reports [10][11][12], recommendations

author were made... or

Recommendations by James [12] were implemented...

the same author has published two or more things in the same year A recent CSIRO study [13] has suggested... or

A recent survey [14] indicated that...

no author's name appears (but there is a sponsoring or body, newspaper or title) A recent survey, Population Projections, [15] shows that...

Current articles [16] on the industry suggest

the editor's name is the identifying feature that... or

University research [1][8] has indicated that... Current publications edited by Harris [16] or suggest that...

University research conducted by Smith [1][8] has indicated...

no date can be established Lewis [17] continued the study... or

The study was continued [17]...

the source is unpublished Nguyen [18] argues for... or

An argument [18] has been made for...

27

reference is made to a personal communication 1994, pers. comm., 17 July).

In a telephone conversation on 17 July 1994, King [personal communications are not included in the revealed the site to be safe. reference list]

or

The site has since proven to be safe (King, B. T.

5.3.2 Numbered reference list

When using the numerical system of referencing, a numbered reference list appears at the end of the report (before the Appendices) listing all sources cited. Personal communications such as conversations or personal email messages need not be included in the reference list, as they cannot be checked.

The reference list is arranged in numerical order according to the sequence in which the authors were cited in the report. When writing the reference list you must take care to include the correct elements, and to use the correct punctuation.

Appendix E summarises the various elements which may be required in a reference list, and highlights the differences between the alphabetical and numerical reference lists. For a numerical reference list, the main differences include:

- the reference number appears first
- the initials or given name come before the surname
- the date appears at the end of the reference before any page numbers

In a numbered reference list, the elements required for a book are: the author's initials and surname(s), title, publisher, place of publication, and year of publication. The correct order of these elements is indicated by the ellipses in Figure 12. The shaded boxes indicate punctuation.

Title Publisher Place of

publication Author's initials Surname(s) Year [5] B. Muller & J. Reinhardt, Neural

Networks: An Introduction, Springer-Verlag, Berlin, 1990.

Full stops Ampersand Capital letters Italics Commas Full stop Figure 12

Book in numerical reference list

The elements required for a journal article are: the author's initials and surname(s), title of the article, title of the journal, volume number, issue number, year of publication, and page numbers. Some elements will not always apply, and sometimes others will be relevant — see Appendix E. The correct order of these

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elements is indicated by the ellipses in Figure 13. The shaded boxes indicate punctuation.

Surname(s) Title of article Quotation

marks Lower case

Author's initials

[6] A. G. Meyers & S. D. Priest, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, 1992, pp. 207–214.

Commas Abbreviations

Italics Year Full stops

Journal title numbers

Volume number Issue number Page

Figure 13 Journal article in numerical reference list

The following aspects of punctuation should be included (as indicated by the shaded boxes in Figures 12 and 13):

- commas used to separate all elements
- full stops used for the abbreviation of words, and the end of the reference
- ampersand (&) used to separate last two authors — instead of 'and'
- italics or underlining used to indicate the title of a book or journal
- single quotation marks used to indicate the title of an article
- titles of unpublished works are not differentiated (by italics, underlining or quotation marks)
- capital letters used for first letters of main words in titles of books and journals, and all other letters are lower case

- capital letters kept to a minimum in titles of articles and unpublished works •
- initial numerical reference with or without square brackets []

The sources referred to in a reference list may be books, articles, reports, journals, conference papers, newspapers, audiovisuals, electronic and other non-print material. For a detailed list of examples, see section 4.2.2 Alphabetical reference list — but note that numerical references are not presented alphabetically, and the sequence of the elements needs to be adapted to the appropriate style.

Numbered reference lists:

1. Dictionary of Scientific and Technical Terms , 5th edn, McGraw-Hill, New York, 1994.
2. J. S. Aukland, A solar airconditioning system, M Eng Thesis, South Australian Institute of Technology, 1989.
3. B. Muller, & J. Reinhardt, Neural Networks: An Introduction, Springer-Verlag, Berlin, 1990.
4. A. G. Meyers, & S. D. Priest, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, 1992, pp. 207– 214.
5. T. E. Browne, Jr (ed.) Circuit Interruption: Theory and Techniques, M. Dekker, New York, 1984.

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or alternatively, using square brackets:

- [1] Dictionary of Scientific and Technical Terms , 5th edn, McGraw-Hill, New York, 1994.
- [2] J. S. Aukland, A solar airconditioning system, M Eng Thesis, South Australian Institute of Technology, 1989.
- [3] B. Muller, & J. Reinhardt, Neural Networks: An Introduction, Springer-Verlag, Berlin, 1990.
- [4] A. G. Meyers, & S. D. Priest, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, 1992, pp. 207– 214.
- [5] T. E. Browne, Jr (ed.) Circuit Interruption: Theory and Techniques, M. Dekker, New York, 1984.

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6 Technical language

Effective use of technical language is a critical part of report writing. The important aspects of technical language to be considered are structure, grammar, vocabulary and punctuation. (The use of punctuation in technical writing is detailed in Appendix D.) A report must be written appropriately for its audience. The purpose of a report must be decided before the writing begins. Once it is clear why a report is being written, then how to write will become more evident. A quality engineering report exhibits these characteristics:

- clear statement of aim or purpose
- concise and precise presentation of detail
- cohesive and logical development of data and ideas
- considered objective separation of fact and opinion
- categorised layout of well-spaced headings, paragraphs and visuals

6.1 Starting writing

When beginning a report, students may find it useful to plan and write the main

sections first. The abstract or summary should be written last as should the conclusion. It can also be useful to write the Introduction later as it includes an outline of the structure of the whole report.

A useful technique for developing a structure is to ask questions. These questions may be: What? Where? When? Why? and How? For example, the planning of an engineering report on solar energy may be assisted by asking such questions as:

- What is solar energy?
- Where and when is solar energy predominantly used?
- Why is usage of solar energy currently limited?
- How is solar energy utilised in engineering contexts?

It is useful to ask as many questions as possible because such questions help to generate ideas and key words for the structure of sections and paragraphs, and for the associated headings. Students may find it useful to refer to report writing resources available from their university library (see Appendix F).

6.2 Structuring writing

Information in a technical report is systematically developed at several levels: sections, paragraphs and sentences. The organisation of this information can be controlled using various techniques including:

- different levels of headings
- summary, overview or key statements indicating purpose •

visual aids such as tables and figures

- bullet points (dots •), dashes and lists

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- linking sentences between sections and paragraphs

6.2.1 Logical sections

Students should consider the overall purpose of the report, and be aware of the specific purpose of individual sections. Using a flow chart or visual representation can assist with planning and structuring the report. (For further information see section 3.2.2 Main sections.) Some possible purposes are to:

- analyse data/literature is investigated; gaps are identified • inform information or research is outlined; results are presented • instruct procedures are explained; strategies are provided • persuade conclusions are discussed; recommendations are made • document written record is created and filed

6.2.2 Cohesive paragraphs

Paragraphs are groups of sentences which organise information into the essential units of the report. Before writing a paragraph, the writer should be clear about the paragraph's purpose(s). A useful technique is to decide what the reader needs to know. The broad and specific purposes of paragraphs in technical writing (see Appendix G) involve:

- describing facts (e.g. paragraphs which define, classify, and describe) • describing

the relationship between facts (e.g. paragraphs which describe, analyse, compare and contrast)

A paragraph should begin with a general overview or summary statement before the specific or exemplifying information is given. Exemplifying information may include examples, illustrations, descriptions, explanations and evidence which expand the ideas. A paragraph is often structured with:

- a key sentence (introducing main idea and/or giving context)
- concept development (expanding main idea/giving specifications etc.)
- a summary statement (concluding with main idea/linking to what follows)

Connectors or linking words help create cohesion within the paragraph and between paragraphs. Information should be organised systematically and one way of achieving this is to use connectors or linking expressions (see Table 5).

Table 5 Connectors / linking words and phrases

Purpose for link Connecting word or phrase

show time/sequence first, second, finally, shortly, immediately, soon, subsequently show importance first, next, most significant, least

add ideas again, furthermore, additionally, indeed, moreover, another show cause and effect thus, since, so, therefore, as a result, hence, consequently, then indicate a summary in brief, finally, to conclude, in fact, indeed, on the whole

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show comparison/contrast on the other hand, though, similarly, however, nevertheless give illustration for example, in illustration, specifically, for instance show concession granted, despite, while, notwithstanding, of course show position next to, beside, in, out, below, juxtaposed

(Adapted from Marshall & Rowland 1993, p. 174)

6.2.3 Clear sentences

Complex sentences are more difficult to read and often contain more grammatical errors. Reading the report aloud is one technique for locating errors. Students should aim to keep sentences as short as possible, although it is effective to vary sentence length. Shorter sentences are useful as summary statements and for emphasis.

6.3 Structuring sentences

6.3.1 Proximity of detail and item

Reading information is easier and the meaning is less confused if the sentence components are kept close together. Care should be given to the proximity of:

- the subject and verb (person and action)

e.g. The test results were confirmed as accurate after a review. ✓ (subject) (verb)

not The test results after a review were confirmed as accurate. ✗ (subject) (verb)

- an item, and its detail or description

e.g. Arrangements were made for a meeting to be held on 19 June. ✓ not On 19

June, arrangements were made for a meeting. ✗

6.3.2 Series of nouns

Reading is improved and less ambiguous when a long series of nouns is avoided.

Also, when verbs or processes are written as nouns (by adding 'ment', 'ance', and 'tion' etc.) the sentence becomes complicated.

e.g. The first test to check the circulating water flow ✓ of the cooling tower...

not The initial performance flow test of the cooling ✗ tower circulating water...

e.g. The fault occurred causing difficulties. ✓ not The occurrence of the fault caused difficulties. ✗

6.3.3 Parallel construction

Items which are listed or grouped should be presented in a parallel form. 'Parallel' means the same grammatical structure, sentence length, capitalisation and layout.

e.g. Reading is easier than writing. ✓ not Reading is easier than to write. ✗

Parallel construction is critical with the use of bullet points or lists (as in instructions, procedures or recommendations).

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e.g. Some aspects of concern are: ✓ not: ✗ • the security of the data; • the need to protect security; • the protection of confidentiality; and • Without doubt, confidentiality; • the costs of such measures. • often these are expensive.

(adapted from Hadgraft et al. 1994, p. 21)

6.4 Grammar

6.4.1 Active and passive voice

The passive voice is usually used in engineering reports where the writing is intended to be impersonal and objective. A passive sentence uses a form of the verb 'to be' (e.g. 'is', 'was', 'has been', 'will be'), plus the past participle of the verb often made by adding 'ed' to the verb (e.g. 'tested', 'identified', 'monitored'). The passive voice is preferred when the human subject or agent is less relevant than the activity.

In a passive sentence, the human subject is placed last or can be removed from the sentence.

e.g. The situation has been monitored by a researcher. (passive) (human subject at end)

e.g. The situation has been monitored. (passive) (no human subject)

The active voice emphasises the human subject or agent by placing it first in the sentence. If no human is directly involved in the action of the sentence (e.g. the subject is a piece of equipment), then it is appropriate to use the active voice.

e.g. A researcher has monitored the situation. (active) (human subject emphasised)

e.g. Temperature gauges monitored the air inside the cylinder. (active) (non-human subject emphasised)

The important issue in the choice between active and passive sentences is placing the key aspect — whether the activity (passive) or the agent (active) — at the beginning of

the sentence.

6.4.2 Tense of verbs

Verb tense changes according to the section in the report, the purpose of the section or sentence and the type of information included. The following suggestions are a general guide only:

- use present tense to explain or discuss
 - continuing objectives e.g. The aim of this report is to... - general principles or laws e.g. Ohm's law states...
 - data or results e.g. These results show... - procedures or instructions e.g. Identify each item by code...
- use past tense (or the past perfect) to state or describe
 - what the objectives were e.g. The main purpose was...
 - what equipment was used e.g. The cylinders were reused... - what procedure was followed e.g. After careful cleaning, the test tubes were... - what happened e.g. The damage to the component had been...

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The review of literature including other research may be written in either the present, past or the present perfect tense. Using the present tense is often less complicated.

e.g. The research by Elberg (1993) shows... (present tense)

e.g. The research by Elberg (1993) showed... (past tense)

e.g. The research by Elberg (1993) has shown... (present perfect tense)

6.4.3 Agreement between noun and number

Nouns need to agree with the number before them. In other words the names of things need to be singular (without an 's') or plural (with an 's') according to whether there is only one or more than one.

e.g. the first check_ ...several checks... ✓ (singular) (plural)

e.g. a pilot program_ ...three pilot programs ✓ (singular) (plural)

Some nouns do not use an 's' to indicate whether they are singular or plural.

e.g. three pieces of equipment_ ; much information_ ✓

6.4.4 Agreement between subject and verb

In a sentence, the subject and verb must agree in relation to number and to person:

e.g. The results ...are available ✓ (plural subject) (plural verb)

e.g. The organisation ...is progressing ✓ (single subject) (single verb)

e.g. I ...am responsible ✓ (first person subject)* (matching verb)

e.g. They ...are responsible ✓ (third person subject) (matching verb)

Mistakes with regard to agreement are common when a single subject is separated from its verb by other plural words, causing the subject to appear to be plural when it is not.

e.g. A list of results ...is available. ✓ (single subject) (single verb)

not A list of results are available. (A list are available??) ✗

The following rule helps students to check if they are using the correct verb (it applies when the present tense is used):

- if the subject is plural (ends in an 's'), then there is no 's' on the verb e.g.

The tests ...demonstrate_ ✓ (subject + 's') (verb with no 's')

- if the subject is single (does not end in an 's'), then the verb does end in an 's' e.g.

The test_ ...demonstrates ✓ (subject with no 's') (verb + 's')

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6.5 Vocabulary

6.5.1 Personal language

Currently in engineering reports, personal language is avoided in preference to impersonal constructions which emphasise scientific objectivity.

e.g. The mining site was investigated. ✓ not I investigated the mining site. ✗

* Check with your lecturer before using personal language such as 'I' or 'we'. Personal language is occasionally used to distinguish between the views or findings of other authors and your own. Alternatively, personal language can be avoided by using objective language.

e.g. Nguyen (1993) determined...whereas my conclusions are... (personal language) e.g. Nguyen (1993) determined...whereas the conclusion of this study is... (objective language)

6.5.2 Gender inclusive language

Gender neutral language is language which includes women and men equally.

Technical reports should avoid using male-only terminology such as 'he'. This can be achieved by:

- using both pronouns

e.g. he/she; s/he; his/her ✓

- using gender non specific terms

e.g. The drafter was responsible for the plans ✓ not The draftsman was responsible for the plans ✗

- restructuring the sentence to avoid personal pronouns

e.g. A report can be completed by the engineer ✓ not The engineer can complete his report ✗

Another method of being inclusive is to use the plural pronoun 'their' with a single subject when it would not be ambiguous. In the past, this was considered to be grammatically incorrect, but the University of South Australia's current Policy on Gender Neutral Language (1991) explains that this is now an option.

e.g. The operator is to check their machine for safety... ✓

6.5.3 Jargon and slang

Clear and simple English should be used rather than jargon (specialised or pretentious language). New terms should be defined for the reader. A glossary of terms or abbreviations may be useful for long, complex reports.

e.g. The use of testing procedures was recommended. ✓ not The utilisation of testing

procedures was recommended. ✗

Informal language or slang should also be avoided. The words which are used should be precise to give an accurate meaning.

e.g. The diode was soldered into the circuit. ✓ not The diode was put into the circuit. ✗

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e.g. Artificial intelligence aims to imitate a brain in a machine. ✓ not Artificial intelligence is like a brain in a machine. ✗

6.5.4 Repetition of words

Often writers have favourite words or expressions. Reports should avoid overuse of particular words, groups of words, or sentence structures.

not Then... Then... Then... ✗ not The test demonstrated... The result demonstrated ✗ not It is a requirement that... It is essential that... This signifies that... ✗

In technical writing, clear expression and attention to detail are essential.

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7 Conclusion

While this document was not developed to be an inflexible set of rules, it is however intended to assist university engineering students in producing professional, well organised reports. Student report writing skills can be enhanced by the application of these guidelines in relation to formatting of documents, components, referencing and technical language.

Students are encouraged to write with attention to the contents of this Style Guide, but where instructed students should follow the directions given by their lecturers or school. Many resources, such as the library, are available to assist students in their report writing. The use of a specific format or style must be followed consistently throughout a report.

By initially planning the presentation of their material, students can concentrate on the content being presented in a report. During the writing and editing process students may find it useful to refer to the Report writing checklist (Appendix H). A well written and logically structured report will result in a quality document which effectively conveys precise information to the reader.

8 Recommendations

With regard to the conclusions detailed above, the following recommendations have been formulated. Students are recommended to:

- identify and follow instructions from particular Schools or employers — particularly where such instructions vary from common practice
- read and use the information in this Style Guide
- ensure internal consistency in formatting (particularly of numbering and

headings)

- give attention to visual impact and readability
- acknowledge sources by referencing both in the text and in a list at the end •

proofread and edit thoroughly all written work

Students should remember that excellent content in a report needs to be supported by high quality presentation, and an accurate use of technical language.

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9 References *

Baylis, P. 1991, Report Writing for Engineers, Engineering Education Australia, Milson's Point.

Eisenberg, A. 1992, Guide to Technical Editing: Discussion Dictionary & Exercises, Oxford University Press, New York.

Equal Opportunity Unit 1991, How to Communicate in Gender Neutral Language, University of South Australia, Adelaide.

Hadgraft, R., Robertson, H. D. & Eaves, D. 1994, Writing engineering reports, Monash University.

Li, X. & Crane, N. B. 1993, Electronic Style: A Guide to Citing Electronic Information, Meckler, London.

Marshall, L. A. & Rowland, F. 1993, A Guide to Learning Independently, Longman Cheshire, Melbourne.

Michaelson, H. B. 1990, How to Write and Publish Engineering Papers and Reports, 3rd edn, Oryx Press, Canada.

National Competency Standards for Professional Engineers (Stages 1 and 2) 1993, 1st edn, The Institution of Engineers, ACT.

Research Degree Student Information Guide 1996, compiler Research Office, University of South Australia, Adelaide.

Style Manual for Authors, Editors and Printers 1994, 5th edn, Australian Government Publishing Service, Canberra.

Schwarz, S. E. & Oldham, W. G. 1984, Electrical Engineering: An Introduction, Holt, Rinehart and Winston, New York.

* This is an example of an alphabetical reference list — used with the author-date (Harvard) system.

(For students who use the numerical referencing system, see section 4.3.2 for an example of a numerical reference list.)

A bibliography (a wider list of all texts that have been read in preparation for writing this document) has not been included in this Style Guide, as it is generally not required for engineering reports.

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Appendix A National Competency Standards for Professional

Engineers

Engineering students should take particular note of competencies 5.1.1 and 5.3.1, (our italics) which are expected of graduate students or Stage 1 Professional Engineers. Graduates are expected to be able to demonstrate competency in these areas with limited guidance.

5. COMMUNICATION

5.1 Communicate effectively in the English language

5.1.1 Communicates fluently in written and oral expression at a professional standard

5.1.2 Contributes to the preparation, interpretation and presentation of information

5.1.3 Liaises with colleagues and experts within the organisation

5.1.4 Correctly interprets engineering instructions received

5.1.5 Issues clear and accurate instructions to subordinates as appropriate

5.1.6 Selects appropriate methods of communication

5.2 Present, report on and advocate engineering ideas

5.2.1 Contributes to the preparation and presentation of lectures at a professional level

5.2.2 Prepares papers for publication in Professional Engineering journals

5.2.3 Communicates engineering information effectively within the engineering work-force, and to others

5.2.4 Conveys engineering information effectively to higher engineering and non engineering levels in the enterprise

5.2.5 Undertakes professional development in negotiation, conflict resolution, counselling, exchanging ideas and conveying convictions and attitudes

5.3 Prepare and comprehend engineering documents

5.3.1 Prepares Professional Engineering reports

5.3.2 Contributes to specifications, standards and graphical representations

5.3.3 Participates in the preparation of more complex documents such as environmental impact statements

5.3.4 Correctly interprets applicable engineering drawings and other graphics, specifications, standards, regulations, codes of practice and environmental impact statements

from National Competency Standards for Professional Engineers (Stages 1 and 2) 1993, 1st edn, The Institution of Engineers, Australia, ACT.

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Appendix B Model title page

University of South Australia

Division of Information Technology, Engineering and the

Environment

School of Electronic and Information Engineering

02478 Communication Systems 2

Computer Simulation Assignment

Student: Kim Citizen

ID Number: 9599999K

Tutor: Mr T. Ferris

Due date: 15/10/95

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Appendix C Plagiarism: Academic Misconduct Policy

Students should be aware of their University's policy on plagiarism. The following definition and policy regarding penalties are taken from the University of South Australia's policy on academic misconduct (from University of SA Policy no. A-11.2 1994). The full policy is located in the Student Diary.

Preamble

With respect to academic misconduct, a distinction will be made between a deliberate act and an inadvertent act occurring through unfamiliarity with academic conventions.

A specific form of academic misconduct is plagiarism. Deliberate plagiarism is regarded as a serious act of academic misconduct.

Definition

1 Academic misconduct, whether inadvertent or deliberate, shall include the following:

1.1 Presentation of data with respect to laboratory work, field trips or other work which has been copied, falsified or in other ways obtained improperly;

1.2 Inclusion of material in individual work which includes significant assistance from another person in a manner unacceptable according to the assessment guidelines for the subject;

1.3 Providing assistance to a student in the presentation of individual work in a manner unacceptable according to the assessment guidelines for the subject;

1.4 Plagiarism, whether inadvertent or deliberate, which includes the following:

1.4.1 Word-for-word copying of sentences or whole paragraphs from one or more sources (the work or data of other persons), or presenting of substantial extracts from books, articles, theses, other unpublished work such as working papers, seminar and conference papers, internal reports, lecture notes or tapes, without clearly indicating their origin. This should be referenced appropriately.

1.4.2 Using very close paraphrasing of sentences or whole paragraphs without due acknowledgment in the form of reference to the original work;

1.4.3 Submitting another student's work in whole or in part;

1.4.4 Use of another person's ideas, work or research data without acknowledgment;

1.4.5 Submitting work which has been written by someone else on the student's behalf;

1.4.6 Copying computer files without clearly indicating their origin;

1.4.7 Submitting work which has been derived, in whole or in part, from another student's work by a process of mechanical transformation (eg changing variable names in computer programs);

1.5 Other actions not covered by the above clauses may be judged by a Dean or the Registrar to be acts of academic misconduct and be dealt with under the provisions of this policy.

Policy on penalties

2 A student found guilty of academic misconduct shall be subject to any of the following penalties:

2.1 failure, that is zero score, in the component of the subject, a note in the student's file and a reprimand from the Head of School;

2.1 failure in the subject, a note in the student's file and a reprimand from the Head of School;

2.3 suspension from the university for a period not exceeding three years;

2.4 expulsion from the university.

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Appendix D Punctuation in technical writing

The purpose of punctuation is to convey ideas clearly and without ambiguity.

Current publishing styles often use minimal or no punctuation, such as in lists.

1 Full stops .

- end a complete idea or sentence

e.g. The machine was operated by a technician for 24 hours a day.

- end an abbreviation where the final letter of the abbreviation is not the last letter of the word (other than discipline-specific abbreviations or units of measurement — check SI units)

e.g. p. for page and ch. for chapter

- show that a word(s) has been omitted using the three points of ellipsis (...)

e.g. Several causes...have been rectified.

2 Colons :

- introduce a list or quotation

e.g. The following factors are critical: cost recovery, priority of tasks and competitiveness. e.g. Farre

(1990, p. 51) stated: 'Education and training are central to the development of industry.'

3 Semicolons ;

- connect two sentences or main clauses which are closely connected, and often not joined with a linking word

e.g. The initial survey revealed a high interest; results showed that further development is valid. •

separate complex or wordy items in a list

e.g. The following factors are critical: the environmental impact statement; the government and union

policies; the approval of business and council; and public opinion.

4 Commas ,

- show a pause or natural separation of ideas

e.g. After the recommendations were implemented, further evaluations were conducted. •

bracket or separate information in a sentence

e.g. The additional results, which were withheld, raised doubts about the initial hypothesis. •

precede linking words, such as 'but', 'so', 'hence', and 'whereas' e.g. The aim was to test strength and density, but the measurements taken were not accurate. • separate information in a list

e.g. The items included: circuit boards, several meters, wires and a power supply.

5 Parentheses ()

• provide information additional to a sentence

e.g. Engineering and Technology courses (as well as Applied Science) are studied at The Levels. •

enclose author-date references in the text

e.g. A number of experiments (Smith 1987; Tan 1990; Wong 1991)...

• enclose the number for an equation, and bracket parts of the equation e.g. $x = G e^{-\omega t} \sin(\omega dt + \phi)$ (1)

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6 Quotation marks ' ' or " " (single quotation marks are more common)

• indicate a shorter quotation as part of a sentence

e.g. 'In information technology, speed is of great importance,' as highlighted by Schwarz and Oldham

(1984, p. 2) in their introduction to the scope of Electrical Engineering. • show the titles of journal articles

e.g. 'Efficient low cost solar cell developed' (1994) is an example of an article without an author.

7 Hyphens and dashes - hyphen - en rule — dash

• join words with a hyphen only when the meaning is changed without them

e.g. liquid-solid separation

• use an en rule (length of the dash equals 'n') to indicate a range e.g. 1939–1945 • use two dashes (em rules) when adding parenthetical information to a sentence e.g.

Understand the uses of the hyphen and the dash — the dash is longer — to avoid confusion.

8 Apostrophes for ownership 's

• place the apostrophe at the end of the owner-word, then add a possessive s

e.g. The researcher's results. (one researcher owns the results) add apostrophe + s

• if the original word ends in an s, place the apostrophe at the end of the owner word without adding a possessive s

e.g. The researchers' results. (more than one researcher owns the results) add apostrophe only (possessive s is invisible)

9 Capital letters XYZ

• use minimally, especially in titles and headings where small words such as 'and', 'in', 'the', and 'by' should not be capitalised

• use only for a specific and formally named item (e.g. 'Ohm's law'), not for a general reference to an item (such as 'a law')

10 Punctuating numerals and abbreviations

- write numbers of ten or less in words, except when followed by units

e.g. nine experiments e.g. 9 mm

- place a space between the numeral and the unit of measurement or operator sign, and do not use full stops with units of measurement e.g. 17 cm e.g. $y = v + 2$
- use SI units (Système International d'Unités)

- do not abbreviate or shorten forms of words in formal reports

e.g. it is a difficult situation so the company does not...

not it's a difficult situation so the company doesn't...

- use the standard or recognised abbreviations of your discipline

e.g. the current was 225 amps when measured...

11 Punctuating bullet point lists

- introduce lists with a colon (:) and either have no punctuation after each item, or punctuate as a normal sentence (with a full stop at the end)

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e.g. The Style Guide includes: or The Style Guide includes: • formatting • formatting;

- report components • report components; • referencing • referencing; and •

technical language • technical language.

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Appendix E Summary of elements in reference lists

This table outlines the elements needed for a reference list, and the order in which those elements appear in both the alphabetical (author-date) reference list, and the numbered reference list. These details are based on the AGPS Style Manual (1994). The type of reference list used must be consistent with the system of referencing selected for the particular report.

Numbered reference list

Alphabetical reference list

(Numerical system)

(Author-date system)

BOOKS – order the information as follows:

(differences highlighted in bold)

- author's surname • author's initials or given name
- author's initials or given name • author's surname
- year of publication
- title of publication • title of publication

(italics or underlined)

(italics or underlined)

- title of series ‡
- title of series ‡
- volume number, or number of volumes ‡

- volume number, or number of volumes
- edition ‡
- ‡ • edition ‡
- editor, reviser, compiler or translator ‡
- editor, reviser, compiler or translator

(if other than the author)

‡ (if other than the author)

- elements of a book ‡
- elements of a book ‡
- publisher
- publisher
- place of publication
- place of publication
- year of publication
- page numbers or numbers ‡

ARTICLES – order the information as

follows: (differences highlighted in bold)

- author's surname Note: ‡ indicates an optional element
- author's initials or given name • author's initials or given name • author's surname
- year of publication
- title of article
- title of article
- title of journal or periodical
- title of journal or periodical (italics or (italics or underlined) underlined)
- title of series ‡
- title of series ‡
- place of publication ‡
- place of publication ‡
- volume number ‡
- volume number ‡
- issue number or other identifier ‡
- issue number or other identifier ‡ • year of publication
- page numbers or numbers
- page numbers or numbers

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Personal communications — such as letters, conversations and email messages — need not be included in a reference list as they cannot be checked, and are usually

incorporated into the text.

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Appendix F Library resources and report writing texts

These notes should be read in conjunction with relevant library publications.

Finding information

In most libraries, materials are arranged in Dewey decimal classification (i.e. resources on the same subject are shelved together). Below is a brief guide to the location of the main engineering subjects:

621 Mechanical Engineering 624 Civil Engineering 621.3 Electrical
Engineering 660 Chemical Engineering 621.38 Electronic Engineering 669
Metallurgy 622 Mining 670 Manufacturing

Use the library catalogue — usually a computerised catalogue — to identify all possible sources of information in the library. Relevant engineering material can often be found by looking on the shelves in different areas of the collection.

Books written by subject experts may provide you with some of the required information.

CD-ROM databases are 'electronic indexes' which help you to identify relevant periodical articles, conference papers, product catalogues and standards. Useful engineering CD-ROMs may include: Applied Science and Technology Index; Compendex (known as Engineering Index in the print version); Engineering and Applied Science (collection of Australian engineering databases); and INSPEC (for electrical and electronic engineering, computing, and manufacturing).

Reference materials such as encyclopedias, dictionaries and handbooks will help you to find brief, factual information such as definitions, constants and equations. Reference books can be found in the Reference Collection, and can only be used in the library.

Standards are very useful sources of practical information for design work. Check which standards are held in your library by using the catalogue or by asking at the help desk. Standards available may include Australian Standards and standards from international organisations such as British Standards, ASTM, ISO and IEEE.

Periodicals or journals provide up-to-date and often very detailed information. The Serials List or the catalogue shows which periodicals are held by the library.

Information desk

At the information desk a member of staff is available to assist you to locate information in your course. Staff can also help you with questions about how to use

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the catalogue, the CD-ROMS and other resources. Library staff are happy to help you with any problems — you only need to ask.

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Appendix G Purposes of paragraphs

A paragraph will be written well if the writer is certain of its purpose(s). Both broad and specific purposes of technical paragraphs are shown in the following table:

Broad purpose Specific purpose Identifying features of the paragraph

(Type of paragraph)

Describing facts Defining • term or name

- group/category/device/type/concept
- distinguishing characteristics or elements

Complex defining

- additional specifications
- explanation of operation
- additional classification
- further examples

Classifying • ordered members or items

- class or principles of arrangement
- definition/basis of classification

(size, shape, type, category etc.)

Describing • item/place

- Physical • developmental stages and connections •
- Process series of instructions relating to a procedure
- use or purpose of an item
- Function • elements involved in a procedure
- Special relationships • connection of two or more items

Describing Analysing cause and effect • statement of cause

the • list of effects - short and long term

relationship • statement of relationship

between facts • statement of reasons for effects

Exemplifying • example(s) or illustration(s)

(Explaining) • evidence

(often used with models, charts, tables etc.)

Comparing/Contrasting • statement of main idea

- development of idea
- statement of two or more

comparative/ contrasted items of information

Adapted from

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Baylis, P. 1991, Report Writing for Engineers, Engineering Education Australia, Milson's Point.

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Appendix H Report writing checklist

Report writing checklist

For effective report writing it is essential to thoroughly edit your work. This checklist may be useful for the editing and redrafting process. Check each of the following:

Formatting improving layout and presentation

- ☐ headings & subheadings developed logically and consistently at each level (e.g. size and style of headings)
- ☐ decimal numbering system used accurately and consistently; fourth-level avoided Tables and figures
- ☐ key tables/figures used in the text (others, if required, in Appendix) ☐ zeros and rounded numbers used for clarity sometimes
- ☐ captions of tables/figures selected for specific differences between tables/figures ☐ caption information consistent with text information
- ☐ data in tables/figures consistent with data in report
- ☐ symbols/labels/signs explained clearly
- ☐ asterisks (*) added explanatory notes of further information, abbreviations, sources etc. that do not fit into rows or columns

Components enhancing logic and readability

Preliminaries

- ☐ names/titles of people etc. spelt correctly & acknowledged fully
- ☐ abstract written to highlight and summarise significant information ☐ disclaimer signed and authorship declared
- ☐ acknowledgments identified all who assisted with material and with editing ☐ contents pages matched exactly to text (e.g. titles of headings with decimal and page numbering)
- ☐ figures & tables lists matched exactly to labels and captions ☐ symbols list included all symbols from the text

Text of report

- ☐ sections divided clearly and developed logically ☐ abbreviations & acronyms written in full when first used
- ☐ definitions of new terms expressed accurately and clearly
- ☐ Appendices (each) referred to in the text of the report Referencing ensuring correct academic standards
- ☐ all sources of information, referenced correctly twice, both in text and in reference list other than your own
- ☐ only one reference system used either author-date or numerical system – not both ☐ punctuation and elements standardised exactly (e.g. order of elements, punctuation, capitalisation and publishing details)

Technical language editing for style and grammatical accuracy

- ☐ repetition of words avoided
- ☐ sentences tightened, focused and varied in length ☐ longer sections or paragraphs introduced and/or summarised to provide an overview of the information given

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- ☐ passive voice used to emphasise the object of action rather than the agent ☐ parallel construction applied accurately in lists of information or sentences ☐ subject and verb agreement related in number and person
- ☐ gender inclusive language used correctly

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GET 322: TECHNICAL WRITING AND REPORTING

...DR C.O. MGBEMENA

Data gathering and presentation

Data can be generated in accordance to known and precise laws or in a purely haphazard way

☐ Deterministic data: Data generated in accordance to known and precise laws

☐ Random data: Data generated with no known precise laws

Exercise 1: Plot a graph in Excel of the data below:

0.00 15.00

0.20 14.80

0.40 14.22

0.60 13.24

0.80 11.86

1.00 10.10

1.20 7.94

1.40 5.40

1.60 2.46

Typical examples of random data are:

- Thermal noise generated in electrical resistances, antennae etc.
- Brownian motion of tiny particles in a fluid.
- Weather variables
- Financial variables such as stock exchange share values
- Gambling game outcomes (dice, lottery, lotto, pool etc.)

1.2 Population, sample and statistics

Consider the following three data sets examples:

Example 1.1

The following table 1.1 lists the number of firms that were established in town X during year 2000 in each of three branches of activity.

Table 1.1 Nature of firms in town X

Branch of activity	Number of firms	frequencies
Commerce	56	$56/109=51.4\%$
Industry	22	$22/109=20.2\%$
Services	31	$31/109=28.4\%$
Total	109	$109/109=100\%$

Example 1.2

The following table 1.2 lists the classifications of a random sample of 50 students in the examination of a certain course evaluated on a scale of 1 to 5.

Table 1.2 Random sample of 50 students

Classification	Number of occurrences	Accumulated frequencies
1	3	$3/50=6.0\%$
2	10	$13/50=26.0\%$
3	12	$25/50=50.0\%$
4	15	$40/50=80.0\%$
5	10	$50/50=100.0\%$
Total	50	100%

*Median=3 Value below which 50% of cases are included

Example 1.3

Table 1.3 lists the measurements performed in a random sample of 10 electrical resistances of nominal value 100 Ω

produced by a machine.

Case # Value (in Ω)

1 101.2

2 100.3

3 99.8

4 99.8

5 99.9

6 100.1

7 99.9

8 100.3

9 99.9

10 100.1

Mean

Table 1.4 Random datasets examples

Dataset Variable Value domain Type

Firms in town X, year Discrete, nominal

2000

Classification of exams Discrete, ordinal

continuous

1.4 Software tools

There are many software tools for statistical analysis covering a broad spectrum of possibilities. Some popular examples are: MINITAB

SPSS,

STATISTICA,

MATLAB,

TABLEAU and

R etc.

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Dr. C.O. Mgbemena

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Topics to consider:

1. Data gathering and presentation
2. Technical correspondence such as letters of inquiry and replies
3. Letters of application and memoranda

4. Illustrating technical writing using tables, graphs, diagrams, equations and appendices
5. Writing progress reports, proposals, student projects, thesis and dissertations
6. Oral and visual presentations
7. Computer aided technical writing and presentation using word processing and presentation software packages

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CHAPTER TWO

2. TECHNICAL CORRESPONDENCE

Letters are used to communicate outside your organization. Whereas the memorandum (memo) is the primarily used for communication within an organization, letters are often used to communicate to individuals outside an organization, especially in formal and semiformal contexts. Letters are an essential part of all business and technical communication because they are more formal and reliable than electronic mail (e-mail) and more precise and permanent than telephone or face-to-face conversations.

2.1 Types of Letters

Like memoranda, letters perform many functions in scientific and technical communication. The following are some of the most common types of letters written by people in technical fields:

- Inquiry letters
- Job application letters
- Acceptance letters
- Transmittal letters
- Technical-information letters
- Letters of recommendation

Format of a Letter

If your organization has a specific style for business letters, follow that format. Otherwise, follow the guidelines provided here.

Business letters are commonly either full-block formatted, with every line starting at the left margin and usually a business letterhead at the top of the page, or modified-block formatted, with the heading and the closing aligned at the centre of the page.

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Elements of a Letter

Business letters have the following elements:

- Heading
- Date
- Recipient's address
- Salutation
- Body
- Closing
- End notations

Heading

If you are using letterhead stationery, include only the date two lines below the bottom of the letterhead. Spell out the name of month.

If you are not using letterhead stationery, begin with your full address (city, street, and zip code) 1 to 1½ inches from the top of the page. Spell out address designations, such as Street, Avenue, and West. The state name may be abbreviated using the two-letter, all-capitals U.S. Postal Service designations. Include the date aligned at left with the address, spelling out the name of the month.

Recipient's Address

Two to four lines below the date, place the following items:

- The recipient's title (such as Mr., Ms., or Dr.) and full name (address a woman who does not have a professional title as Ms. unless you know she prefers Miss or Mrs.; if the recipient does not have a title and you are unsure of his or her gender, omit the title).
- The recipient's job title, if appropriate.
- The name of the company or institution, if appropriate.
- The full address, following the same format as for the address in the heading.

The recipient's address is always aligned on the left margin.

Salutation

Place the salutation two lines below the recipient's address. The salutation begins with the word Dear, continues with the recipient's title and last name, and ends with a colon. If you are unsure of the recipient's gender and the recipient does not have a professional title, omit the title and, instead, use both the first and the last names in the salutation (Dear Leslie Perelman:). If you do not know the name of the recipient

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of the letter, refer to the department you are writing to (Dear Technical Support:).

Avoid salutations such as Dear Sir or Madam:

Body

Start the letter two lines after the salutation. Body paragraphs should be single spaced with a double space between paragraphs. (Indenting the first line of each

paragraph is acceptable but is more informal than the unindented style.)

Be brief, direct, and considerate. State the letter's purpose in the opening paragraph. Include supporting information in a middle paragraph or two, and conclude your letter with a brief paragraph that both establishes goodwill and expresses what needs to be done next.

If a letter requires more than one page, make sure there are at least two lines of body text on the final page. Never use an entire page for just the closing. The second page and all subsequent pages must include a heading with the recipient's name, the date, and the page number.

Closing Phrase

Write a complimentary closing phrase two lines below the final body paragraph. Yours truly, Sincerely, or Sincerely yours are common endings for professional letters. Capitalize the first letter of the first word of your complimentary closing, and end the complimentary closing with a comma.

Four lines below the closing phrase, write your full name. If you are writing in an official capacity that is not included in the stationery's letterhead, write your title on the next line. Your signature goes above your typed name.

End Notations

At the bottom of the last page of a business letter, end notations may show who typed the letter, whether any materials are enclosed with the letter, and who is receiving a copy of the letter.

The typist's initials, in lowercase letters, follow the initials of the author, in capital letters, and a colon or a front-slash (LCP:ecb or LCP/ecb).

An enclosure notation--Enclosure:, Encl., or Enc.--alerts the recipient that additional material (such as a résumé or a technical article) is included with the letter. You can either identify the enclosure or indicate how many pieces there are.

Enclosure: Article by I. W. Waitz

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Encl. (2)

Enc. (2)

In addition to the enclosure notation, always refer to your enclosures explicitly within the text of the letter.

A copy notation (cc:) lets the recipient of the letter know who else is receiving a copy. Put each recipient of a copy on a separate line.

cc: Dr. Maria Lopez

Mr. William Astley

2.1.1 Letters of inquiry

A letter of inquiry seeks answer(s) to specific information. In certain cases, such as a request for promotional material, the recipient will have a clear interest in responding to your letter. In other cases, such as a request for specific information on a product, the recipient may or may not be as motivated to respond quickly. Consequently, always make the tone of the letter friendly and make it easy for the recipient to identify and provide the information you need.

Format of a Letter of Inquiry

The following format is employed in writing a letter of inquiry:

1. In the first paragraph, identify yourself and, if appropriate, your position, and your institution or firm.
2. In the second paragraph, briefly explain why you are writing and how you will use the requested information. Offer to keep the response confidential if such an offer seems reasonable.

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3. List the specific information you need. You can phrase your requests as questions or as a list of specific items of information. In either case, make each item clear and discrete.
4. Conclude your letter by offering your reader some incentive for responding.

The following letter of inquiry is written by a computer programmer requesting specific information about an upcoming release of a software product.

2.1.2 Job Application Letters

Job Application letters are usually just one page and consist of three sections:

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1. Front matter. State that you are applying for a specific job title or field. Also, mention the person who referred you or told you about the job. If you learned of the job from an advertisement, mention that.
2. Body. Explain specifically why you are qualified for the job. Describe education and work experience and any other activities that display relevant talents, such as foreign-language proficiencies and leadership or supervisory experience.
3. End matter. Refer to your enclosed résumé or curriculum vitae and express your desire for an interview, stating when and where you will be available for one. In addition, invite further inquiries, and state how you can be contacted.

Following is an example of an application letter.

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2.1.3 Acceptance Letters

Letters of acceptance not only convey information, but also establish a contractual agreement between you and the organization or person offering you the position. Consequently, acceptance letters should accurately restate the key terms of employment. Acceptance letters should be sent as soon as you are sure that you want to accept a job offer. If you need more than two weeks to decide if to accept an offer, telephone or write the person making the offer and ask for an extension of time to decide.

Format for writing an acceptance letter

- i. Begin the first paragraph by enthusiastically accepting the job. Be sure to state the exact title of the job and the salary.
- ii. The second paragraph usually discusses details about the position, especially the date that you will begin work.
- iii. The last paragraph is short statement of goodwill, usually indicating that you are looking forward to starting your new job.

The following is an example of a standard letter of acceptance.

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2.1.4 Technical-Information Letters and Memoranda

Technical-information letters and memoranda are short documents that announce new technical information, such as a software bug and its solution, or a new feature. Use the memorandum format if the information is being sent inside an organization. Use the letter format if the document will be sent to outside individuals.

The following example is adapted from a technical-information memorandum written by Information Systems at the Massachusetts Institute of Technology. The document informs users of MIT's Athena network of a change in the protocols of

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MIT's main World Wide Web (WWW) server that allows users to place shorter Uniform Resource Locators (URLs) in their WWW pages.

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2.1.5 References and Letters of Recommendation

References are an essential component of almost all admissions, grant, and employment processes. Hiring managers, human resources (HR) and other individuals in any selection process need evaluations from individuals who have had long and close contact with applicants in order to assess accurately their abilities and accomplishments. The number of recommendations required by employers and universities usually ranges from one to five, with three being the most common

number.

There are two basic forms of references: oral and written. Most often, oral references are conducted by telephone. The name, professional title and affiliation,

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address, and phone number of each reference are included in a list of references, which is often mentioned in résumés and application letters but always kept separate from them.

Give your list of references to a potential employer or to a selection committee only when it is requested. Your recommenders are doing you a favour. If you subject them to too many intrusions, you may find them less willing to help you in the future.

Written letters of recommendation are more formal. Often, they are written to a specific individual for a specific position. In other cases, however, letters of recommendation are written for a general type of job or for graduate school and are included in a collection of letters of recommendation called a dossier.

Obtaining Letters of Recommendation

Make your candidacy as strong as possible by carefully soliciting references and letters of recommendation. Letters of recommendation are extremely important in decisions to hire or admit an individual or to award a scholarship or grant. Consequently, be sure to spend adequate time and effort to ensure their effectiveness.

In some cases, you list an individual as a reference, and the organization or individual considering your application will contact that person, often only if you are a finalist for the position. Other situations, however, require a formal letter of recommendation from each reference.

General Guidelines for Requesting References or Letters of Recommendation

- If possible, always ask one or two more persons than the minimum number you need.
- Ask a range of persons who can testify to different abilities and accomplishments relevant to the position or award.
- If possible, ask someone in person to write a recommendation or to be a reference. If the individual is far away, ask him or her by phone or through a written letter. Generally, unless you know the person extremely well, do not ask someone to be a reference or write a letter of recommendation through e-mail.
- Never list a person as a reference or as someone who will provide a letter of recommendation until the person has agreed to do so.
- Give each person serving as a reference or writing a letter of recommendation the following written information:

- o a description of the position or award you are applying for
- o a current résumé or curriculum vitae

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- o a short statement of your professional goals longer than a one-sentence career-objective statement included as part of your résumé
- o background material to refresh the recommender's memory of specific work you did with him or her that you feel should be included in a letter of recommendation

In addition, follow these guidelines for obtaining written letters of recommendation:

- Generally, people give more weight to recommendations that contain a waiver of your right to see the recommendation. Sometimes, however, it may be prudent not to waive your right to review the recommendation, especially if you are unsure of the recommender's attitude toward you or of his or her writing ability.
- Recommenders are busy people. Ask a person to write the recommendation and give him or her all the necessary material at least three weeks before the recommendation is due.
- Sometimes recommendations are sent to a central office, such as a university career center, which then forwards the recommendations to specific individuals. In these cases, you may be able to ask someone in the office to review the recommendations on file and suggest which combination of recommendations best present an overall picture of your abilities and accomplishments related to your career objectives.
- If appropriate, provide each recommender with
 - o all required recommendation forms; be sure you have completed all parts of the form you are required to fill out.
 - o a sheet stating to whom or where the recommendation should be sent.
 When appropriate, include addressed envelopes and, in some cases, postage.
- One week before the recommendations are due, tactfully ask each recommender if he or she has sent the recommendation.
- After the letter of recommendation has been sent or the reference has been given, write a short letter thanking the recommender. The recommender has spent time helping you, and such a recognition of his or her efforts not only is polite but also may make it easier for you to request future references or recommendations.
- Keep all letters of recommendation and references current. In general, written recommendations that are older than three or four years carry little weight.

Guidelines for Writing Letters of Recommendation

Before writing the letter:

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- In most cases, agree to write a letter of recommendation only if you can honestly write a supportive letter. If you cannot portray an individual positively, decline to write the recommendation.
- Ask for a current résumé and as complete a description as possible of the position or program to which the person is applying.
- Assemble and review all other relevant information you may have about the person you are recommending. It is often easy to overlook some important accomplishment.

Writing the letter:

- Present the person truthfully but positively. A recommendation that paints an unrealistic picture of a candidate may be discounted. A recommendation that focuses on negative qualities may do more harm than intended.
- Tailor the recommendation to the position. A letter recommending an individual for a job as a camp counsellor should contain different information from that in a letter recommending the same individual for a job as a computer programmer.
- Begin the letter by describing how you know the individual you are recommending and the specific contexts upon which you are basing your evaluation. In what situations have you known the individual? For how long? How closely?
- Present the individual's general qualities relevant to the position along with one or two detailed examples. Including vivid detail will make the recommendation much more effective.
- In most cases, a letter of recommendation should consist of three or four paragraphs and not be over one page in length.

2.2 Memoranda

Memoranda are brief, informal reports used to establish a record. They generalize the communication process by transmitting the message from one or more authors to one or more recipients. E-mail messages typically take the form of memoranda.

The memorandum is among the most versatile of organizational documents. From brief research reports and progress reports to trip reports and thumbnail proposals, the memo form is widely used to communicate technical and administrative information. Memoranda are written for numerous internal purposes--for example, to request information, to make announcements, to outline policies, and to transmit meeting minutes. Thus, in most organizations, memos play a crucial role in establishing a record of decisions, requests, responsibilities, results, and concerns.

The Memo Heading

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The distinctive element of the memorandum is its heading, which is used to frame the message in a very accessible and transparent manner.

This information sets out the context of the message and should be detailed enough to make the context very clear.

The Memo Body

Generally, organize the topics of the memorandum in order of importance, with the key statements first and the details further on. The memorandum should normally begin with a brief summary statement, in one or two sentences, identifying the key topic and the scope of the memorandum.

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ASSIGNMENT #3

(DO NOT COPY OR CHEAT IN THIS ASSIGNMENT)

1. Write a letter of recommendation to EFCC Nigeria providing relevant information and presenting an individual truthfully and positively.
2. Write an internal memo as the Course representative to your class informing them of the Eid-El- Kabir Holiday and postponement of a GET 322 lecture.

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University of South Australia

Division of Information Technology, Engineering and the Environment

School of Natural and Built Environments

Report Writing

Style Guide

for Engineering Students

4th edition

by Anne Winckel and Bonnie Hart

revised and updated by Monica Behrend and Bev Kokkinn

Prepared in association with Engineering staff and students, and the

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June 2002

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- inclusion of a section of Writing a section
- updating of the Referencing section

Major revisions in the third edition:

- inclusion of examples of electronic referencing • additions to Appendix D, Punctuation in technical writing • inclusion of examples in Appendix G, Purposes of paragraphs

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- addition of an index
- variation to the headings for lists of Figures, Tables and Symbols
- addition of a table on the report writing process and related resources (Table 1)
- addition of a section on Starting writing (section 5.1) • inclusion of all referencing examples in section 4 instead of the appendices
- addition of new diagrams (Figs 10–13) to illustrate the elements involved in a reference list

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Abstract

Students need to develop the skills associated with report writing to communicate appropriately at University and to fulfil the obligations of their future employment. The provision of this Report Writing Style Guide for Engineering Students is to assist students in developing these writing skills. The initial project was funded by the University of South Australia Foundation Inc, the Faculty of Engineering and the Flexible Learning Centre (formerly the Centre for University Teaching and Learning) at the University of South Australia. Learning Advisers, Engineering Staff and personnel from other units collaborated to produce this document which is written in the form of a model report. The Style Guide specifically deals with: formatting guidelines, components of a report, writing a section, referencing of sources, and the technical language appropriate to a quality report. Style is often a matter of personal preference. Report writing styles will sometimes differ according to the purpose of the report and the requirements of a particular school, employer or publisher. It is recommended that particular attention be given to visual impact and readability; referencing of sources; proofreading and editing; and that the selection of style be consistent and appropriate to both audience and purpose.

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Disclaimer

We declare the following to be our own work, unless otherwise referenced, as defined by the University's policy on plagiarism.

Anne Winckel Bonnie Hart Monica Behrend

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Symbols

G = amplitude of oscillation

t = time

x = displacement

X_{st} = static deflection of the system under the influence of the imposed force β =

frequency ratio

ϕ = phase angle

ω = natural circular frequency

ω_d = damped natural circular frequency

ζ = damping ratio

x

1 Introduction

Report writing is an essential skill for engineering students. Engineering reports analyse data, present results and conclusions, and make recommendations in a logical, precise and accessible manner. Report writing is both a requirement for completing an engineering qualification at university and a central part of succeeding in future employment. The ability to prepare professional engineering reports is one of the competency standards specified by The Institution of Engineers, Australia (1993) (see Appendix A). Having recognised the need for students to write quality reports, the University of South Australia Foundation Inc, the Faculty of Engineering, and the Flexible Learning Centre initially provided funding to produce this Report Writing Style Guide for Engineering Students (now referred to as Style Guide). Learning Advisers developed the Style Guide in consultation with University of South Australia Engineering staff, staff from other units within the University and other personnel from various external organisations.

While this Style Guide has been prepared to assist engineering students with their report writing, it is not intended to be an inflexible set of rules but rather guidelines. Style is often a personal choice, and report writing styles and content will differ according to the type of report, the purpose, and the audience. For example, engineering companies have their own 'in-house' style. For each report, students should consult their lecturer or school to determine if any specific requirements apply — just as engineers who publish articles and technical reports must conform to the style required by their particular publisher or employer. (Staff may vary guidelines from this document according to school requirements.)

This Style Guide is one of many resources that a student may use when writing a report (see Table 1). In the writing process often the planning, writing and revising stages need to be repeated to achieve a quality document.

Table 1 Resources for report writing process

Report writing process Possible resources

Plan • Analyse requirements/topic $\Rightarrow$ lecturers; topic guidelines and objectives •

Brainstorm/preliminary plan $\Rightarrow$ prior knowledge

Study • Research/investigate $\Rightarrow$ library; computer data; field work; experiments

Plan • Revise plan $\Rightarrow$ newly informed knowledge; notes • Outline sections and headings $\Rightarrow$

Style Guide: structuring technical writing

Write • Draft report $\Rightarrow$ Style Guide: formatting, components, referencing, and technical language

Revise • Edit and proofread $\Rightarrow$ Style Guide: recommendations, technical language, report

writing checklist

Write • Revise draft report ⇒ Style Guide: formatting, components, referencing, and technical language

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Publish • Publish final report ⇒ other students; staff

Where possible, this Style Guide models the format of an engineering report. At times, however, this document varies from report styles expected of students. (For instance, as this document is intended to provide information in an easily accessible format, there are numerous bullet points.) This Style Guide suggests guidelines for engineering reports in relation to formatting, components, writing a section, referencing of sources, and technical language. It is essential that aspects of style are consistent throughout a report.

1.1 Project reports

Primarily this Style Guide will be useful to undergraduate engineering students who are writing project reports. Project reports must be presented in a well structured and visually attractive manner; the competent use of technical language, and accurate referencing of all sources is also a requirement.

To logically structure a report, students should understand the purpose of each component. The preliminary pages of a report include an Abstract which summarises the purpose, methods, results and conclusions of the project. The Introduction provides the background, objective(s), approach, scope and limitations of the report. The main sections of the report include the work or research undertaken and the associated information. This must be arranged logically with headings to guide the reader. Visual aids such as figures and tables can assist with the clear communication of material. The Conclusion summarises and identifies the important findings, placing them in context. Recommendations may be included indicating future directions. A report ends with a list of references, and any required appendices.

1.2 Other uses of the Style Guide

Students will be required to write a variety of other reports while at university, such as laboratory/practical reports, field reports, industrial experience and vacation reports. Postgraduate students will also have to produce a thesis. Not every aspect of style detailed in this document will be relevant to all of these reports. The Technical language section however, may assist all students with their writing, while the Referencing and Formatting sections may also be of use to postgraduate students. An example of the requirements of a thesis are set out in the University of South Australia's publication, Research Degree Student Information Guide (1996).

Writers should strive to present their work with clear expression and within an organised structure. Such writing will lead to a professional result which communicates effectively.

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2 Formatting guidelines

Appropriate formatting of reports improves the readability and accessibility of information. This section outlines essential formatting guidelines as required by your lecturers. An important principle underlying all formatting choices is consistency. Therefore each report should maintain the same internal style and structure, which can be achieved. This Style Guide is formatted as a model and the guidelines assume that reports will be word processed on a computer. If you are unsure about how to apply the formatting guidelines which follow, refer to the Help menu within the word processor. If reports are to be hand-written, students should consult their lecturers.

2.1 Templates

One way in which consistency of a word processed report can be achieved is through the use of a template. The template forms the basis of a word processed document as it determines the basic structure and formatting for the entire document. You can use templates provided with your word processor or use the ones provided by your lecturers.

When using a template, styles can be created and modified to suit your purposes. For example, you can create styles for the several levels of headings. Creating a style saves you the tedium of applying the same font size, appearance, numbering and white space every time you type a heading or sub-heading. The single most significant advantage is that using styles allows you to automatically create a Contents page.

2.2 Format of pages

2.2.1 Paper and margins

- white A4 paper is appropriate, and printing on one side only is often preferred for assessment purposes
- each major section of the report begins on a new page; this may be inappropriate for very short sections or short reports
- wide margins are recommended; for example, allowing 2.5 cm on all sides • bound reports have a left-hand margin of at least 3.5 cm, and 1.5 on the other three sides

To change these features using MSWord 2000, go to the File menu and access Page Setup.

2.2.2 Page numbering

- the title page should not be numbered
- all other pages may be numbered either in the right-hand upper corner, or in the centre at the bottom of the page
- roman numerals — i, ii, iii, iv etc. — are usually used to number the preliminary pages (Abstract, Contents etc.)

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- arabic numerals — 1, 2, 3, 4 etc. — are used to number the text from the Introduction page to the end of the Appendices
- alternatively, pages in the Appendices can be numbered internally, according to the letter of the individual appendix and the number of pages within each appendix — A1, A2, B1, C1, C2, C3 etc. (where Appendix A has two pages; Appendix B has one page, and so on)

To change these features using MSWord 2000, go to the View menu and access Header and Footer.

2.2.3 Headers and footers

- headers or footers position page numbers automatically
- two different headers or footers must be created so that the preliminary pages are numbered separately from the text of the report
- student name(s) and ID number(s) can appear in a footer placed against the right margin in 9 point size (see the bottom of this page)

2.3 Format of type and headings

2.3.1 Font

- a serif font (such as Times or Palatino) is usually chosen for the text of a report, as serif fonts tend to be easier to read
- a different font (such as the sans serif fonts, Helvetica or Arial) may be used for headings and tables
- the same font should be used throughout the whole report, unless a second font is chosen for headings and tables

To change these features using MSWord 2000, go to the Format menu and access Font.

2.3.2 Size and style of type

- 12 point type is the normal size for the text of a report
- 10 point may be used for long, indented quotations, figures and tables
- Table 2 suggests a style of headings with specific sizes
- italics can be used to emphasise text where necessary (but it should not be overused, particularly as there may be confusion with italicised titles)
- underlining or bold is avoided in emphasis of text as it is distracting
- the use of precise words may be sufficient to create the required emphasis

2.3.3 Headings

- headings are numbered according to their importance, using the decimal numbering system
- some writers do not use numbering, but differentiate between levels of headings through size and style (readability and access to information are the key considerations)
- headings without text should never appear on the bottom line of a page

- size and style of headings should vary according to the importance of the heading; the suggested style in Table 2 may prove helpful (to change the style of headings, go to the Format menu and access Style)
- headings can be differentiated using bold type, size and italics (overuse of underlining should be avoided as it makes reading more difficult)
- capital letters are for the first letter of the first word in each heading only; except for any acronyms (e.g. IEEE), trade names, or personal names and places, which may require more than one capital (aim for minimal capitalization in headings as overuse of capital letters slows reading)

Table 2 Size and style of headings

Heading Point Size Style Example

Acknowledgments

First-level (includes

Preliminaries) 6 Environmental

18 bold

impact

Second-level 14 bold

6.1 Chemical waste Third-level 12 bold

6.1.1 Cost analysis

Fourth-level 12 bold 6.1.1.1 Equipment costs

italics

- the best headings are:
 - summary statements rather than questions
 - not too brief (e.g. 'Inspection of contaminated site' is more explicit than 'Results') – not judgmental (e.g. 'Equipment for review' is more neutral than 'Poor equipment') – not too long (e.g. 'The equipment on the Whyalla factory floor' is unhelpfully long)

2.3.4 Decimal numbering system

- decimal numbering of headings is normally used throughout the text up to and including the reference list (excluding the preliminaries)
- Introduction is numbered 1; References have the final number (e.g. 8)
- too many levels of subheadings should be avoided: third-level headings are the preferred limit (e.g. 7.2.1)
- fourth-level headings (e.g. 7.2.1.1) should only be used if necessary, but not if there is only one heading at that fourth-level
- decimal points are placed between numerals, but not after them (e.g. 7.2)
- Appendices are not numbered using the decimal system; they can be internally labelled with letters (Appendix A, Appendix B etc.)
- omission of decimal numbering in shorter reports or where there are multiple subheadings, may increase readability

2.4 Format of text

2.4.1 Justification

- all text and headings can be aligned against the left margin except for where indentation is appropriate (see 2.3.3 Indentation and centring)
- text may also be justified on the right-hand margin, and full justification (right and left margins) is often preferred for formal documents and theses

2.4.2 Line spacing

- 1.5 or double line spacing is generally used for work submitted for assessment; automatic line spacing (as provided with word processing) may be acceptable
- line spacing should be greater at the end of a section so it assists in separating sections
- line spacing may be decreased between headings and text so it assists in grouping information; the smallest line space (i.e. a four point space) may be appropriate before text such as a list of bullet points
- extra line spacing should be used in the following cases: – to separate paragraphs
- to separate figures, tables, equations and long quotations from the text – to separate a main heading from the section of text which comes before

To change line spacing using MSWord 2000, go to the Format menu and access Paragraph.

2.4.3 Indentation and centring

- indenting text at the beginning of a paragraph is unnecessary when an extra line space is being allowed between paragraphs
- indentation is appropriate for long quotations, complex mathematical expressions, lists with bullet points (dots •) and the list of headings on the contents page
- long quotations are usually indented from both the left and right margins

figures and tables may be centred in the text

- students may choose to indent each subheading with its associated text, as shorter lines of text are easier to read; overall length is then affected

To change indentation using MSWord 2000, go to the Format menu and access Paragraph.

2.4.4 Figures and tables

- figures include diagrams, graphs, sketches, photographs and maps
- tables summarise data in rows and columns; a dash (—) indicates no data
- all figures and tables may be centred
- all should be labelled in bold with the appropriate consecutive number (figures are generally labelled at the bottom, and tables at the top)

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- another method of labelling figures and tables is to number them according to the major section where they appear (as Table 4 is the second table in section 3

of this document, it would become Table 3.2)

- each should have a simple, descriptive caption which explains the figure or table; arrows, symbols, or abbreviations should be explained in the text
- all figures and tables must be referred to in the text (e.g. 'see Figure 14'); avoid using words such as 'Figure above' or 'Table below'
- all should be referenced (either author-date or numerical) if copied or adapted from another source
- figures and tables can be separated from the text with extra line spacing; labels and captions can be separated using four point spacing

To design a table using MSWord 2000, go to the Table menu.

2.4.5 Quotations

- quotes of less than two lines (or 30 words) in length should be inserted 'within your text within single inverted commas' (Hadgraft, Robertson & Eaves 1994, p. 23).
- quotes of more than two lines (or 30 words) should be:
set off from the body of the text in a separate block, indented...and single spaced. Do not use inverted commas around block quotations. When using the block format, place the [author-date] citation on the line below the last line of the quote [or if using the numerical system of referencing, place the number directly after the quotation].
(Hadgraft, Robertson & Eaves 1994, p. 23)
- changes or additions to a quote should be placed in square brackets e.g. 'Ambiguity was caused by [financial] statistics being excluded from the report.'
- words left out of a quote are indicated by the three points of ellipsis (...) e.g. 'Ambiguity was caused by...statistics being excluded...'

2.4.6 Equations and formulae

- equations should be numbered consecutively as they appear, with arabic numerals in parentheses () on the right margin, in order to distinguish them from any reference numbers which may appear in square brackets
- equations can begin on a new line and should be centred in the text, e.g.
$$x = G e^{-\omega t} \sin(\omega t + \phi) \quad (1)$$
- in a series of consecutive equations, the equal signs (=) should be vertically aligned, e.g.
$$0 = -C_1\beta^2 + C_2(2\zeta\beta) + C_1 \quad (2) \quad X_{st} = -C_2\beta^2 + C_1(2\zeta\beta) + C_2 \quad (3)$$
- equations should then be referred to by either their number only, e.g. (2); or with the abbreviation Eq. and the number, e.g. A combination of Eq. 2 and Eq. 3 allows the evaluation of constants C_1 and C_2 .

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3 Components of a report

An engineering report can be divided into three main sections:

- preliminary pages

- text of the report
- supplements if appropriate

The components of a report will be determined by the purpose and complexity of the report. The possible components and sequence of a report are illustrated in Figure 1, and the significance of the core of the report is highlighted by the double-lined box.

- Symbols ‡

Preliminaries Text of report

- Introduction
- Main sections organised under appropriate headings and subheadings
- Conclusion
- Recommendations ‡
- References
- Title Pa
- Appendices ‡
- Ab
- Disclaim
- Acknow
- Content
- Figures
- Tables ‡

Supplements (‡ indicates an optional section) Figure 1 Components of a report

3.1 Preliminary pages

The preliminary pages are all numbered with roman numerals (i, ii, iii, iv, v, vi etc.) except for the title page which is generally not numbered. Not all the components described below will be required in every report; for example, students may choose

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to omit a list of figures if only one or two diagrams appear in the report (for other optional preliminary components, see Figure 2).

Also, a glossary of technical terms or acronyms might be added to a technical report, or an executive summary may be required in certain cases. Students can adapt the order of their preliminary pages to meet any specific instructions given by lecturers or particular schools. Postgraduate students should order the preliminary pages of a thesis according to the sequence required by their university.

- Title Page

Preliminaries • Abstract

- Disclaimer
- Acknowledgments ‡

- Contents
- Figures ‡
- Tables ‡
- Symbols ‡

(‡ indicates an optional section)

Figure 2 Preliminary components of a report

3.1.1 Title page

A model title page can be seen in Appendix B. A title should be brief but informative.

A title page should be simple, attractive, and include the following information:

- name of the university
- name of the particular engineering school
- name and code of the subject
- title of the report (which is precise and indicates the topic and scope)
- name of author(s) and ID number(s) (with the family name underlined)
- name of tutor/supervisor
- date of submission and/or date of experiment as applicable

3.1.2 Disclaimer

A 'disclaimer' or declaration of authenticity is often required in major pieces of work such as large reports, projects and theses. It is a signed statement declaring that the report is the work of the stated author(s). The disclaimer should appear on a page following the abstract, and the acknowledgments may be on the same page. This is a possible disclaimer statement:

I declare the following to be my own work, unless otherwise referenced, as defined by the University's policy on plagiarism ... followed by the author's signature.

Students are not guilty of plagiarism if they have correctly referenced the sources of all material which is not their own work (see Appendix C for a university policy on plagiarism). Reference must be made both in the text of the report and in the

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reference list. A disclaimer may also be used to indicate that the report's results, conclusions and recommendations are those of the author, and not necessarily a reflection of the opinions or policies of the university or sponsoring organisation.

3.1.3 Abstract

The abstract is also known as an overview, synopsis or summary. The abstract is often written last as its purpose is to provide a summary of the report's essential information. All material in the abstract will also be in the report, particularly the Introduction. The abstract should appear on a separate page after the title page, and it is usually about 100–200 words in length. The abstract should include the following elements:

- why? background problem and purpose of the report
- how? brief details

of the approach/procedure/methods • what? important results/findings

- so what? major conclusion(s) and recommendation(s)

In professions such as engineering, the abstract may be the key part of the report that is read. A manager may use the abstract to decide whether or not the content of the report is relevant. For academic purposes, an abstract allows researchers to quickly review literature and significant findings on a particular topic. An executive summary can be required for a more complex report. It is a detailed overview of the essential content, and its length can be several pages or more.

3.1.4 Acknowledgments

The acknowledgments section is optional; however, it is essential that significant assistance, editing or work carried out by another person or organisation be acknowledged. Also, students may wish to thank colleagues or supervisors. In a group-writing situation where different sections of a report are written by various people, writers of separate sections must be clearly identified.

3.1.5 Contents

The heading for the list of contents is Contents. The list should clearly include:

- all major section/subdivision headings; numbered and worded exactly as in the text of the report (minor headings are optional)
- page numbers for each section/subsection

Some students choose to include the preliminary pages on the contents list; however, the title page should be excluded. The contents page may be set out with each level of subheading indented by a tab space. This allows the reader to understand at a glance the structure of the report, and to differentiate between important and less significant information. Figure 3 gives a possible format for a contents list.

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5 Modelling of linear and non-linear network parts 8 5.1 Linear network 8 5.2

Non-linear network 9

5.2.1 The function of saturation 9 5.2.2 The function of hysteresis 9 5.2.2.1 Only use fourth-level headings if essential 10

Figure 3 Indentation of headings in a contents list

3.1.6 Figures

The heading for the list of figures is Figures. The list is only necessary if more than a few figures appear in the main text of the report. The list includes the figure number, caption, and page number, ordered as in the text. For a model of a list of figures, see preliminary page viii.

3.1.7 Tables

The heading for the list of tables is Tables. The list is only necessary if more than a few tables appear in the main text of the report. The list includes the table number, caption, and page number, ordered as in the text. For a model of a list of tables, see

preliminary page viii.

3.1.8 Symbols

Where symbols are used extensively, a list of definitions should appear at the beginning of the report. If there is no list, symbols should be defined in the text when first used. The heading for the list is Symbols. The list of symbols should include appropriate information such as the symbol, definition, quantity to which the symbol refers, and the unit of measurement. Use an appropriate number of significant figures or level of accuracy when presenting measurements.

All units of measurement should be in the metric form given by the International System of Units (SI: *Système International d'Unités*). Appendix D covers Punctuating numerals and abbreviations, and care should be taken to include the space between the numeral and the unit (e.g. 55 cm). Table 3 indicates SI base units. If you are using another system of units, convert these into SI units.

Table 3 Base SI units

Quantity	Unit	Symbol
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K

(from Style Manual for Authors, Editors and Printers 2002, p. 178)

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3.2 Text of report

The pages of the text of the report (beginning with the Introduction) are numbered with arabic numerals (1, 2, 3 etc.).

An effective report has clear objectives and presents information in a logical sequence (see section 5.2 Structuring writing). A structured approach assists the reader by providing a framework for reading the report. An effective structure includes an introduction, followed by material organised under appropriate headings, and ends with a conclusion and possibly recommendations (see Figure 4).

Care should be taken to refer to the sources of all information both in the text and in the reference list at the end (see section 4 Referencing of sources and originality).

- Introduction
- Main sections organised under

Text of report

appropriate headings and subheadings

- Conclusion
- Recommendations ‡
- References

(‡ indicates an optional section)

Figure 4 Components of the text of a report

3.2.1 Introduction

The Introduction gives the reader the necessary background information. It can include:

- a description of purpose(s) and objective(s)/topic(s)
- a statement of the problem(s)
- a survey of background information
- a review of previous work/research and the relationship to the current project •

the method(s) of approach

- an indication of the scope and limitations of study
- an outline of material presented in the rest of the report

(adapted from Michaelson 1990, p. 38)

3.2.2 Main sections

The main sections of the report expand and develop the material in a logical and coherent manner, reflecting the structure outlined in the Introduction. Information in the report should be placed in context, and well supported with evidence and documentation by using references. In Engineering you will be asked to write two types of reports, scientific and analytical.

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In scientific reports based on experiments, the following areas, with appropriate headings, may be included:

- methods/procedures and materials used
- data collected/information presented
- findings or results/comparisons with previous work
- discussion or analysis of the material

In comparison, analytical reports are based on a review of literature and other evidence. Therefore these reports involve a presentation of research and analysis of the literature. The structure of these analytical reports is determined by the type and purpose of the report (e.g. a feasibility study identifies a problem and evaluates alternative solutions). The structure is also determined by the key ideas or themes which emerge from your thinking about what you have read. Therefore a key skill is to logically order information into sections by using headings and subheadings. In the sections information which came from your reading is explained, expanded on and supported with evidence by using references, figures and tables. How to construct and write a well organized section in a report is explained in detail in Chapter 3. Another key skill is referencing, where you indicate your source of information in the text and in a reference list at the end of the text (Windshuttle 1999, p.295). The skill of referencing is explained in detail in Chapter 4.

The headings and subheadings should divide the report into well defined sections. A

careful division of the report and choice of headings creates an outline or framework for the report which allows the reader to progress logically through the material. Lengthy reports are improved by introductions and summaries in each major section. For information about the format of headings, see section 2.3.3 Headings. While words such as 'Introduction' and 'Conclusion' may be appropriate headings, students should never use 'Main body' or 'Main sections' as headings.

Figures (including diagrams, graphs, sketches, photographs and maps) and tables (involving summarised data in rows and columns) can be incorporated into the text to give clear and concise information. Such visual aids support the content of the report and can communicate information more effectively. They must be located in the text as close as possible after the first reference to them. A sentence referring to the figure or table is necessary in the report. For example, 'Over a five year period the performance of groups A, B and C have varied markedly, with no group indicating constant improvement (refer to Table 4)'. For formatting and labelling requirements, see section 2.3.4 Figures and tables.

Consideration should be given as to whether the data is better communicated to the reader by a table or a figure. Using tables or dotpoint lists will focus the reader on relevant and needed information. This technique is easier for the reader than reading another paragraph. For example, write an introductory comment and then list advantages and disadvantages of two types of dwellings in a table. Also consider that sometimes a figure will demonstrate a numerical trend more effectively than a

table. In the following example, the significance of the data in Table 4 may be more clearly presented in the form of a graph, as can be seen in Figures 5 and 6.

Table 4 Sample Table (from Hadgraft et al. 1994, p. 6) *

Year A B C

1961 97.70 89.93 7.77

1962 1.20 7.74 -6.54

1963 36.10 39.24 -3.14

1964 31.00 30.11 0.89

1965 55.90 13.64 42.26

* Asterisks or other note identifiers (e.g. double daggers ‡) can be used in a footnote to add detail, e.g. to a table.

100 100

C

A

80

80 A

B 60

C B

60 40

20

40 0

-20

20

1961

0 AB

1961 1962 1963 1964 1965 1963

-20

1965C

Figure 5 Sample of 3D columns

Figure 6 Sample of 3D

lines

(from Hadgraft et al. 1994, p. 7)

3.2.3 Conclusion

The conclusion(s) of a report must be related to, and resulting from, the material which appears in the report. The content of the Conclusion will be linked to the Introduction. The Conclusion places findings in perspective without introducing any new material, and it may include:

- a clear and concise summary of the main points
- the context and significance of the information
- a reference to the original aim(s)/purpose(s) of the report
- the application(s) of the results
- the limitations and advantages of the findings
- the writer's judgment/evaluation

3.2.4 Recommendations

Not all reports include recommendations, but if they are required recommendations should emerge from the conclusions of the report. This section is important to those who must act on the findings. The writer may include a brief, persuasive statement before presenting the recommendations clearly listed in numbered or bullet points. A series of recommendations may be worded in instructional language; for example, each beginning with a verb. There are styles of reports where the Recommendations follow the Introduction. Recommendations may involve:

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- strategies, procedures or techniques for solving the problem(s) •
- an indication of further work which needs to be completed

3.2.5 References

A reference list (not to be confused with a bibliography) must appear at the end of a report, listing all sources that have been referred to in the text. The heading for this list will be References. (Students should ensure that all sources are referenced in the text as well as in the reference list at the end of the report.)

The format of the reference list will depend on the system of referencing chosen for the report. There are two different types of reference lists used in engineering:

- alphabetical reference list according to author – used with the author-date (Harvard) system (see section 4.2.2 Alphabetical reference list)
- numbered reference list in order of their appearance in the text – used with the numerical (endnote) system (see section 4.3.2 Numbered reference list)

The reference list only includes the sources referred to in the report. A bibliography is a wider list of all texts that have been read in preparation for writing. A bibliography is not usually included in an engineering report (check with your lecturer).

3.3 Appendices

Not all engineering reports will need appendices, which are supplements to the text of the report (see Figure 7). The Appendices may include material which is incidental to the report, supportive of the report, or too long or technical to include in the text; for example, maps, folded diagrams, tables of results, letters, questionnaires, statistics, and computer program information. Large maps or figures should be reduced to A3 size where possible, and multiple folding is discouraged.

- Appendices ‡

Supplements (‡ indicates an optional section)

Figure 7 Supplementary components of a report

Each separate appendix should be lettered (Appendix A, Appendix B, Appendix C etc.). The text page numbering can continue into the Appendices, or alternatively, the Appendices can be numbered internally using the letter of the individual appendix and the number of pages within each appendix (A1, B1, B2, B3, C1, C2 etc. where Appendix A has one page, Appendix B has three pages, and so on).

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4 Writing a section of a report

An approach to writing a section of a report is explained in this chapter. This process involves planning, gathering information, reading and noting, further planning, writing the first draft, and finally revising and redrafting. This process is facilitated by the use of many resources, as summarised in Table 1 (page 1).

4.1 Planning

The first step of the writing process always involves clarifying the following expectations: message; purpose; audience, and scope of the report. This also applies when writing a section.

Message: Begin by clarifying the message of the section, i.e. write a precise heading or a summary sentence that summarises the section. This will enable you to focus on the information to communicate.

Purpose: The general purpose of the writing a section of a report is for you to gather information on a specific topic and the using this information to present a concise,

organised, logical section. The specific purpose of writing a section relates to the message. For example, you might explain the concept of 'Solar cells' by providing a definition, detailing how they operate, providing an application and discussing research findings.

Audience: Each report is written for a specific person or group of people in mind. For example, often the audience is colleagues and professionals, such as engineers.

Scope: The scope of a section limits the information within the section. You can do this by stating exactly what you will or will not cover in the section. For example, in the section on 'Solar cells' you will not discuss solar energy in general.

Once you have clarified the expectations, you can now plan your section. Useful initial techniques are brainstorming, mindmapping and developing a Contents page.

4.1.1 Brainstorming

Literally allow your brain to storm or generate many ideas. One way to start doing this is by listing all the ideas or keywords that come into your mind, as shown in the example. Do not organise ideas at this stage as that is the next step of mindmapping.

Brainstorm example – Solar energy

Solar cells Why? Suitability of location Weather Financial Amount
of sunshine Advantages Businesses Infrastructure
Disadvantages

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4.1.2 Mindmapping

Now try to develop a logical organisation to the ideas. One way to achieve this is by developing a mindmap, a diagram that shows the linking of all the ideas listed.

Figure 8 An example of a section of a contents page

4.2 Developing a contents page

From the organisation from the mindmap you are now able to prioritise the ideas and generate a series of headings and subheadings. If you can achieve three levels of headings and subheadings, your section is starting to develop a logical argument.

Contents

1 Solar energy

1.1 Sunshine

1.1.1 Amount

1.1.2 Location

1.1.3 Conversion

1.2 Meteorological variables

1.2.1 Cloud cover

1.2.2 Wind

1.2.3 Rain

1.2.4 Seasons

Figure 8 An example of a section of a contents page

Organising information into logical sections is challenging This process can be assisted by having read widely and having obtained more information about the topic.

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5 Referencing of sources and originality

References must be provided in your report if you use someone else's opinions, theories, data or organisation of material, and this includes all occasions when you:

- paraphrase (use their same idea in your own words) • summarise (use their main points)
- quote (use their exact words)
- copy (use their figures, tables or structure)

It is expected at university that students will present original work for assessment. Where any ideas or written work from other sources/authors appear in your report, you must acknowledge the sources by making reference to them. These references must be in the text as well as in the reference list. Failure to indicate the sources of your ideas may suggest that you are wrongfully claiming their ideas or words to be your own, which is known as 'plagiarism'. Universities impose heavy penalties on students who plagiarise (see Appendix C for an example of a plagiarism policy). Referencing of sources is also encouraged because it strengthens a writer's argument. Referencing shows wide reading and identifies the critical aspects of other research.

5.1 Two systems of referencing

There are two main ways of referencing sources in engineering reports:

- author-date referencing (Harvard system) – see section 4.2 •
- numerical referencing (endnote system) – see section 4.3

In a report, students must use either the author-date or the numerical system, not both. The main differences between the two systems are described in Figure 8.

Numerical referencing

Author-date referencing In the text of report:

In the text of report:

- consecutive numbers in
 - author and date in parentheses square brackets
- e.g. ...validation of results (Smith 1992). e.g. ...validation of results [4].

- page number included if needed

Reference list at the end of report:

Reference list at the end of report:

- numbered list of references
- alphabetical list of references – author's initials or name can be
 - author's surname first before the surname
 - date follows author – title follows author, with date

– no page numbers of quotations – appearing later
 where more than one line in length, – page numbers are last if needed
 the second line is indented

Figure 8 Differences between referencing systems

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5.1.1 Choosing between referencing systems

When deciding on which referencing system to use, students should consider:

- the audience of the report
- the purpose of the report

An author's referencing style may be determined by the audience of the report — as specific referencing styles are often required by different lecturers, employers and publishers of journals and conference papers. For example, many international journals in Electrical Engineering use the numerical system, while academic journals in Mechanical Engineering often use the author-date system.

The selection of an appropriate system of referencing will also depend on the purpose of the report. Numerical referencing may be used where the purpose of a report is to communicate technical information and the conclusions are of more interest to the reader than the names of cited sources. The author-date system is useful where the purpose is to present scholarly research, such as in a research or project report, or in an academic article or thesis. The names of those who have completed previous work in the area will be of interest to the reader, and direct textual references to sources will therefore be useful, rather than having the reader turn to a list at the end.

Possible advantages and disadvantages of the two referencing systems are summarised in Figure 9.

Numerical referencing

Author-date referencing Advantages

Advantages

- prevents the text of the report from
- allows author and date to be seen being interrupted by wordy references
- in context within the text of report • prevents constant repetition of the
- saves turning to a list at the end to same references as only a number
- find the name of a cited source needs repeating
- provides an alphabetical reference

list at the end

- means that inserting extra
- references into the text is easy

Disadvantages

Disadvantages

- creates a non-alphabetical reference
- list at the end

- creates very long author-date entries
- if there are multiple authors and • means turning to reference list to match a numerical reference to its sources
- creates repetition and source
- may create complications if an disruption to the text when the same extra reference needs inserting later source is used repeatedly

Figure 9 Advantages and disadvantages of referencing systems

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5.1.2 Consistency in referencing

The numerical system and author-date system outlined below are based on the style that is common for engineering report writing, and the Australian Government Publishing Service (AGPS) Style Manual for Authors, Editors and Printers (1994). Other methods of each referencing system are used by individuals, faculties and publishers. It is therefore important for internal consistency that only one version of a referencing system is used within any report. When referencing sources, writers should also take particular care to maintain consistency with punctuation. For further guidelines, students can refer to a librarian or the AGPS Style Manual (1994).

5.2 Author-date referencing (Harvard system)

There are two major elements involved in the author-date system of referencing:

- in the text – the author's surname and date appear in parentheses () •
- at the end – an alphabetical reference list is included

5.2.1 In-text author and date

To acknowledge a source within the text of your report using the author-date system of referencing, you should state the following items:

- author's surname
- date of the publication
- page numbers, if necessary (e.g. when material is quoted or copied)

This information should appear in parentheses or round brackets () directly after the material quoted (see the first example below). If the author's name has already been used in the text of the report then it does not need to be placed in the parentheses, as seen in the alternative example below.

Author-date references in the text:

a quotation is used It has been argued that 'the relative seriousness of the two kinds of errors differs

from situation to situation' (Smith 1982, p. 47).

or

Smith (1982, p. 47) has argued that 'the relative seriousness of the two kinds of errors differs from situation to situation'.

the author's words are paraphrased A recent study (Jones & Chan 1991) has shown that testing procedures are inadequate.

the same author is cited again Another argument raised by Smith (1982) relates to the error evident

in the conclusion.

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several different sources need to be cited at the same or

time Recent studies (Jones & Chan 1991; Kuwlesky

Jones and Chan (1991), Kuwlesky (1988) and Tan 1988; Tan 1992) all agree...

(1992), all agree that testing procedures are [names of first authors are alphabetical; and separated

inadequate. by semicolons]

Author-date references in the text cont.

there are two or three authors A recent study (Lim, Thompson & King 1993) found... or

Lim, Thompson and King (1993) found the analysis...

[use an ampersand (&) when in the brackets; or 'and' outside]

more than three authors appear A recent study (Boyd et al. 1994) has indicated... or

Boyd et al. (1994) have indicated...

['et al.' means 'and the others']

there are different authors with the same surname idea.

[differentiate between authors by using initials]

Marini (Tan 1992) stated that...

you are referring to an author (primary reference) [Marini is the primary reference]

which you read about in another book or article or

(secondary reference) Marini's study in 1975 (cited in Tan 1992) stated

that... or

Tan (1992), in reporting Marini's 1975 study, states

that...

reference is made to more than one work by the same

author University research (Smith 1982, 1988) has

indicated that... or

University research conducted by Smith (1982,

1988) has indicated that...

the same author has published two or more things in

the same year In recent reports (James 1992, 1993a, 1993b),

recommendations were made...

or

Recommendations by James (1993b) were

implemented... [add a, b, c etc. to distinguish

no author's name appears (but there is a sponsoring between texts in the same year]

body, newspaper or title)

A recent study (CSIRO 1990) has suggested...

or

A recent survey (Advertiser 24 Jan. 1994, p. 5)

the editor's name is the identifying feature indicated that... or

A recent survey (Population Projections 1996)

shows that...

The theory was first put forward in 1970 (Jones,

C. L. 1971), but since then many researchers, Current articles (ed. Harris 1994) on the industry including A. G. Jones (1983), have rejected the suggest that... or

Current publications edited by Harris (1994) suggest that...

no date can be established Lewis (n.d.) continued the study... or

The study was continued (Lewis n.d.)...

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the source is unpublished Nguyen (unpub.) argues for...

or

An argument (Nguyen, unpub.) has been made for...

or

reference is made to a personal communication The site's safety is proved (King, B. T. 1994, pers. In a telephone conversation on 17 July 1994, King comm., 17 July). [personal communications are not

revealed the site to be safe. included in the reference list]

5.2.2 Alphabetical reference list

When using the author-date system of referencing, you must include an alphabetical reference list at the end of the report (before the Appendices) listing all sources cited. Personal communications such as conversations or personal email messages need not be included in the reference list, as they cannot be checked. When writing the reference list you must take care to include the correct elements, and to use the correct punctuation.

The elements required for a book are: the author's surname(s) and initials, year of publication, title, publisher, and place of publication (other elements may also be relevant — see Appendix E). The correct order of these elements is indicated by the ellipses in Figure 10. The shaded boxes indicate punctuation.

Author's surname(s)Initials Title Publisher Place of publication Year

Muller, B. & Reinhardt, J. 1990, Neural Networks: An Introduction, Springer-Verlag, Berlin.

Ampersand Full stops Commas Capital letters Italics Commas Full stop Figure

10 Book in author-date reference list

The elements required for a journal article are: the author's surname(s) and initials, year of publication, title of the article, title of the journal, volume number, issue number, and page numbers. Some elements will not always apply, and sometimes

others will be relevant — see Appendix E. The correct order of these elements is indicated by the ellipses in Figure 11. The shaded boxes indicate punctuation.

Author's surname(s)Initials Title of article

Year

Quotation marks Lower case

Meyers, A. G. & Priest, S. D. 1992, 'A technique for moulding cylindrical discontinuous models', *Rock Mechanics and Rock Engineering*, Technical Note, vol. 25, no. 3, pp. 207–214.

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Indent Italics Abbreviations Full stops

Commas Issue number

Volume number Page numbers

Journal title

Figure 11 Journal article in author-date reference list

The following aspects of punctuation should be included (as indicated by the shaded boxes in Figures 10 and 11):

- commas used to separate all elements
 - full stops used for the abbreviation of words, and the end of the reference
 - ampersand (&) used to separate last two authors — instead of 'and'
 - italics or underlining used to indicate the title of a book or journal
 - single quotation marks used to indicate the title of an article
 - titles of unpublished works — such as theses — are not differentiated (by italics, underlining or quotation marks)
 - capital letters used for first letters of main words in titles of books and journals, and all other letters are lower case
 - capital letters kept to a minimum in titles of articles and unpublished works
- short indentation of the second and subsequent lines of a reference is helpful to show the alphabetical order

A reference list is arranged in alphabetical order according to the authors' surnames:

- if there is no author, then they are listed alphabetically using the first word of the title, excluding 'a' or 'the'
- if there are two or more references by the same author(s), they are listed in order of publication date, with the oldest work first
- if works by the same author(s) are published in the same year, they are arranged alphabetically by title of the work; lower-case letters are added to the publication dates to distinguish each work (e.g. 1992a)
- works where the author is the sole author are listed before works where the author is the senior author with co-authors, regardless of date of publication

For example, this is the correct alphabetical order for an author-date reference list:

Aukland, J. S. 1989, A solar airconditioning system, M Eng Thesis, South Australian Institute of Technology.

Browne, T. E. Jr (ed.) 1984, *Circuit Interruption: Theory and Techniques*, M. Dekker, New York.

Commission of Inquiry into Poverty 1975, Poverty in Australia, First Main Report (Prof. R. F. Henderson, Chairman), AGPS, Canberra.

Dictionary of Scientific and Technical Terms 1994, 5th edn, McGraw-Hill, New York.

Henderson Report. See Commission of Inquiry into Poverty 1975.

Meyers, A. G. & Priest, S. D. 1992, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, pp. 207-214.

Muller, B. & Reinhardt, J. 1990, Neural Networks: An Introduction, Springer-Verlag, Berlin.

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Subramanian, C. 1992, 'Some considerations towards the design of a wear resistant aluminium alloy',

Wear, vol. 155, pp. 193-205.

Subramanian, C. 1993, 'On mechanical mixing during dry sliding of aluminium-12.3 wt. % silicon alloy against copper', Wear, vol. 161, pp. 53-60.

Subramanian, C. & Strafford, K. N. 1993a, 'Review of multicomponent and multilayer coatings for tribological applications', Wear, vol. 165, pp. 85-95.

Subramanian, C. & Strafford, K. N. 1993b, 'Towards optimization in the selection of surface coatings and treatments to control wear in metal-forming dies and tools', Materials & Design, vol. 14, pp. 291-298.

Subramanian, C., Strafford, K. N., Wilks, T. P., Ward, L. P. & McMillan, W. 1993, 'Influence of substrate roughness on the scratch adhesion of titanium nitride coatings', Surface & Coatings Technology, vol. 62, pp. 529-535.

The sources referred to in a reference list may be books, articles, reports, journals, conference papers, newspapers, audiovisuals, electronic and other non-print material. Referencing of electronic sources is a relatively new activity, and the following examples are adapted from the style suggested by Li and Crane (1993).

Author-date reference lists:

Books:

One or two authors Muller, B. & Reinhardt, J. 1990, Neural Networks: An Introduction, Springer-Verlag, Berlin.

3 or more authors Arrillaga, J., Arnold, C. P. & Harker, B. J. 1983, Computer Modelling of Electrical Power Systems, Wiley, Chichester.

No author, 2nd or later edition Computer Integrated Manufacturing 1990, 2nd edn, Architecture Technology Corporation, Minneapolis.

Books sponsored by an institution, corporation or other Australian Manufacturing Council 1993, Emerging

organisation Exporters: Australia's High Value-added

Manufacturing Exporters: Final Report of the Study,

Australian Manufacturing Council, Melbourne.

The Electrical Engineering Handbook 1993, ed.

Edited work (role of editor not paramount Richard C. Dorf, CRC Press, Boca Raton.

Edited work (role of editor paramount) Browne, T. E. Jr (ed.) 1984, Circuit Interruption: McKinsey & Company & the Australian Theory and Techniques, M. Dekker, New York.

Manufacturing Council Secretariat to the

One volume of multi-volume work Gopel, W., Hesse, J. & Zemel, J. N. (eds) 1989, Sensors: A Comprehensive Survey, vol. 1, Fundamentals and General Aspects, eds T. Grandke & W. H. Ko, VCH, Weinheim, Germany.

Chapter in edited book Stead, J. P. & Strutt, J. E. 1987, 'Marine corrosion on offshore pipelines: predicting the problem using computer-aided risk evaluation', in Plant Corrosion: Prediction of Materials Performance, eds J. E. Strutt & J. R. Nicholls, Ellis Horwood, Chichester.

Journals:

One or two authors Albrecht, P. & Shabshab, C. 1994, 'Fatigue strength of weathered rolled beam made of A588 steel', Journal of Materials in Civil Engineering, vol. 6, pp. 407–428.

Issue number included (if each issue is paginated Mechanics and Mining Sciences & Geomechanics separately) Abstracts, Technical Note, vol. 29, no. 2, pp.

Meyers, A. G. & Priest, S. D. 1992, 'A 187–191. micro-processor controlled pump for triaxial cell pressure control', International Journal of Rock

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No author 'Efficient low cost solar cell developed' 1994, Search, vol. 25, p. 309.

Specialised sources:

Conference papers Hunt, F. L. 1990, 'Ballarat East — from Eureka to the 80's', Proceedings of the Pacific Rim Congress 90, Gold Coast, Australia, Vol. III, Australasian Institute of Mining and Metallurgy, Parkville, Victoria, pp. 473–480.

Maps Mullineaux, D. R. 1976, Preliminary Overview Map of Volcanic Hazards in the 48 Conterminous United States, U. S. Geological Survey Miscellaneous Field Studies Map MF-786, scale 1:7 500 000.

Standards American Society for the Testing of Materials 1988, Standard Specification for Cast Steel Wheels for Railway Service, (A 583 – 88), ASTM, Philadelphia.

Standards Association of Australia 1991, Australian Standard: Electrical Installations — Buildings, Structures and Premises, known as the SAA Wiring Rules, (AS 3000 – 1991), Standards Australia, North Sydney.

Patents Tan, I. S. & Arnold, F. F. (US Air Force) 1993, In situ Molecular Composites Based on Rigid-rod Polyamides, US patent 5 247 057.

Author-date reference lists cont.

Government publications Bureau of Transport Economics 1986, Demand for Australian Domestic Aviation Services, Occasional Paper no. 79, AGPS, Canberra.

Australian Bureau of Statistics 1994, Building Approvals Australia, Cat. no. 8731.0, ABS, Canberra.

Ergas, H. 1986, Telecommunications and the Australian Economy, Report to the Department of Communications, AGPS, Canberra.

Government reports Commission of Inquiry into Poverty 1975, Poverty in Australia, First Main Report (Prof. R. F. Henderson, Chairman), AGPS, Canberra.

Henderson Report. See Commission of Inquiry into Poverty 1975.

Newspaper articles Fist, S. 1994, 'The call to disconnect analogue phones', Australian, 2 Aug., p. 25.

Pamphlet Quit: Give Smoking Away in 5 Days 1987, Victorian Smoking and Health Program booklet, Health Department Victoria, Anti Cancer Council and National Heart Foundation, May.

Microfiche Herbert, W. G. 1987, The Australian Beef Industry: An Overview, Australian Livestock Council, Canberra, microfiche.

Video What Are We Going to Do With the Mo9ney? (video recording) 8 August 1982, ABC Television.

Unpublished material:

Thesis Wenjiong, Y. 1994, Efficient coding scheme for video-conference applications, M Eng Thesis, University of South Australia.

Unpublished report Janz, J. K. 1993, Orroroo bedrock drilling 1993, South Australian Department of Mines & Energy, Unpublished Report RB 93/042.

Electronic media:

25

Document on World Wide Web Brown, H. 1995, Citing Computer References [Online]. URL: <http://neal.ctstateu.edu/history/cite.html>

Document via FTP or Telnet King, M. L. 1963 August, I Have a Dream [Online]. Available FTP: mrcnext.cso.uiuc.edu Directory: gutenberg/freenet File: i-have-a-dream

Document via email Harnad, S. 1992, Post-Gutenberg galaxy: The fourth revolution in the means of production of knowledge, in Directory of Electronic Journals, Newsletters and Academic Discussion Lists ,

2nd edn [Online]. Available email: LISTSERV@UOTTAWA

Message: Get EJOURNL1 DIRECTORY

Journal article on CD Rom Girotti, T. B., Tweed, N. B. & Houser, N. R. 1990 February, Realtime var control by SCADA, IEEE Transactions on Power

Systems [CD Rom] vol. 5, no. 1, pp. 61-64. Available: UMI

File: IPO (IEEE/IEE Publications Ondisc) Item: 3631953

Journa article via FTP Drew, J. R. 1989 October, So that's why they call it the Big Apple, Qyabta [Online] vol. 1, no. 1. Available FTP: [export.acs](ftp://export.acs.cmu.edu)

.cmu.edu Directory: [pub/quanta](ftp://pub/quanta) File: [quanta-oct.ps.Z](ftp://pub/quanta)

Online databases The Educational Directory [Online] 1992. Available: Knowledge Index File: The Educational Directory (EDUC6)

Lists or bulletin boards Alcohol and Drug Studies [Online]. Available email: ALCO

HOL@LMUACAU.BITNET

Computer programs Wu, A. H. & Jenkins, B. 1990 May1, Diagnostic Ordering in Clinical Medicine (DOS version 1.0) [Computer program].

Available Distributor: Health Sciences Consortium, Chapel

Hill, NC (Address: 201 Silver Cedar Court, Zip: 27514)

5.3 Numerical referencing (endnote system)

There are two major elements involved in the numerical system of referencing:

- in the text – consecutive numbers appear in square brackets [] • at the end – a numbered reference list is included

5.3.1 In-text numbers

To acknowledge a source within the text of your report using the numerical referencing system, you should place an arabic number in square brackets after the quotation, relevant material, or author's name. This number will then be repeated in the reference list with the corresponding details of the source (e.g. the reference list will include elements such as the author, title, and page numbers).

Consecutive numbers should be used for each new reference. However, where you cite the same source or reference more than once in the report, you can use the same number again each time. However, if a second citing of a source involves a different page number, then a new numerical reference is used — and a corresponding new entry will be included in the reference list.

Numerical references in the text:

a quotation is used It has been argued that 'the relative seriousness of the two kinds of errors differs

from situation to situation' [1].

or

Smith [1] has argued that 'the relative seriousness of the two kinds of errors differs from situation to situation'.

the author's words are paraphrased A recent study [2] has shown that testing procedures are inadequate.

the same author is cited again Another argument raised by Smith [1] relates to the error evident in the conclusion.

several different sources need to be cited at the same Jones and Chan [2], Kuwlesky [9] and Tan [7], all

time agree...

Recent studies [2][7][9] all agree that testing procedures are... or

there are two or three authors A recent study [3] found the analysis... or
Lim, Thompson and King [3] found the analysis...

more than three authors appear A recent study [4] has indicated... or
Boyd et al. [4] have indicated...

there are different authors with the same surname
Marini [7] stated that...

you are referring to an author (primary reference) or
which you read about in another book or article Marini's study in 1975 [7] stated that...
(secondary reference) or

The theory was first put forward in 1970 [5], but Tan [7], in reporting Marini's 1975 study, states
since then many researchers, including A. G. that...
Jones [6], have rejected the idea.

Numerical references in the text cont.

reference is made to more than one work by the same In recent reports [10][11][12],
recommendations

author were made... or

Recommendations by James [12] were
implemented...

the same author has published two or more things in
the same year A recent CSIRO study [13] has suggested...

or

A recent survey [14] indicated that...

no author's name appears (but there is a sponsoring or
body, newspaper or title) A recent survey, Population Projections, [15] shows
that...

Current articles [16] on the industry suggest

the editor's name is the identifying feature that... or

University research [1][8] has indicated that... Current publications edited by Harris [16]
or suggest that...

University research conducted by Smith [1][8]
has indicated...

no date can be established Lewis [17] continued the study... or

The study was continued [17]...

the source is unpublished Nguyen [18] argues for... or

An argument [18] has been made for...

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reference is made to a personal communication 1994, pers. comm., 17 July).

In a telephone conversation on 17 July 1994, King [personal communications are not included in the reference list]

or

The site has since proven to be safe (King, B. T.

5.3.2 Numbered reference list

When using the numerical system of referencing, a numbered reference list appears at the end of the report (before the Appendices) listing all sources cited. Personal communications such as conversations or personal email messages need not be included in the reference list, as they cannot be checked.

The reference list is arranged in numerical order according to the sequence in which the authors were cited in the report. When writing the reference list you must take care to include the correct elements, and to use the correct punctuation.

Appendix E summarises the various elements which may be required in a reference list, and highlights the differences between the alphabetical and numerical reference lists. For a numerical reference list, the main differences include:

- the reference number appears first
- the initials or given name come before the surname
- the date appears at the end of the reference before any page numbers

In a numbered reference list, the elements required for a book are: the author's initials and surname(s), title, publisher, place of publication, and year of publication. The correct order of these elements is indicated by the ellipses in Figure 12. The shaded boxes indicate punctuation.

Title Publisher Place of

publication Author's initials Surname(s) Year [5] B. Muller & J. Reinhardt, Neural

Networks: An Introduction, Springer-Verlag, Berlin, 1990.

Full stops Ampersand Capital letters Italics Commas Full stop Figure 12

Book in numerical reference list

The elements required for a journal article are: the author's initials and surname(s), title of the article, title of the journal, volume number, issue number, year of publication, and page numbers. Some elements will not always apply, and sometimes others will be relevant — see Appendix E. The correct order of these

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elements is indicated by the ellipses in Figure 13. The shaded boxes indicate

punctuation.

Surname(s) Title of article Quotation

marks Lower case

Author's initials

[6] A. G. Meyers & S. D. Priest, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, 1992, pp. 207–214.

Commas Abbreviations

Italics Year Full stops

Journal title numbers

Volume number Issue number Page

Figure 13 Journal article in numerical reference list

The following aspects of punctuation should be included (as indicated by the shaded boxes in Figures 12 and 13):

- commas used to separate all elements
 - full stops used for the abbreviation of words, and the end of the reference
 - ampersand (&) used to separate last two authors — instead of 'and'
 - italics or underlining used to indicate the title of a book or journal
 - single quotation marks used to indicate the title of an article
 - titles of unpublished works are not differentiated (by italics, underlining or quotation marks)
 - capital letters used for first letters of main words in titles of books and journals, and all other letters are lower case
 - capital letters kept to a minimum in titles of articles and unpublished works
- initial numerical reference with or without square brackets []

The sources referred to in a reference list may be books, articles, reports, journals, conference papers, newspapers, audiovisuals, electronic and other non-print material. For a detailed list of examples, see section 4.2.2 Alphabetical reference list — but note that numerical references are not presented alphabetically, and the sequence of the elements needs to be adapted to the appropriate style.

Numbered reference lists:

1. Dictionary of Scientific and Technical Terms , 5th edn, McGraw-Hill, New York, 1994.
2. J. S. Aukland, A solar airconditioning system, M Eng Thesis, South Australian Institute of Technology, 1989.
3. B. Muller, & J. Reinhardt, Neural Networks: An Introduction, Springer-Verlag, Berlin, 1990.
4. A. G. Meyers, & S. D. Priest, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, 1992, pp. 207– 214.
5. T. E. Browne, Jr (ed.) Circuit Interruption: Theory and Techniques, M. Dekker, New York, 1984.

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or alternatively, using square brackets:

[1] Dictionary of Scientific and Technical Terms , 5th edn, McGraw-Hill, New York, 1994.

- [2] J. S. Aukland, A solar airconditioning system, M Eng Thesis, South Australian Institute of Technology, 1989.
- [3] B. Muller, & J. Reinhardt, Neural Networks: An Introduction, Springer-Verlag, Berlin, 1990.
- [4] A. G. Meyers, & S. D. Priest, 'A technique for moulding cylindrical discontinuous models', Rock Mechanics and Rock Engineering, Technical Note, vol. 25, no. 3, 1992, pp. 207– 214.
- [5] T. E. Browne, Jr (ed.) Circuit Interruption: Theory and Techniques, M. Dekker, New York, 1984.

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6 Technical language

Effective use of technical language is a critical part of report writing. The important aspects of technical language to be considered are structure, grammar, vocabulary and punctuation. (The use of punctuation in technical writing is detailed in Appendix D.) A report must be written appropriately for its audience. The purpose of a report must be decided before the writing begins. Once it is clear why a report is being written, then how to write will become more evident. A quality engineering report exhibits these characteristics:

- clear statement of aim or purpose
- concise and precise presentation of detail
- cohesive and logical development of data and ideas
- considered objective separation of fact and opinion
- categorised layout of well-spaced headings, paragraphs and visuals

6.1 Starting writing

When beginning a report, students may find it useful to plan and write the main sections first. The abstract or summary should be written last as should the conclusion. It can also be useful to write the Introduction later as it includes an outline of the structure of the whole report.

A useful technique for developing a structure is to ask questions. These questions may be: What? Where? When? Why? and How? For example, the planning of an engineering report on solar energy may be assisted by asking such questions as:

- What is solar energy?
- Where and when is solar energy predominantly used?
- Why is usage of solar energy currently limited?
- How is solar energy utilised in engineering contexts?

It is useful to ask as many questions as possible because such questions help to generate ideas and key words for the structure of sections and paragraphs, and for the associated headings. Students may find it useful to refer to report writing resources available from their university library (see Appendix F).

6.2 Structuring writing

Information in a technical report is systematically developed at several levels: sections, paragraphs and sentences. The organisation of this information can be controlled using various techniques including:

- different levels of headings
- summary, overview or key statements indicating purpose •

visual aids such as tables and figures

- bullet points (dots •), dashes and lists

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- linking sentences between sections and paragraphs

6.2.1 Logical sections

Students should consider the overall purpose of the report, and be aware of the specific purpose of individual sections. Using a flow chart or visual representation can assist with planning and structuring the report. (For further information see section 3.2.2 Main sections.) Some possible purposes are to:

- analyse data/literature is investigated; gaps are identified • inform information or research is outlined; results are presented • instruct procedures are explained; strategies are provided • persuade conclusions are discussed; recommendations are made • document written record is created and filed

6.2.2 Cohesive paragraphs

Paragraphs are groups of sentences which organise information into the essential units of the report. Before writing a paragraph, the writer should be clear about the paragraph's purpose(s). A useful technique is to decide what the reader needs to know. The broad and specific purposes of paragraphs in technical writing (see Appendix G) involve:

- describing facts (e.g. paragraphs which define, classify, and describe) • describing the relationship between facts (e.g. paragraphs which describe, analyse, compare and contrast)

A paragraph should begin with a general overview or summary statement before the specific or exemplifying information is given. Exemplifying information may include examples, illustrations, descriptions, explanations and evidence which expand the ideas. A paragraph is often structured with:

- a key sentence (introducing main idea and/or giving context) • concept development (expanding main idea/giving specifications etc.) • a summary statement (concluding with main idea/linking to what follows)

Connectors or linking words help create cohesion within the paragraph and between paragraphs. Information should be organised systematically and one way of achieving this is to use connectors or linking expressions (see Table 5).

Table 5 Connectors / linking words and phrases

Purpose for link Connecting word or phrase

show time/sequence first, second, finally, shortly, immediately, soon, subsequently show importance first, next, most significant, least

add ideas again, furthermore, additionally, indeed, moreover, another show cause and effect thus, since, so, therefore, as a result, hence, consequently, then indicate a summary in brief, finally, to

conclude, in fact, indeed, on the whole

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show comparison/contrast on the other hand, though, similarly, however, nevertheless give illustration for example, in illustration, specifically, for instance show concession granted, despite, while, notwithstanding, of course show position next to, beside, in, out, below, juxtaposed

(Adapted from Marshall & Rowland 1993, p. 174)

6.2.3 Clear sentences

Complex sentences are more difficult to read and often contain more grammatical errors. Reading the report aloud is one technique for locating errors. Students should aim to keep sentences as short as possible, although it is effective to vary sentence length. Shorter sentences are useful as summary statements and for emphasis.

6.3 Structuring sentences

6.3.1 Proximity of detail and item

Reading information is easier and the meaning is less confused if the sentence components are kept close together. Care should be given to the proximity of:

- the subject and verb (person and action)

e.g. The test results were confirmed as accurate after a review. ✓ (subject) (verb)

not The test results after a review were confirmed as accurate. ✗ (subject) (verb)

- an item, and its detail or description

e.g. Arrangements were made for a meeting to be held on 19 June. ✓ not On 19

June, arrangements were made for a meeting. ✗

6.3.2 Series of nouns

Reading is improved and less ambiguous when a long series of nouns is avoided. Also, when verbs or processes are written as nouns (by adding 'ment', 'ance', and 'tion' etc.) the sentence becomes complicated.

e.g. The first test to check the circulating water flow ✓ of the cooling tower...

not The initial performance flow test of the cooling ✗ tower circulating water...

e.g. The fault occurred causing difficulties. ✓ not The occurrence of the fault caused difficulties. ✗

6.3.3 Parallel construction

Items which are listed or grouped should be presented in a parallel form. 'Parallel' means the same grammatical structure, sentence length, capitalisation and layout.

e.g. Reading is easier than writing. ✓ not Reading is easier than to write. ✗

Parallel construction is critical with the use of bullet points or lists (as in instructions, procedures or recommendations).

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e.g. Some aspects of concern are: ✓ not: ✗ • the security of the data; • the need to protect security; • the protection of confidentiality; and • Without doubt, confidentiality; • the costs of such measures. • often these are expensive.

(adapted from Hadgraft et al. 1994, p. 21)

6.4 Grammar

6.4.1 Active and passive voice

The passive voice is usually used in engineering reports where the writing is intended to be impersonal and objective. A passive sentence uses a form of the verb 'to be' (e.g. 'is', 'was', 'has been', 'will be'), plus the past participle of the verb often made by adding 'ed' to the verb (e.g. 'tested', 'identified', 'monitored'). The passive voice is preferred when the human subject or agent is less relevant than the activity. In a passive sentence, the human subject is placed last or can be removed from the sentence.

e.g. The situation has been monitored by a researcher. (passive) (human subject at end)

e.g. The situation has been monitored. (passive) (no human subject)

The active voice emphasises the human subject or agent by placing it first in the sentence. If no human is directly involved in the action of the sentence (e.g. the subject is a piece of equipment), then it is appropriate to use the active voice.

e.g. A researcher has monitored the situation. (active) (human subject emphasised)

e.g. Temperature gauges monitored the air inside the cylinder. (active) (non-human subject emphasised)

The important issue in the choice between active and passive sentences is placing the key aspect — whether the activity (passive) or the agent (active) — at the beginning of the sentence.

6.4.2 Tense of verbs

Verb tense changes according to the section in the report, the purpose of the section or sentence and the type of information included. The following suggestions are a general guide only:

- use present tense to explain or discuss
 - continuing objectives e.g. The aim of this report is to... – general principles or laws e.g. Ohm's law states...
 - data or results e.g. These results show... – procedures or instructions e.g. Identify each item by code...
- use past tense (or the past perfect) to state or describe
 - what the objectives were e.g. The main purpose was...
 - what equipment was used e.g. The cylinders were reused... – what procedure was followed e.g. After careful cleaning, the test tubes were... – what happened e.g. The damage to the component had been...

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The review of literature including other research may be written in either the present, past or the present perfect tense. Using the present tense is often less complicated. e.g. The research by Elberg (1993) shows... (present tense)

e.g. The research by Elberg (1993) showed... (past tense)

e.g. The research by Elberg (1993) has shown... (present perfect tense)

6.4.3 Agreement between noun and number

Nouns need to agree with the number before them. In other words the names of things need to be singular (without an 's') or plural (with an 's') according to whether there is only one or more than one.

e.g. the first check_ ...several checks... ✓ (singular) (plural)

e.g. a pilot program_ ...three pilot programs ✓ (singular) (plural)

Some nouns do not use an 's' to indicate whether they are singular or plural.

e.g. three pieces of equipment_ ; much information_ ✓

6.4.4 Agreement between subject and verb

In a sentence, the subject and verb must agree in relation to number and to person:

e.g. The results ...are available ✓ (plural subject) (plural verb)

e.g. The organisation ...is progressing ✓ (single subject) (single verb)

e.g. I ...am responsible ✓ (first person subject)* (matching verb)

e.g. They ...are responsible ✓ (third person subject) (matching verb)

Mistakes with regard to agreement are common when a single subject is separated from its verb by other plural words, causing the subject to appear to be plural when it is not.

e.g. A list of results ...is available. ✓ (single subject) (single verb)

not A list of results are available. (A list are available??) ✗

The following rule helps students to check if they are using the correct verb (it applies when the present tense is used):

- if the subject is plural (ends in an 's'), then there is no 's' on the verb e.g.

The tests ...demonstrate_ ✓ (subject + 's') (verb with no 's')

- if the subject is single (does not end in an 's'), then the verb does end in an 's' e.g.

The test_ ...demonstrates ✓ (subject with no 's') (verb + 's')

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6.5 Vocabulary

6.5.1 Personal language

Currently in engineering reports, personal language is avoided in preference to impersonal constructions which emphasise scientific objectivity.

e.g. The mining site was investigated. ✓ not I investigated the mining site. ✗

* Check with your lecturer before using personal language such as 'I' or 'we'. Personal language is occasionally used to distinguish between the views or findings of other authors and your own. Alternatively, personal language can be avoided by using objective language.

e.g. Nguyen (1993) determined...whereas my conclusions are... (personal language) e.g. Nguyen (1993) determined...whereas the conclusion of this study is... (objective language)

6.5.2 Gender inclusive language

Gender neutral language is language which includes women and men equally. Technical reports should avoid using male-only terminology such as 'he'. This can be achieved by:

- using both pronouns

e.g. he/she; s/he; his/her ✓

- using gender non specific terms

e.g. The drafter was responsible for the plans ✓ not The draftsman was responsible for the plans ✗

- restructuring the sentence to avoid personal pronouns

e.g. A report can be completed by the engineer ✓ not The engineer can complete his report ✗

Another method of being inclusive is to use the plural pronoun 'their' with a single subject when it would not be ambiguous. In the past, this was considered to be grammatically incorrect, but the University of South Australia's current Policy on Gender Neutral Language (1991) explains that this is now an option.

e.g. The operator is to check their machine for safety... ✓

6.5.3 Jargon and slang

Clear and simple English should be used rather than jargon (specialised or pretentious language). New terms should be defined for the reader. A glossary of terms or abbreviations may be useful for long, complex reports.

e.g. The use of testing procedures was recommended. ✓ not The utilisation of testing procedures was recommended. ✗

Informal language or slang should also be avoided. The words which are used should be precise to give an accurate meaning.

e.g. The diode was soldered into the circuit. ✓ not The diode was put into the circuit. ✗

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e.g. Artificial intelligence aims to imitate a brain in a machine. ✓ not Artificial intelligence is like a brain in a machine. ✗

6.5.4 Repetition of words

Often writers have favourite words or expressions. Reports should avoid overuse of particular words, groups of words, or sentence structures.

not Then... Then... Then... ✗ not The test demonstrated... The result demonstrated ✗ not It is a requirement that... It is essential that... This signifies that... ✗

In technical writing, clear expression and attention to detail are essential.

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7 Conclusion

While this document was not developed to be an inflexible set of rules, it is however intended to assist university engineering students in producing professional, well

organised reports. Student report writing skills can be enhanced by the application of these guidelines in relation to formatting of documents, components, referencing and technical language.

Students are encouraged to write with attention to the contents of this Style Guide, but where instructed students should follow the directions given by their lecturers or school. Many resources, such as the library, are available to assist students in their report writing. The use of a specific format or style must be followed consistently throughout a report.

By initially planning the presentation of their material, students can concentrate on the content being presented in a report. During the writing and editing process students may find it useful to refer to the Report writing checklist (Appendix H). A well written and logically structured report will result in a quality document which effectively conveys precise information to the reader.

8 Recommendations

With regard to the conclusions detailed above, the following recommendations have been formulated. Students are recommended to:

- identify and follow instructions from particular Schools or employers — particularly where such instructions vary from common practice
- read and use the information in this Style Guide
- ensure internal consistency in formatting (particularly of numbering and headings)
- give attention to visual impact and readability
- acknowledge sources by referencing both in the text and in a list at the end •

proofread and edit thoroughly all written work

Students should remember that excellent content in a report needs to be supported by high quality presentation, and an accurate use of technical language.

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9 References *

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Research Degree Student Information Guide 1996, compiler Research Office, University of South Australia, Adelaide.

Style Manual for Authors, Editors and Printers 1994, 5th edn, Australian Government Publishing Service, Canberra.

Schwarz, S. E. & Oldham, W. G. 1984, Electrical Engineering: An Introduction, Holt, Rinehart and Winston, New York.

* This is an example of an alphabetical reference list — used with the author-date (Harvard) system.

(For students who use the numerical referencing system, see section 4.3.2 for an example of a numerical reference list.)

A bibliography (a wider list of all texts that have been read in preparation for writing this document) has not been included in this Style Guide, as it is generally not required for engineering reports.

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Appendix A National Competency Standards for Professional Engineers

Engineering students should take particular note of competencies 5.1.1 and 5.3.1, (our italics) which are expected of graduate students or Stage 1 Professional Engineers. Graduates are expected to be able to demonstrate competency in these areas with limited guidance.

5. COMMUNICATION

5.1 Communicate effectively in the English language

5.1.1 Communicates fluently in written and oral expression at a professional standard

5.1.2 Contributes to the preparation, interpretation and presentation of information

5.1.3 Liaises with colleagues and experts within the organisation

5.1.4 Correctly interprets engineering instructions received

5.1.5 Issues clear and accurate instructions to subordinates as appropriate

5.1.6 Selects appropriate methods of communication

5.2 Present, report on and advocate engineering ideas

5.2.1 Contributes to the preparation and presentation of lectures at a professional level

5.2.2 Prepares papers for publication in Professional Engineering journals 5.2.3

Communicates engineering information effectively within the engineering

work-force, and to others

5.2.4 Conveys engineering information effectively to higher engineering and non engineering levels in the enterprise

5.2.5 Undertakes professional development in negotiation, conflict resolution, counselling, exchanging ideas and conveying convictions and attitudes

5.3 Prepare and comprehend engineering documents

5.3.1 Prepares Professional Engineering reports

5.3.2 Contributes to specifications, standards and graphical representations

5.3.3 Participates in the preparation of more complex documents such as environmental impact statements

5.3.4 Correctly interprets applicable engineering drawings and other graphics, specifications, standards, regulations, codes of practice and environmental impact statements

from National Competency Standards for Professional Engineers (Stages 1 and 2) 1993, 1st edn, The Institution of Engineers, Australia, ACT.

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Appendix B Model title page

University of South Australia

Division of Information Technology, Engineering and the Environment

School of Electronic and Information Engineering

02478 Communication Systems 2

Computer Simulation Assignment

Student: Kim Citizen

ID Number: 9599999K

Tutor: Mr T. Ferris

Due date: 15/10/95

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Appendix C Plagiarism: Academic Misconduct Policy

Students should be aware of their University's policy on plagiarism. The following definition and policy regarding penalties are taken from the University of South Australia's policy on academic misconduct (from University of SA Policy no. A-11.2 1994). The full policy is located in the Student Diary.

Preamble

With respect to academic misconduct, a distinction will be made between a deliberate act and an inadvertent act occurring through unfamiliarity with academic conventions.

A specific form of academic misconduct is plagiarism. Deliberate plagiarism is regarded as a serious

act of academic misconduct.

Definition

1 Academic misconduct, whether inadvertent or deliberate, shall include the following:

1.1 Presentation of data with respect to laboratory work, field trips or other work which has been copied, falsified or in other ways obtained improperly;

1.2 Inclusion of material in individual work which includes significant assistance from another person in a manner unacceptable according to the assessment guidelines for the subject;

1.3 Providing assistance to a student in the presentation of individual work in a manner unacceptable according to the assessment guidelines for the subject;

1.4 Plagiarism, whether inadvertent or deliberate, which includes the following:

1.4.1 Word-for-word copying of sentences or whole paragraphs from one or more sources (the work or data of other persons), or presenting of substantial extracts from books, articles, theses, other unpublished work such as working papers, seminar and conference papers, internal reports, lecture notes or tapes, without clearly indicating their origin. This should be referenced appropriately.

1.4.2 Using very close paraphrasing of sentences or whole paragraphs without due acknowledgment in the form of reference to the original work;

1.4.3 Submitting another student's work in whole or in part;

1.4.4 Use of another person's ideas, work or research data without acknowledgment;

1.4.5 Submitting work which has been written by someone else on the student's behalf;

1.4.6 Copying computer files without clearly indicating their origin;

1.4.7 Submitting work which has been derived, in whole or in part, from another student's work by a process of mechanical transformation (eg changing variable names in computer programs);

1.5 Other actions not covered by the above clauses may be judged by a Dean or the Registrar to be acts of academic misconduct and be dealt with under the provisions of this policy.

Policy on penalties

2 A student found guilty of academic misconduct shall be subject to any of the following penalties:

2.1 failure, that is zero score, in the component of the subject, a note in the student's file and a reprimand from the Head of School;

2.1 failure in the subject, a note in the student's file and a reprimand from the Head of School;

2.3 suspension from the university for a period not exceeding three years;

2.4 expulsion from the university.

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Appendix D Punctuation in technical writing

The purpose of punctuation is to convey ideas clearly and without ambiguity.

Current publishing styles often use minimal or no punctuation, such as in lists.

1 Full stops .

- end a complete idea or sentence

e.g. The machine was operated by a technician for 24 hours a day.

- end an abbreviation where the final letter of the abbreviation is not the last letter of the word (other than discipline-specific abbreviations or units of measurement — check SI units)

e.g. p. for page and ch. for chapter

- show that a word(s) has been omitted using the three points of ellipsis (...)

e.g. Several causes...have been rectified.

2 Colons :

- introduce a list or quotation

e.g. The following factors are critical: cost recovery, priority of tasks and competitiveness. e.g. Farre

(1990, p. 51) stated: 'Education and training are central to the development of industry.'

3 Semicolons ;

- connect two sentences or main clauses which are closely connected, and often not joined with a linking word

e.g. The initial survey revealed a high interest; results showed that further development is valid. • separate complex or wordy items in a list

e.g. The following factors are critical: the environmental impact statement; the government and union

policies; the approval of business and council; and public opinion.

4 Commas ,

- show a pause or natural separation of ideas

e.g. After the recommendations were implemented, further evaluations were conducted. • bracket or separate information in a sentence

e.g. The additional results, which were withheld, raised doubts about the initial hypothesis. • precede linking words, such as 'but', 'so', 'hence', and 'whereas' e.g. The aim was to test strength and density, but the measurements taken were not accurate. • separate information in a list

e.g. The items included: circuit boards, several meters, wires and a power supply.

5 Parentheses ()

- provide information additional to a sentence

e.g. Engineering and Technology courses (as well as Applied Science) are studied at The Levels. • enclose author-date references in the text

e.g. A number of experiments (Smith 1987; Tan 1990; Wong 1991)...

- enclose the number for an equation, and bracket parts of the equation e.g. $x = G e^{-\omega t} \sin(\omega t + \phi)$ (1)

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6 Quotation marks ' ' or " " (single quotation marks are more common)

- indicate a shorter quotation as part of a sentence

e.g. 'In information technology, speed is of great importance,' as highlighted by Schwarz and Oldham

(1984, p. 2) in their introduction to the scope of Electrical Engineering. • show the titles of journal

articles

e.g. 'Efficient low cost solar cell developed' (1994) is an example of an article without an author.

7 Hyphens and dashes - hyphen - en rule — dash

- join words with a hyphen only when the meaning is changed without them

e.g. liquid-solid separation

- use an en rule (length of the dash equals 'n') to indicate a range e.g. 1939–1945
- use two dashes (em rules) when adding parenthetical information to a sentence e.g.

Understand the uses of the hyphen and the dash — the dash is longer — to avoid confusion.

8 Apostrophes for ownership 's

- place the apostrophe at the end of the owner-word, then add a possessive s

e.g. The researcher's results. (one researcher owns the results) add apostrophe + s

- if the original word ends in an s, place the apostrophe at the end of the owner word without adding a possessive s

e.g. The researchers' results. (more than one researcher owns the results) add apostrophe only (possessive s is invisible)

9 Capital letters XYZ

- use minimally, especially in titles and headings where small words such as 'and', 'in', 'the', and 'by' should not be capitalised
- use only for a specific and formally named item (e.g. 'Ohm's law'), not for a general reference to an item (such as 'a law')

10 Punctuating numerals and abbreviations

- write numbers of ten or less in words, except when followed by units

e.g. nine experiments e.g. 9 mm

- place a space between the numeral and the unit of measurement or operator sign, and do not use full stops with units of measurement e.g. 17 cm e.g. $y = v + 2$
- use SI units (Système International d'Unités)

- do not abbreviate or shorten forms of words in formal reports

e.g. it is a difficult situation so the company does not...

not it's a difficult situation so the company doesn't...

- use the standard or recognised abbreviations of your discipline

e.g. the current was 225 amps when measured...

11 Punctuating bullet point lists

- introduce lists with a colon (:) and either have no punctuation after each item, or punctuate as a normal sentence (with a full stop at the end)

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e.g. The Style Guide includes: • formatting • formatting;

- report components • report components; • referencing • referencing; and •

technical language • technical language.

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Appendix E Summary of elements in reference lists

This table outlines the elements needed for a reference list, and the order in which those elements appear in both the alphabetical (author-date) reference list, and the numbered reference list. These details are based on the AGPS Style Manual (1994). The type of reference list used must be consistent with the system of referencing selected for the particular report.

Numbered reference list

Alphabetical reference list

(Numerical system)

(Author-date system)

BOOKS – order the information as follows:

(differences highlighted in bold)

- author's surname • author's initials or given name

- author's initials or given name • author's surname

- year of publication

- title of publication • title of publication

(italics or underlined)

(italics or underlined)

- title of series ‡

- title of series ‡

- volume number, or number of volumes ‡

- volume number, or number of volumes

- edition ‡

‡ • edition ‡

- editor, reviser, compiler or translator ‡

- editor, reviser, compiler or translator

(if other than the author)

‡ (if other than the author)

- elements of a book ‡

- elements of a book ‡

- publisher

- publisher

- place of publication

- place of publication

- year of publication

- page numbers or numbers ‡

ARTICLES – order the information as

follows: (differences highlighted in bold)

- author's surname Note: ‡ indicates an optional element

- author's initials or given name • author's initials or given name • author's surname

- year of publication
- title of article
- title of article
- title of journal or periodical
- title of journal or periodical (italics or underlined)
- title of series ‡
- title of series ‡
- place of publication ‡
- place of publication ‡
- volume number ‡
- volume number ‡
- issue number or other identifier ‡
- issue number or other identifier ‡ • year of publication
- page numbers or numbers
- page numbers or numbers

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Personal communications — such as letters, conversations and email messages — need not be included in a reference list as they cannot be checked, and are usually incorporated into the text.

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Appendix F Library resources and report writing texts

These notes should be read in conjunction with relevant library publications.

Finding information

In most libraries, materials are arranged in Dewey decimal classification (i.e. resources on the same subject are shelved together). Below is a brief guide to the location of the main engineering subjects:

621 Mechanical Engineering 624 Civil Engineering 621.3 Electrical Engineering 660 Chemical Engineering 621.38 Electronic Engineering 669 Metallurgy 622 Mining 670 Manufacturing

Use the library catalogue — usually a computerised catalogue — to identify all possible sources of information in the library. Relevant engineering material can often be found by looking on the shelves in different areas of the collection.

Books written by subject experts may provide you with some of the required information.

CD-ROM databases are 'electronic indexes' which help you to identify relevant periodical articles, conference papers, product catalogues and standards. Useful engineering CD-ROMs may include: Applied Science and Technology Index;

Compendex (known as Engineering Index in the print version); Engineering and Applied Science (collection of Australian engineering databases); and INSPEC (for electrical and electronic engineering, computing, and manufacturing).

Reference materials such as encyclopedias, dictionaries and handbooks will help you to find brief, factual information such as definitions, constants and equations. Reference books can be found in the Reference Collection, and can only be used in the library.

Standards are very useful sources of practical information for design work. Check which standards are held in your library by using the catalogue or by asking at the help desk. Standards available may include Australian Standards and standards from international organisations such as British Standards, ASTM, ISO and IEEE.

Periodicals or journals provide up-to-date and often very detailed information. The Serials List or the catalogue shows which periodicals are held by the library.

Information desk

At the information desk a member of staff is available to assist you to locate information in your course. Staff can also help you with questions about how to use

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the catalogue, the CD-ROMS and other resources. Library staff are happy to help you with any problems — you only need to ask.

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Books on technical writing for engineers

Report and assignment writing

Baylis, P. 1991, *Report Writing for Engineers*, Engineering Education Australia, Milson's Point.

Blicq, R. 1995, *Writing Reports to Get Results: Quick Effective Results using the Pyramid Method*, 2nd edn, IEEE Press, New York.

Elton Mayo School of Management 1995, *Guidelines for Essay and Report Writing*, 13th edn, ed. A. Kay, EMSM Publications, Adelaide.

McEvedy, M. R. & Wyatt, P. 1990, *Presenting an Assignment*, Nelson, South Melbourne.

Michaelson, H. B. 1990, *How to Write and Publish Engineering Papers and Reports*, 3rd edn, Oryx Press, Phoenix.

Neufeld, J. K. 1987, *A Handbook for Technical Communication*, Prentice Hall, Englewood Cliffs, New Jersey.

Roze, M. 1994, *Technical Communication: The Practical Craft*, 2nd edn, Merrill, New York.

Turk, C. & Kirkman, J. 1989, *Effective Writing: Improving Scientific, Technical, and Business Communication*, 2nd edn, E. & F. N. Spon, London.

Weissberg, R. & Buker, S. 1990, *Writing Up Research: Experimental Research Report Writing for Students of English*, Prentice Hall Regents, New Jersey.

Thesis writing

Anderson, J. & Poole, M. 1994, Thesis and Assignment Writing, 2nd edn, John Wiley & Sons, Brisbane.

Lewins, F. W. 1993, Writing a Thesis: A Guide to its Nature and Organisation, 4th edn, Bibliotech, ANUTECH, Canberra.

Rudestam, K. E. & Newton, R. R. 1992, Surviving your Dissertation: A Comprehensive Guide to Content and Process, SAGE, Newbury Park, California.

Style guides and referencing

Kirkman, J. 1992, Good Style: Writing for Science and Technology, E. & F. N. Spon, London.

Li, X. & Crane, N. B. 1993, Electronic Style: A Guide to Citing Electronic Information, Meckler, Westport.

Style Manual for Authors, Editors and Printers 1994, 5th edn, Australian Government Publishing Service, Canberra.

Young, M. 1989, The Technical Writer's Handbook: Writing with Style and Clarity, University Science Books, California.

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Appendix G Purposes of paragraphs

A paragraph will be written well if the writer is certain of its purpose(s). Both broad and specific purposes of technical paragraphs are shown in the following table:

Broad purpose	Specific purpose	Identifying features of the paragraph (Type of paragraph)
Describing facts	Defining • term or name	• group/category/device/type/concept • distinguishing characteristics or elements
Complex defining	• additional specifications • explanation of operation • additional classification • further examples	
Classifying	• ordered members or items	• class or principles of arrangement • definition/basis of classification
(size, shape, type, category etc.)		
Describing	• item/place	
- Physical	• developmental stages and connections	
- Process	series of instructions relating to a procedure	
• use or purpose of an item		
- Function	• elements involved in a procedure	

- Special relationships • connection of two or more items

Describing Analysing cause and effect • statement of cause

the • list of effects - short and long term

relationship • statement of relationship

between facts • statement of reasons for effects

Exemplifying • example(s) or illustration(s)

(Explaining) • evidence

(often used with models, charts, tables etc.)

Comparing/Contrasting • statement of main idea

• development of idea

• statement of two or more

comparative/ contrasted items of

information

Adapted from

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Baylis, P. 1991, Report Writing for Engineers, Engineering Education Australia, Milson's Point.

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Appendix H Report writing checklist

Report writing checklist

For effective report writing it is essential to thoroughly edit your work. This checklist may be useful for the editing and redrafting process. Check each of the following:

Formatting improving layout and presentation

☐ headings & subheadings developed logically and consistently at each level (e.g. size and style of headings)

☐ decimal numbering system used accurately and consistently; fourth-level avoided Tables and figures

☐ key tables/figures used in the text (others, if required, in Appendix) ☐ zeros and rounded numbers used for clarity sometimes

☐ captions of tables/figures selected for specific differences between tables/figures ☐ caption information consistent with text information

☐ data in tables/figures consistent with data in report

☐ symbols/labels/signs explained clearly

☐ asterisks (*) added explanatory notes of further information, abbreviations, sources etc. that do not

fit into rows or columns

Components enhancing logic and readability

Preliminaries

☐ names/titles of people etc. spelt correctly & acknowledged fully

☐ abstract written to highlight and summarise significant information ☐ disclaimer signed and

authorship declared

- ☐ acknowledgments identified all who assisted with material and with editing ☐ contents pages matched exactly to text (e.g. titles of headings with decimal and page numbering)
- ☐ figures & tables lists matched exactly to labels and captions ☐ symbols list included all symbols from the text

Text of report

- ☐ sections divided clearly and developed logically ☐ abbreviations & acronyms written in full when first used
- ☐ definitions of new terms expressed accurately and clearly
- ☐ Appendices (each) referred to in the text of the report Referencing ensuring

correct academic standards

- ☐ all sources of information, referenced correctly twice, both in text and in reference list other than your own
- ☐ only one reference system used either author-date or numerical system – not both ☐ punctuation and elements standardised exactly (e.g. order of elements, punctuation, capitalisation and publishing details)

Technical language editing for style and grammatical accuracy

- ☐ repetition of words avoided
- ☐ sentences tightened, focused and varied in length ☐ longer sections or paragraphs introduced and/or summarised to provide an overview of the information given

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- ☐ passive voice used to emphasise the object of action rather than the agent ☐ parallel construction applied accurately in lists of information or sentences ☐ subject and verb agreement related in number and person
- ☐ gender inclusive language used correctly

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